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# Regime-Aware Tactical Asset Allocation for Equities and Fixed Income: Independent and Coupled Jump-Model Labels

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## Abstract

We study whether forecasted equity and bond return regimes improve monthly allocation across equities, bonds, and cash after transaction costs. The empirical design estimates realised regimes, forecasts next-month regimes, and maps forecast probabilities into portfolio weights. Unsupervised jump-model teachers estimate equity and bond return environments from realised return-based features. Student models are supervised only by these teacher labels and forecast next-month regimes from lagged macro-financial predictors; the COUPLED student also uses current-regime indicators and regime run length. The resulting forecast probabilities are converted into regime-conditional mean–variance allocations.

We compare two regime representations. The INDEPENDENT specification estimates separate two-state favourable and unfavourable labels for equities and bonds. The COUPLED specification estimates a three-state joint stock–bond label with calm, normal, and stressed regimes. Both representations distinguish the central portfolio mechanism: bonds support equity drawdowns in disinflationary episodes such as the dot-com bust and the global financial crisis, while inflation-driven sell-offs such as 1994 and 2022 weaken the bond hedge. In the student stage, regularised logistic regression and random forests produce informative next-month regime probabilities. In the allocation stage, selected dynamic strategies improve on the 60/40 benchmark along distinct margins, including lower drawdowns, higher Sharpe ratios, and higher terminal wealth, depending on risk aversion and turnover control.

## 1 Introduction

Balanced portfolios depend on how equity and bond returns interact across market environments. In disinflationary equity sell-offs, bond returns often rise and reduce portfolio losses. In inflation-driven sell-offs, equities and bonds can both lose value. A tactical allocation rule therefore needs information about the prevailing stock–bond environment when choosing next-month equity, bond, and cash exposures.

This paper studies monthly regime-aware allocation for a broad equity leg and a broad fixed-income leg. The equity leg represents the stock component of a balanced portfolio, and the bond leg represents the fixed-income component. The empirical design has three stages. First, unsupervised jump-model teachers estimate regime labels from equity and bond returns derived features. Second, student models forecast the next-month teacher labels from lagged macro-financial predictors; the COUPLED student also uses current-regime indicators and regime run length. Third, the allocation stage maps the forecast probabilities into regime-conditional return moments and monthly equity, bond, and cash weights.

We compare two regime-label structures. The INDEPENDENT approach estimates separate binary favourable-state labels for the equity and bond legs. The COUPLED approach estimates

one joint stock–bond label with calm, normal, and stressed states. The independent structure gives asset-level signals; the coupled structure gives a direct signal for the joint diversification environment.

The rest of the paper follows the same sequence. Section 2 describes the return series and predictors. Section 3 reports the realised regime labels. Section 4 reports next-month forecast performance. Section 5 evaluates the allocation results. The appendices report formulas, grids, diagnostics, and sensitivity checks.

## 2 Data

The empirical analysis uses monthly excess returns for the equity and bond legs; cash is the zero-excess-return asset in the allocation stage. The equity leg is the S&P 500. The bond leg is the Bloomberg aggregate bond-index. The one-month risk-free rate defines excess returns,

$$r_t^{\text{eq}} = \text{equity return}_t - rf_t, \quad r_t^{\text{bd}} = \text{bond return}_t - rf_t.$$

**Teacher features.** The INDEPENDENT equity teacher uses daily equity exponentially weighted returns and Sortino ratios over 21, 63, 126, and 252 trading days. The INDEPENDENT bond teacher uses monthly bond exponentially weighted returns, downside risk, Sharpe ratios, and volatility over one- to twelve-month horizons. The COUPLED teacher uses monthly equity and bond Sortino, drawdown, and trend features. Appendix A reports the formulas, horizons, and transformations.

**Student panel.** The student panels are monthly and lagged. The INDEPENDENT panel combines equity, Treasury, credit, commodity, volatility, macro, sentiment, and stock–bond correlation predictors, with causal winsorisation, rolling z-scores, and exponential smoothing. The COUPLED panel uses a screened monthly predictor set for the joint equity–bond label: lagged macro-financial, Treasury, commodity, and regime-context variables, together with current-regime indicators and regime run length. Appendices A and G report the predictor definitions and transformations.

## 3 Teacher: regime labelling with statistical jump models

The teacher stage defines the realised regimes that the student models later forecast. The jump-model teachers are unsupervised: they estimate regimes from return-based features. A regime label must be persistent and must separate different return outcomes. Persistence measures whether a state represents a sustained environment. State-conditional returns show whether the states distinguish favourable from adverse outcomes. Crisis-window diagnostics compare the labels with known stock–bond episodes.

### 3.1 The jump model

A jump model assigns each observation to a finite set of states while penalising frequent state changes. For a standardised feature vector  $x_t$ , state  $s_t$ , and state centres  $\{c_k\}$ , the state sequence solves

$$\min_{s_1, \dots, s_T, \{c_k\}} \sum_{t=1}^T d(x_t, c_{s_t}) + \lambda \sum_{t=2}^T \Lambda(s_{t-1}, s_t), \quad (1)$$

where  $d(\cdot, c_k)$  measures distance from state centre  $c_k$ ,  $\Lambda$  penalises switches, and  $\lambda$  controls persistence. Higher values of  $\lambda$  produce longer spells. Using an iterative approach, the model uses dynamic programming to find the state sequence from the current centers, then updates the centers based on that sequence. To avoid local minima, this loop is run from several different initializations.

### 3.2 Two regime parameterisations

While both regime specifications rely on the jump-model penalty, they define distinct state spaces. Both teachers are fit causally; thus, the label for month  $t$  is derived strictly from

observations available at or before that point.

**Independant: two independent binary labels.** The INDEPENDENT teacher estimates separate two-state jump models for the equity and bond legs. For asset  $a \in \{\text{eq}, \text{bd}\}$ , feature vector  $x_i^a$ , and teacher member  $p$ , the state sequence solves

$$\min_{s_1^a, \dots, s_{T_a}^a, \{c_0^a, c_1^a\}} \sum_{i=1}^{T_a} d_p(x_i^a, c_{s_i^a}^a) + \lambda_p \sum_{i=2}^{T_a} \mathbf{1}\{s_i^a \neq s_{i-1}^a\}. \quad (2)$$

The equity teacher relies on daily observations, whereas the bond teacher uses monthly data. Both models employ an identical 12-year historical training window before starting and update at each month-end. During these updates, features are standardized using only the training sample available up to that specific date.

The teacher ensemble aggregates models across a grid of persistence penalties ( $\lambda_p$ ) using both  $L^2$  (squared Euclidean loss with mean centers) and  $L^1$  (absolute loss with medoid centers) specifications. Appendix B details the parameter grids and ensemble construction.

After each fit, hidden states are ranked by their realized in-sample Sharpe ratio, and the higher-Sharpe state ( $g^{a,p}$ ) is labeled “favorable.” Because the equity teacher runs on daily data while the bond teacher runs on monthly data, their monthly target labels are defined as:

$$S_m^{\text{eq},p} = \mathbf{1}\left\{\frac{1}{n_m} \sum_{i \in m} \mathbf{1}\{s_i^{\text{eq},p} = g^{\text{eq},p}\} \geq \frac{1}{2}\right\}, \quad S_m^{\text{bd},p} = \mathbf{1}\{s_m^{\text{bd},p} = g^{\text{bd},p}\}. \quad (3)$$

Thus, the monthly equity target is a majority vote of its daily states, whereas the bond target is simply that month’s state.

**coupled: three joint regimes with a coupling probability.** The COUPLED approach represents the equity–bond environment with three joint regimes. Each leg has a benign or adverse bit, and the admissible joint states are

$$\mathcal{S} = \{\text{calm} = (0, 0), \text{ normal} = (1, 0), \text{ stressed} = (1, 1)\},$$

where the pair is the equity bit and the bond bit. Calm means that both legs are benign; normal means that the equity leg is in its high-uncertainty state—positive on average but with the widest dispersion of the three regimes, spanning the disinflationary drawdowns such as the dot-com bust and the global financial crisis—while bonds remain benign and provide the hedge; and stressed means that both legs are genuinely adverse.

For a joint state  $j$  with bits  $(j_E, j_B)$ , the per-month cost combines the two leg-specific distances with a coupling term:

$$e_t(j) = d(z_t^{\text{eq}}, \mu_{j_E}^{\text{eq}}) + d(z_t^{\text{bd}}, \mu_{j_B}^{\text{bd}}) + \eta \left[ \kappa_t \mathbf{1}\{j \in D\} + (1 - \kappa_t) \mathbf{1}\{j \in A\} \right], \quad (4)$$

where  $A$  denotes the agree states, calm and stressed, and  $D$  denotes the disagree state, normal. The scalar  $\eta > 0$  sets the strength of the coupling term, and  $\kappa_t \in [0, 1]$  is a learned *co-movement probability*: the model-implied probability that, given current market conditions, the equity and bond legs occupy the same (agree) regime rather than the split (disagree) regime. Because  $\kappa_t$  multiplies the disagree state and  $(1 - \kappa_t)$  the agree states in (4), a high  $\kappa_t$  makes the agree states (calm, stressed) cheaper and hence more likely, while a low  $\kappa_t$  makes the disagree state (normal) cheaper; the coupling term is thus a soft, data-driven prior on whether the two legs move together in month  $t$ . Given the current state path,  $\kappa_t$  is fitted by an  $L^2$ -penalised logistic regression of the leg-agreement indicator  $y_t = \mathbf{1}\{s_t^{\text{eq}} = s_t^{\text{bd}}\}$  on equity–bond correlation, equity and bond trend, and equity and bond drawdown:

$$\kappa_t = \sigma(\beta_0 + \beta^\top x_t^{\text{mkt}}), \quad x_t^{\text{mkt}} = \{\rho_{12,t}^{\text{eq,bd}}, \text{trend}_t^{\text{eq}}, \text{trend}_t^{\text{bd}}, \text{dd}_t^{\text{eq}}, \text{dd}_t^{\text{bd}}\}. \quad (5)$$

The transition penalty is applied separately to the equity and bond bits,  $\Lambda(i, j) = \lambda \mathbf{1}\{i_E \neq j_E\} + \lambda \mathbf{1}\{i_B \neq j_B\}$ . The coupling term acts as a soft bias in the state cost. It raises the cost of states that conflict with the fitted equity–bond dependence signal and lowers the cost of states supported by that signal.

The coupling logistic (5) is re-fitted at *every* labelled month, with no delay: at month  $t$  it uses only the expanding history through  $t$  (refit cadence one month), and it is estimated *jointly* with the state path inside the coordinate descent that defines the teacher. Each iteration re-estimates the leg centres, refits the logistic of the current leg-agreement indicator  $y_t = \mathbf{1}\{s_t^{\text{eq}} = s_t^{\text{bd}}\}$  on  $x_t^{\text{mkt}}$ , and updates the path by dynamic programming under (4). This also fixes the supervised/unsupervised status of the method: the jump model is unsupervised, and although the coupling logistic is a supervised classifier, its target  $y_t$  is generated by that same unsupervised model rather than by any external or forward-looking label. The teacher stage is therefore unsupervised with respect to future returns and strictly causal. Appendix D specifies the coupling logistic, its inputs, its estimation, and its effect on the state cost in full.

**Relationship.** The two parameterisations define different regime objects. The INDEPENDENT labels keep equity and bond regimes separate, so the final state can take any of the four favourable–unfavourable combinations. The COUPLED label is a single joint state. Calm corresponds to strongly favourable equity and favourable bonds; normal to uncertain equity (positive on average but spanning equity drawdowns) with favourable, hedging bonds; and stressed to unfavourable equity and unfavourable bonds. The INDEPENDENT representation gives two asset-level regime signals. The COUPLED representation gives one regime signal for the joint equity–bond environment.

### 3.3 Ensembling and Voting

The final regime labels are aggregated across a grid of persistence penalties and specifications. For the INDEPENDENT approach, equity and bond paths are combined independently via a majority vote across all constituent models. For the COUPLED approach, the ensemble aggregates 22 joint-regime specifications formed by crossing distinct feature families with the  $L^1$  and  $L^2$  loss configurations. In both settings, the resulting vote shares quantify model consensus. Appendix B and Appendix D detail the explicit voting formulas, transition matrices, and model-level diagnostics.

### 3.4 Label results

The two teacher specifications produce persistent regimes with distinct return profiles. Figure 1 compares their label histories: the INDEPENDENT panel marks favourable-regime months for each leg, while the COUPLED panel shades the same stock–bond history by calm, normal, and stressed joint regimes. Appendix D reports coupled vote shares and the coupling-probability path.



**Figure 1:** Causal teacher-label representations of the stock–bond history. Both panels report labels formed using information available through the labelled month. Left (INDEPENDENT): equity and bond excess-return indexes with favourable-regime months marked for each leg. Right (COUPLED): cumulative excess-return indexes shaded by calm, normal, and stressed joint regimes.



**State economics.** Table 1 reports label-conditional return statistics. In the INDEPENDENT panel, favourable states have positive Sharpe ratios and unfavourable states have negative Sharpe ratios for both equity and bonds. In the COUPLED panel, calm is the strongest equity state; normal is the high-uncertainty equity state—positive on average but the widest-dispersion regime, and the one containing the disinflationary equity drawdowns—where bonds provide reliable support; and stressed records losses in both risky legs.

**Table 1:** Label-conditional economics. The table reports same-month annualised excess return, annualised volatility, Sharpe ratio, sample share, and average spell duration. The INDEPENDENT panel conditions on each leg’s binary label. The COUPLED panel conditions on the three joint regimes. Panel headers report the causal label sample used for each representation.

| Independent: two binary labels (1988–2024) |              |       |        |       |       |      | Coupled: three joint regimes (1989-01 to 2024-11) |          |       |        |       |       |      |
|--------------------------------------------|--------------|-------|--------|-------|-------|------|---------------------------------------------------|----------|-------|--------|-------|-------|------|
| Class                                      | Regime       | Share | Ret.   | Vol.  | Sh.   | Dur. | Class                                             | Regime   | Share | Ret.   | Vol.  | Sh.   | Dur. |
| Equity                                     | Favourable   | 76.3% | +13.8% | 12.1% | +1.14 | 14.3 | Equity                                            | Calm     | 14.2% | +21.0% | 8.1%  | +2.60 | 4.7  |
| Equity                                     | Unfavourable | 23.7% | −15.4% | 19.8% | −0.78 | 4.6  | Equity                                            | Normal   | 76.6% | +10.1% | 14.9% | +0.68 | 11.8 |
| Bond                                       | Favourable   | 67.1% | +5.7%  | 3.2%  | +1.81 | 13.8 | Equity                                            | Stressed | 9.3%  | −40.9% | 13.7% | −2.99 | 2.2  |
| Bond                                       | Unfavourable | 32.9% | −4.1%  | 5.1%  | −0.81 | 6.8  | Bond                                              | Calm     | 14.2% | +0.8%  | 3.5%  | +0.22 | 4.7  |
|                                            |              |       |        |       |       |      | Bond                                              | Normal   | 76.6% | +4.1%  | 3.9%  | +1.06 | 11.8 |
|                                            |              |       |        |       |       |      | Bond                                              | Stressed | 9.3%  | −8.4%  | 5.0%  | −1.69 | 2.2  |

**Persistence.** Persistent labels make the forecast horizon economically usable: a one-month-ahead signal has allocation value only if regimes last long enough for portfolio weights to adjust without being dominated by turnover costs. Table 1 reports average spell durations, and Appendices B and D report the transition matrices and switch-count diagnostics.

**Crisis windows.** Table 2 compares the labels across historical stress windows. The comparison separates equity drawdowns with bond support from episodes in which equity and bond returns fall together.

**Table 2:** Crisis-window behaviour under both labellings. The INDEPENDENT columns report the bond label’s favourable share and Sharpe ratio. The COUPLED columns report the joint regime mix within each window and the equity and bond Sharpe ratios. Sharpe ratios use same-month annualised excess returns over each window.

| Window                  | INDEPENDENT bond |        | COUPLED regime share |        |         | COUPLED Sharpe |       |
|-------------------------|------------------|--------|----------------------|--------|---------|----------------|-------|
|                         | Fav.             | Sharpe | Calm                 | Normal | Stress. | Eq.            | Bd.   |
| 1990 recession          | 100.0%           | +1.33  | 0%                   | 100%   | 0%      | +0.05          | +1.41 |
| 1994 bond shock         | 8.3%             | −1.55  | 8%                   | 33%    | 58%     | −0.48          | −1.61 |
| Dot-com bust            | 87.5%            | +1.81  | 0%                   | 97%    | 3%      | −1.07          | +1.97 |
| Global financial crisis | 72.2%            | +0.74  | 0%                   | 94%    | 6%      | −1.58          | +0.74 |
| Euro sovereign 2011     | 100.0%           | +3.46  | 12%                  | 88%    | 0%      | −0.61          | +3.69 |
| Covid crash             | 100.0%           | +2.25  | 0%                   | 100%   | 0%      | −0.88          | +2.76 |
| Inflation 2022          | 0.0%             | −1.82  | 0%                   | 33%    | 67%     | −0.93          | −1.90 |

The crisis evidence links the regime labels to the allocation question. The disinflationary equity sell-offs sit in the normal regime, where the bond hedge stays intact, while the inflation-led sell-offs (1994, 2022) sit in the stressed regime, where both legs fall. Disinflationary sell-offs therefore favour bond exposure, while inflation-led sell-offs call for lower exposure to both risky legs.

**Feature separation.** Feature-separation diagnostics identify the inputs that define the teacher states. The INDEPENDENT equity labels load most on medium-horizon Sortino ratios and return trends; the bond labels load most on Sharpe ratios and downside-risk measures. In the COUPLED teacher, equity trend distinguishes calm months, and bond trend and bond Sortino measures distinguish stressed months. Full rankings are reported in Appendices B and D.

## 4 Student: forecasting the next-month regime

The student stage turns realised teacher labels into out-of-sample forecast probabilities. At the end of month  $t$ , the predictors  $X_t$  are observed and the target is the teacher label in month  $t + 1$ . The INDEPENDENT student estimates two binary favourable-state probabilities, one for equity and one for bonds:

$$\hat{p}_{t+1|t}^a = \Pr(S_{t+1}^a = 1 \mid X_t), \quad a \in \{\text{eq, bd}\}. \quad (6)$$

The COUPLED student estimates the corresponding three-class probability vector over calm, normal, and stressed joint regimes. The student models use regularised logistic regression and random forests; Appendix E and Appendix G report the predictor designs, validation protocol, model grids, and full forecast diagnostics.

**Forecast quality.** Table 3 reports out-of-sample forecast performance. The INDEPENDENT forecasts are evaluated as two binary favourable-state predictions. The COUPLED forecast is evaluated as a three-class probability forecast over the joint stock–bond regimes. AUC values are therefore compared within panels, not as direct cross-panel levels: the INDEPENDENT tasks are binary per-leg forecasts, while the COUPLED task is a three-class joint-regime forecast. The allocation section tests whether these probability forecasts improve net portfolio outcomes.

**Table 3:** Out-of-sample student forecast quality. The INDEPENDENT rows report binary diagnostics for the equity and bond favourable-state forecasts. The full independent predictor-design results are in Table F.20. The COUPLED rows report diagnostics for the joint three-class forecast; AUC is the macro one-vs-rest average, and Brier and log loss are reported in their multiclass forms.

| Approach    | Task   | Model               | OOS sample | AUC   | Bal. acc. | Brier | Log loss |
|-------------|--------|---------------------|------------|-------|-----------|-------|----------|
| INDEPENDENT | Equity | Logistic regression | 1996–2024  | 0.929 | 0.846     | 0.088 | 0.296    |
| INDEPENDENT | Equity | Random forest       | 1996–2024  | 0.903 | 0.831     | 0.107 | 0.361    |
| INDEPENDENT | Bond   | Logistic regression | 1996–2024  | 0.892 | 0.831     | 0.128 | 0.462    |
| INDEPENDENT | Bond   | Random forest       | 1996–2024  | 0.873 | 0.800     | 0.147 | 0.471    |
| COUPLED     | Joint  | Logistic regression | 1996–2024  | 0.724 | 0.620     | 0.553 | 1.074    |
| COUPLED     | Joint  | Random forest       | 1996–2024  | 0.759 | 0.593     | 0.367 | 0.747    |

The INDEPENDENT results favour logistic regression on the headline binary metrics, while the COUPLED results show a trade-off between class accuracy and probabilistic scoring. Full predictor-design comparisons and probability-path diagnostics are reported in Appendices F and G.

**Predictor relevance.** Predictor-relevance diagnostics are reported in Appendices F and G. Equity-regime forecasts rely more on equity cross-section, industry, and style information. Bond-regime forecasts use a broader mix of equity, Treasury, and credit information. The COUPLED forecast draws on a more dispersed set of regime-context, commodity, Treasury, and macro-financial blocks, consistent with a joint stock–bond target.

## 5 Regime-aware allocation

The allocation stage assesses whether student forecast probabilities enhance a monthly equity–bond–cash portfolio after accounting for transaction costs. At the end of each month  $t$ , these forecasts are mapped into regime-conditional return moments, which then serve as inputs for a mean-variance optimization to determine target portfolio weights. We evaluate the realized returns net of one-way transaction costs, comparing the performance against a monthly rebalanced 60/40 and additional benchmarks.

For the INDEPENDENT allocator,  $\hat{p}_{t+1|t}^a$  is the end-of-month- $t$  probability that asset  $a \in \{\text{eq, bd}\}$  is favourable in month  $t + 1$ . State-specific means and variances are estimated from past returns assigned to each state and shrunk toward pre-out-of-sample unconditional moments with weight  $n_{s,t}^a / (n_{s,t}^a + 36)$ . The probability-mixture mean is

$$\hat{\mu}_t^a = (1 - \hat{p}_{t+1|t}^a) \hat{\mu}_{0,t}^a + \hat{p}_{t+1|t}^a \hat{\mu}_{1,t}^a. \quad (7)$$

The variance uses the corresponding second-moment mixture. The covariance matrix uses an expanding stock–bond correlation estimate shrunk toward the pre-prediction stock–bond correlation. Appendix H reports the full probability-to-moment and realised net-return formulas.

For the INDEPENDENT allocator, the monthly portfolio is long-only in equity, bonds, and cash and solves

$$w_t = \arg \max_{w \in \mathcal{W}} \left\{ w^\top \hat{\mu}_t - \gamma w^\top \hat{\Sigma}_t w - \tau c \|w - w_{t-1}\|_1 \right\}, \quad (8)$$

where  $\mathcal{W}$  is the long-only equity–bond–cash simplex,  $\gamma$  is risk aversion,  $c$  is the one-way transaction cost, and  $\tau$  scales turnover control. Appendix H reports the independent grid, selected-strategy settings, additional benchmarks, and parameter sweeps.

The COUPLED allocator uses the predicted probabilities of calm, normal, and stressed joint regimes in the same portfolio objective. Its conditional moments are formed as a probability mixture over the three joint regimes, estimated from past returns assigned to each joint regime using the same shrinkage, probability smoothing, and covariance construction as the INDEPENDENT allocator. The two allocators are otherwise identical: they share the long-only equity–bond–cash universe, the 0.02 weight grid, the mean–variance objective (8), the identical risk-aversion, turnover-penalty, transaction-cost, and probability-smoothing grids, and the 10 bps headline one-way cost, and they are evaluated on the same out-of-sample window (1996-02 to 2024-10). They differ only in the regime representation that supplies the conditional moments, which isolates the value of the representation itself. Appendix I reports the coupled allocation details, benchmark rules, cost sensitivity, and parameter sweeps. The headline results for both approaches are evaluated against the 60/40 benchmark.

### 5.1 Allocation trade-off

The allocation results show that regime forecasts have different portfolio roles across the selected dynamic strategies. Table 4 reports three roles from each allocation grid: a defensive allocation, a low-turnover allocation, and a high-return allocation. Both approaches define these roles mechanically on the same headline-cost grid: the defensive role maximises the Sharpe ratio subject to a maximum-drawdown floor, the low-turnover role minimises annual turnover subject to a Sharpe floor, and the high-return role maximises annualised excess return subject to a maximum-drawdown floor (Appendices H.6 and I.2). The selected rows summarise the main trade-off between drawdown control, trading intensity, and terminal wealth. Appendices H and I report the selection protocol, full allocation tables, cost sensitivity, and parameter sweeps. Additional independent benchmarks, including equity-only, bond-only, and volatility-targeted 60/40 allocations, are reported in Appendix H.5.

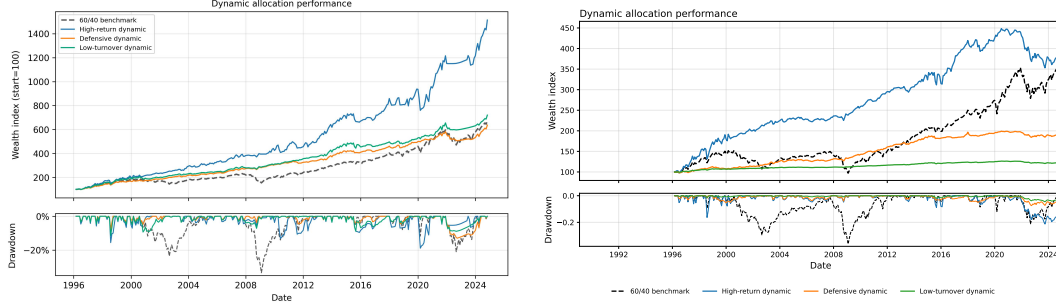
**Table 4:** Selected allocation results, out of sample. The table reports representative dynamic roles and the 60/40 benchmark, which is reported once in the INDEPENDENT panel. Returns are annualised excess returns net of 10 bps one-way transaction costs; “Final” indexes wealth to 100 at sample start. Both approaches use long-only equity–bond–cash weights and monthly rebalancing; the appendices report approach-specific moment estimates, grids, and turnover-penalty conventions.

| Approach    | Role         | Strategy                                  | Ann. exc. | Vol.  | Sharpe | Max DD | Final  |
|-------------|--------------|-------------------------------------------|-----------|-------|--------|--------|--------|
| INDEPENDENT | Defensive    | Random forest, smoothed predictors        | +4.4%     | 5.8%  | 0.76   | −13.0% | 636.6  |
| INDEPENDENT | Low-turnover | Logistic, z-score + smoothed z-score      | +4.8%     | 7.2%  | 0.67   | −11.5% | 720.0  |
| INDEPENDENT | High-return  | Logistic, smoothed predictors             | +7.6%     | 10.6% | 0.71   | −18.9% | 1513.6 |
| INDEPENDENT | Benchmark    | 60/40                                     | +4.5%     | 9.6%  | 0.47   | −36.2% | 359.4  |
| COUPLED     | Defensive    | Logistic, prob. mixture ( $\gamma=3$ )    | +2.3%     | 3.2%  | 0.72   | −7.5%  | 188.8  |
| COUPLED     | Low-turnover | Logistic, prob. mixture ( $\gamma=12$ )   | +0.7%     | 1.1%  | 0.61   | −4.6%  | 121.9  |
| COUPLED     | High-return  | Logistic, prob. mixture ( $\gamma=0.25$ ) | +4.9%     | 8.1%  | 0.61   | −21.3% | 372.9  |

Table 4 shows that the selected strategies improve on the 60/40 benchmark along different margins. Defensive allocations mainly reduce drawdowns and raise Sharpe ratios. Low-turnover allocations preserve part of this risk control with less aggressive rebalancing.

High-return allocations accept larger drawdowns in exchange for greater participation in favourable regimes. The full metric set is reported in Tables I.41, H.24, and H.25.

Figure 2 reports the corresponding wealth and drawdown paths.



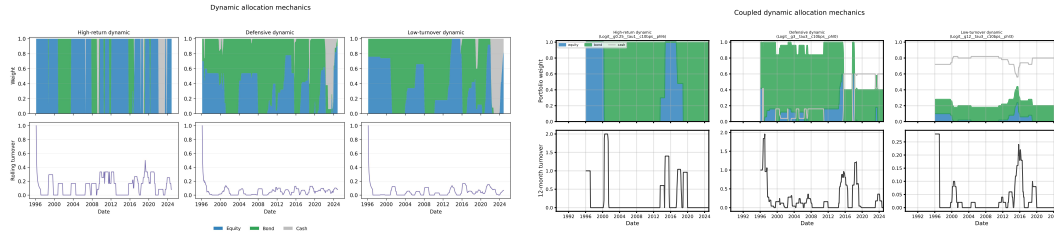
**Figure 2:** Dynamic allocation performance, out of sample. The panels report net wealth indexes and drawdowns for selected dynamic allocations and the 60/40 benchmark, with wealth indexed to 100 at the start of each evaluation sample.

## 5.2 Portfolio-weight behaviour

Figure 3 illustrates the empirical allocation mechanics and the resulting trade-offs between position smoothing and turnover. As visible in the structural profiles across weight plots, the optimization behavior splits along strategic objectives:

The high-return rules frequently trigger discrete, binary switches between asset classes, hitting corner solutions where the portfolio shifts entirely into a single dominant asset class. This behavior occurs because the optimization targets maximum conditional returns, causing aggressive portfolio rebalancing and sharp spikes in rolling turnover whenever a regime shift is predicted.

Conversely, the defensive and low-turnover configurations exhibit highly continuous, smooth weight trajectories. Rather than abruptly abandoning an asset class, these strategies scale exposures gradually in response to shifting forecast probabilities.



**Figure 3:** Portfolio weights and turnover, out of sample. The panels report equity, bond, and cash weights and turnover for the selected dynamic allocations.

## 5.3 Sensitivity

Appendices H and I report sensitivity to risk aversion, turnover penalties, transaction costs, probability smoothing, and moment shrinkage. The patterns are stable: higher risk aversion lowers risky exposure, smoothing and turnover penalties reduce trading, and higher transaction costs lower net performance.

## 6 Discussion and conclusion

This paper asked whether regimes estimated from stock and bond return behaviour can be forecast one month ahead and used in allocation. The answer is yes, but in a benchmark-relative sense. Regime forecasts create useful portfolio choices along different margins: drawdown control, Sharpe improvement, turnover reduction, and terminal wealth.

The first contribution is the construction of economically meaningful teacher labels. The INDEPENDENT labels separate favourable and unfavourable environments for each asset class: favourable equity and bond states have positive Sharpe ratios, while unfavourable states have negative Sharpe ratios. The COUPLED labels separate the stock–bond environment directly. Calm and normal regimes describe periods in which at least one risky leg remains useful, whereas stressed regimes identify months in which both equity and bond returns are adverse. This matters because the allocation problem is not only whether equities are attractive, but whether bonds still diversify equity risk.

The crisis-window evidence gives the clearest economic interpretation of the labels. During the dot-com bust, the global financial crisis, the euro-sovereign episode, and the Covid crash, bond exposure remained favourable and helped the balanced portfolio. During the 1994 bond shock and the 2022 inflation episode, bond exposure became part of the adverse environment. The regime models therefore recover the distinction that motivated the project: some equity sell-offs preserve the stock–bond hedge, while inflation-led sell-offs can damage both legs at the same time.

The second contribution is the separation between regime identification and regime forecasting. The teacher stage uses realised information to define the states. The student stage then asks whether those states are predictable from lagged macro-financial and regime-context variables. The out-of-sample diagnostics show that this step is feasible: the INDEPENDENT equity and bond labels are forecast with strong binary performance, and the COUPLED labels retain useful information despite the harder three-class joint-regime task. This makes the exercise more demanding than an ex-post regime interpretation.

The allocation results show where the forecasts have economic value. Defensive rules use adverse-regime probabilities to reduce drawdowns relative to 60/40. Low-turnover and balanced rules retain part of this risk control while limiting rebalancing. High-return rules use favourable forecasts to keep larger risky exposure and improve terminal wealth. The evidence therefore supports regime forecasts as allocation inputs, not as stand-alone market-timing signals.

## A Data Sources and Feature Construction

**Table A.5:** Teacher features by approach. The INDEPENDENT teacher uses daily equity features and monthly bond features. The COUPLED teacher uses a monthly joint stock–bond construction. Horizons are reported in trading days for daily equity features and in months for monthly features.

| Approach               | Feature                                                                | Definition                                                                                                                                         | Horizons                      |
|------------------------|------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| INDEPENDENT (equity)   | Exponentially weighted return; Sortino ratio                           | Daily equity excess-return features based on recent mean performance and downside-risk-adjusted performance                                        | 21, 63, 126, 252 trading days |
| INDEPENDENT (bond)     | Exponentially weighted return; downside risk; Sharpe ratio; volatility | Monthly bond excess-return features based on recent mean performance, negative-return magnitude, rolling risk-adjusted performance, and volatility | 1, 3, 6, 12 months            |
| COUPLED (equity, bond) | Sortino ratio                                                          | Recent average excess return over downside risk (EWMA of squared negative excess returns)                                                          | 1, 2, 4, 12 m                 |
| COUPLED (equity, bond) | Drawdown level / term structure                                        | Log downside deviation; short minus long log downside deviation                                                                                    | 6 m; 2 vs 12 m                |
| COUPLED (equity, bond) | MA crossover / trend                                                   | Short minus long average excess return, scaled by downside risk; EWMA-standardised trend                                                           | 3 vs 12 m; 12, 24 m           |

### A.1 Time series

| Time series                                                                                                                       | Frequency used | Source    |
|-----------------------------------------------------------------------------------------------------------------------------------|----------------|-----------|
| S&P 500 Index level                                                                                                               | Daily          | Bloomberg |
| Bloomberg US Aggregate Bond Index level                                                                                           | Monthly        | Bloomberg |
| Fama–French market excess return, size, value, profitability, investment, momentum, and risk-free return                          | Monthly        | WRDS FF   |
| Six Fama–French size and book-to-market portfolio returns: small low, small medium, small high, big low, big medium, and big high | Monthly        | WRDS FF   |
| CRSP common-stock returns, prices, shares outstanding, SIC classifications, share classifications, and exchange identifiers       | Monthly        | WRDS CRSP |
| CRSP Treasury security returns, total amount outstanding, and maturity dates                                                      | Monthly        | WRDS Tsy  |
| CRSP fixed-term Treasury nominal yields at 1 month, 3 months, 1 year, 2 years, 5 years, 7 years, 10 years, 20 years, and 30 years | Monthly        | WRDS Tsy  |
| Equal-weight commodity portfolio excess return from <i>Commodities for the Long Run</i>                                           | Monthly        | AQR       |
| National Financial Conditions Index                                                                                               | Monthly        | FRED      |
| Adjusted National Financial Conditions Index                                                                                      | Monthly        | FRED      |
| Investment-grade corporate bond fund adjusted close                                                                               | Monthly        | Yahoo     |
| High-yield corporate bond fund adjusted close                                                                                     | Monthly        | Yahoo     |
| One-year Treasury yield                                                                                                           | Monthly        | FRED      |
| Moody’s Baa and Aaa corporate bond yields                                                                                         | Monthly        | FRED      |
| Sticky-price core consumer inflation                                                                                              | Monthly        | FRED      |
| CBOE VIX Index                                                                                                                    | Monthly        | FRED      |
| Chicago Fed National Activity Index                                                                                               | Monthly        | FRED      |
| University of Michigan Consumer Sentiment Index                                                                                   | Monthly        | FRED      |
| CRSP value-weighted equity market return                                                                                          | Monthly        | WRDS CRSP |
| S&P 500 return                                                                                                                    | Monthly        | WRDS CRSP |
| Aggregate bond return                                                                                                             | Monthly        | Bloomberg |
| Ten-year Treasury bond return                                                                                                     | Monthly        | WRDS Tsy  |

## A.2 Teacher-feature definitions

**Common return definitions.**

$$\alpha_h = 1 - \exp\left(\frac{\log(0.5)}{h}\right), \quad (9)$$

$$\text{EWM}_{h,t}(r) = \alpha_h r_t + (1 - \alpha_h) \text{EWM}_{h,t-1}(r), \quad (10)$$

$$r_t^- = \min\{r_t, 0\}, \quad (11)$$

$$D_{h,t}(r) = [\text{EWM}_{h,t}((r_t^-)^2)]^{1/2}. \quad (12)$$

**Equity teacher features.** Daily equity excess return:  $r_t^E$ . Horizons:  $h \in \{21, 63, 126, 252\}$ .

$$\text{EWMRet}_{h,t}^E = \text{EWM}_{h,t}(r^E), \quad (13)$$

$$\text{Sortino}_{h,t}^E = \frac{\text{EWM}_{h,t}(r^E)}{D_{h,t}(r^E) + 10^{-12}}. \quad (14)$$

**Bond teacher features.** Monthly bond excess return:  $r_t^B$ .

$$\text{EWMRet}_{h,t}^B = \text{EWM}_{h,t}(r^B), \quad h \in \{3, 6, 12\}, \quad (15)$$

$$\text{Sharpe}_{h,t}^B = \frac{\text{mean}(r_{t-h+1}^B, \dots, r_t^B)}{\text{sd}(r_{t-h+1}^B, \dots, r_t^B) + 10^{-12}}, \quad h \in \{3, 6, 12\}, \quad (16)$$

$$\text{LogDownside}_{h,t}^B = \log(D_{h,t}(r^B) + 10^{-12}), \quad h \in \{1, 3, 6, 12\}, \quad (17)$$

$$\text{Vol}_{6,t}^B = \text{sd}(r_{t-5}^B, \dots, r_t^B). \quad (18)$$

### A.3 Student base-feature definitions

**Rolling scalar features.**

$$\text{Sum}_{h,t}(x) = \sum_{j=0}^{h-1} x_{t-j}, \quad (19)$$

$$\text{Mean}_{h,t}(x) = \frac{1}{h} \sum_{j=0}^{h-1} x_{t-j}, \quad (20)$$

$$\text{Vol}_{h,t}(x) = \text{sd}(x_{t-h+1}, \dots, x_t). \quad (21)$$

**Return-block features.** For  $x_{1,t}, \dots, x_{K,t}$ ,

$$\text{Disp}_t = \text{sd}(x_{1,t}, \dots, x_{K,t}), \quad (22)$$

$$\text{BMW}_t = \max_k x_{k,t} - \min_k x_{k,t}. \quad (23)$$

**Correlation-matrix features.** For the 12-month rolling correlation matrix  $C_t$ ,

$$\bar{\rho}_t = \text{mean}_{i < j}(C_{ij,t}), \quad (24)$$

$$|\bar{\rho}|_t = \text{mean}_{i < j} |C_{ij,t}|, \quad (25)$$

$$\text{PC1}_t = \frac{\lambda_{\max}(C_t)}{\sum_j \lambda_j(C_t)}, \quad (26)$$

$$\text{Rotation}_t = \|C_t - C_{t-1}\|_F. \quad (27)$$

**Fisher correlation changes.**

$$\rho_t^c = \min\{\max\{\rho_t, -0.999\}, 0.999\}, \quad (28)$$

$$F_t = \frac{1}{2} \log\left(\frac{1 + \rho_t^c}{1 - \rho_t^c}\right), \quad (29)$$

$$\Delta F_t = F_t - F_{t-1}. \quad (30)$$

**Non-commutativity features.**

$$\Delta \Sigma_t = \Sigma_t - \Sigma_{t-1}, \quad (31)$$

$$\Lambda_t^\Sigma = \log\left(1 + 100 \frac{\|\Sigma_t \Delta \Sigma_t - \Delta \Sigma_t \Sigma_t\|_F}{\|\Sigma_t\|_F \|\Delta \Sigma_t\|_F}\right). \quad (32)$$

The same formula defines  $\Lambda_t^C$  after replacing  $\Sigma_t$  by  $C_t$ . Here  $\|\cdot\|_F$  denotes the Frobenius norm.

**Differences.**

$$\Delta x_t = x_t - x_{t-1}, \quad (33)$$

$$\Delta \log x_t = \log(x_t) - \log(x_{t-1}). \quad (34)$$

#### A.4 Student-feature transformations

**Causal winsorisation.**

$$L_t = Q_{0.01}(x_{t-120}, \dots, x_{t-1}), \quad (35)$$

$$U_t = Q_{0.99}(x_{t-120}, \dots, x_{t-1}), \quad (36)$$

$$\tilde{x}_t = \min\{\max(x_t, L_t), U_t\}. \quad (37)$$

Window: 120 months. Minimum prior observations: 60.

**Causal z-score.**

$$z_t = \frac{\tilde{x}_t - \text{mean}(\tilde{x}_{t-60}, \dots, \tilde{x}_{t-1})}{\text{sd}(\tilde{x}_{t-60}, \dots, \tilde{x}_{t-1}) + 10^{-12}}. \quad (38)$$

Window: 60 months. Minimum prior observations: 36.

**Smoothed z-score.**

$$z_t^s = \text{EWM}_{12,t}(z). \quad (39)$$

Half-life: 12 months.

**Student feature vector.** For base features  $j = 1, \dots, 134$ ,

$$Z_t = (z_{1,t}, z_{1,t}^s, \dots, z_{134,t}, z_{134,t}^s). \quad (40)$$

#### A.5 Coupled approach: return definitions and features

For half-life  $h$ , let  $\text{EWM}_h(\cdot)$  denote the exponentially weighted moving average with that half-life. For an excess-return series  $x_t$ , define downside deviation and the Sortino ratio as

$$\text{dd}_{h,t}(x) = \left[ \text{EWM}_h(\min(x_t, 0)^2) \right]^{1/2}, \quad (41)$$

$$\text{Sortino}_{h,t}(x) = \frac{\text{EWM}_h(x)}{\text{dd}_{h,t}(x) + \varepsilon}, \quad h \in \{1, 2, 4, 12\}. \quad (42)$$

With  $m_h = \text{EWM}_h(x)$ , the coupled teacher uses four additional return-based features:

$$\text{dd\_lvl}_t = \log(\text{dd}_{6,t}(x) + \varepsilon), \quad \text{dd\_diff}_t = \log(\text{dd}_{2,t} + \varepsilon) - \log(\text{dd}_{12,t} + \varepsilon), \quad (43)$$

$$\text{ma\_cross}_t = \frac{m_3 - m_{12}}{\text{dd}_{6,t} + \varepsilon}, \quad \text{trend}_t = \text{EWM}_6\left(\frac{x_t - \text{EWM}_{12}(x)}{\text{EWM-sd}_{24}(x_t - \text{EWM}_{12}(x)) + \varepsilon}\right). \quad (44)$$

The market-derived block for the coupling logistic (5) uses the trailing 12-month equity-bond correlation  $\rho_{12,t}^{\text{eq,bd}}$ , together with each leg's  $\text{trend}_t$  and  $\text{dd\_lvl}_t$ .

**Table A.7:** Coupled teacher feature families. Each family is estimated under both the  $L^1$  and  $L^2$  specifications.

| Family               | Features                                                |
|----------------------|---------------------------------------------------------|
| <i>all</i>           | Sortino <sub>1,2,4,12</sub> , dd_lvl, dd_diff, ma_cross |
| <i>sortino</i>       | Sortino <sub>1,2,4,12</sub>                             |
| <i>sortino_short</i> | Sortino <sub>1,2,4</sub>                                |
| <i>dd</i>            | dd_lvl, dd_diff                                         |
| <i>trend</i>         | ma_cross                                                |
| <i>risk</i>          | dd_lvl, Sortino <sub>4</sub>                            |
| <i>sortino_trend</i> | Sortino <sub>2,4</sub> , ma_cross                       |
| <i>dd_trend</i>      | dd_lvl, ma_cross                                        |
| <i>sortino_dd</i>    | Sortino <sub>2,4,12</sub> , dd_lvl, dd_diff             |
| <i>core</i>          | Sortino <sub>4</sub> , dd_lvl, ma_cross                 |
| <i>mom</i>           | Sortino <sub>2</sub> , ma_cross                         |

Crossing the eleven feature families with the  $L^1$  and  $L^2$  specifications gives 22 coupled teacher members. At each month, features are z-scored against the trailing 120-month window after



winsorising that window at the 1%/99% quantiles. The current observation is clipped to the same bounds before scaling, so the transformation is strictly causal. Figures D.15 and D.16 plot the resulting standardised inputs.

## B Teacher Label Estimation Details

### B.1 Teacher panels and refitting schedule

**Table B.8:** Teacher panels

| Label               | Teacher features                         | Frequency | Initial sample |
|---------------------|------------------------------------------|-----------|----------------|
| Equity regime label | Daily equity return and Sortino features | Daily     | 12 years       |
| Bond regime label   | Monthly bond return and risk features    | Monthly   | 12 years       |

$$x_{t,j}^c = \min\{\max(x_{t,j}, \mu_j - 3\sigma_j), \mu_j + 3\sigma_j\}, \quad (45)$$

$$z_{t,j} = \frac{x_{t,j}^c - \mu_j}{\sigma_j}. \quad (46)$$

Daily equity models are refitted at month ends. Monthly bond models are refitted each month. Each refit uses only observations available at the refit date. In each refit,  $\mu_j$  and  $\sigma_j$  are computed from that training sample.

### B.2 Jump-model objective

For standardised teacher features  $z_t$ , the two-state objective is

$$\min_{s_1, \dots, s_T, c_0, c_1} \sum_{t=1}^T d(z_t, c_{s_t}) + \lambda \sum_{t=2}^T \mathbf{1}\{s_t \neq s_{t-1}\}, \quad s_t \in \{0, 1\}. \quad (47)$$

### B.3 $L^2$ jump model

$$\mathcal{T}_k = \{t : s_t = k\}, \quad k \in \{0, 1\}, \quad (48)$$

$$c_k = \frac{1}{|\mathcal{T}_k|} \sum_{t \in \mathcal{T}_k} z_t, \quad (49)$$

$$d_2(z_t, c_{s_t}) = \frac{1}{2} \|z_t - c_{s_t}\|_2^2. \quad (50)$$

### B.4 $L^1$ medoid jump model

$$\tilde{z}_k = \text{median}\{z_t : t \in \mathcal{T}_k\}, \quad (51)$$

$$m_k = \arg \min_{z_i : i \in \mathcal{T}_k} \|z_i - \tilde{z}_k\|_1, \quad (52)$$

$$d_1(z_t, m_{s_t}) = \|z_t - m_{s_t}\|_1 = \sum_{j=1}^p |z_{t,j} - m_{s_t,j}|. \quad (53)$$

### B.5 Penalty grids and optimisation

$$\lambda^E \in \{2, 4, 8, 16, 32, 64, 128\}, \quad (54)$$

$$\lambda^B \in \{0, 10^{-5}, 10^{-4}, 0.001, 0.00390625, 0.0078125, 0.015625, 0.03125, 0.0625, 0.125, 0.25, 0.5, 1, 2, 4\}. \quad (55)$$

Each fit uses five initialisations and at most 100 coordinate-descent iterations. Conditional on the state centres, the state path is updated by dynamic programming.

## B.6 State ordering

For fitted raw state  $k$ , define

$$\hat{\mu}_k = \text{mean}\{r_t : s_t = k\}, \quad (56)$$

$$\hat{\sigma}_k = \text{sd}\{r_t : s_t = k\}, \quad (57)$$

$$\widehat{\text{SR}}_k = \frac{\hat{\mu}_k}{\hat{\sigma}_k}. \quad (58)$$

The favourable state is

$$g = \arg \max_{k \in \{0,1\}} \widehat{\text{SR}}_k. \quad (59)$$

If the Sharpe-ratio comparison is tied,

$$g = \arg \max_{k \in \{0,1\}} \hat{\mu}_k. \quad (60)$$

## B.7 Monthly labels

For daily equity paths, the monthly favourable-state share is

$$p_m^E = \frac{1}{N_m} \sum_{t \in m} \mathbf{1}\{s_t = g\}, \quad (61)$$

and the monthly equity label is

$$S_m^E = \mathbf{1}\{p_m^E \geq 0.5\}. \quad (62)$$

For monthly bond paths,

$$S_m^B = \mathbf{1}\{s_m = g\}. \quad (63)$$

For asset  $a \in \{E, B\}$ , majority-vote labels are

$$\bar{S}_m^a = \frac{1}{K_a} \sum_{k=1}^{K_a} S_{m,k}^a, \quad (64)$$

$$S_m^{a,MV} = \mathbf{1}\{\bar{S}_m^a \geq 0.5\}. \quad (65)$$

The one-month-ahead student target is

$$Y_{m+1}^a = S_{m+1}^{a,MV}. \quad (66)$$

## C Teacher Label and Feature Diagnostics

This appendix reports diagnostics for the equity and bond teacher labels. The diagnostics cover individual  $L^2$  and  $L^1$ -medoid teacher paths, jump-penalty sensitivity, majority-vote construction, transition behaviour, crisis-window behaviour, and teacher-feature state separation. These diagnostics document whether the labels are persistent, economically separated, and sufficiently stable to serve as supervised targets in the student stage. All return statistics use the monthly excess return of the labelled asset. The reported independent prediction and allocation results use the equity return–Sortino teacher label and the canonical bond teacher label; volatility/downside equity labels were candidate diagnostics and are not used in the reported allocation results.

### C.1 Individual teacher paths

Figures C.4–C.7 show the fitted teacher paths across jump-penalty values. Each panel plots the labelled asset’s excess-return index and shades months assigned to the favourable or unfavourable regime. The panel title reports the jump penalty, favourable-state share, number of switches, and average spell duration.

Equity regime label:  $L^2$  teacher paths

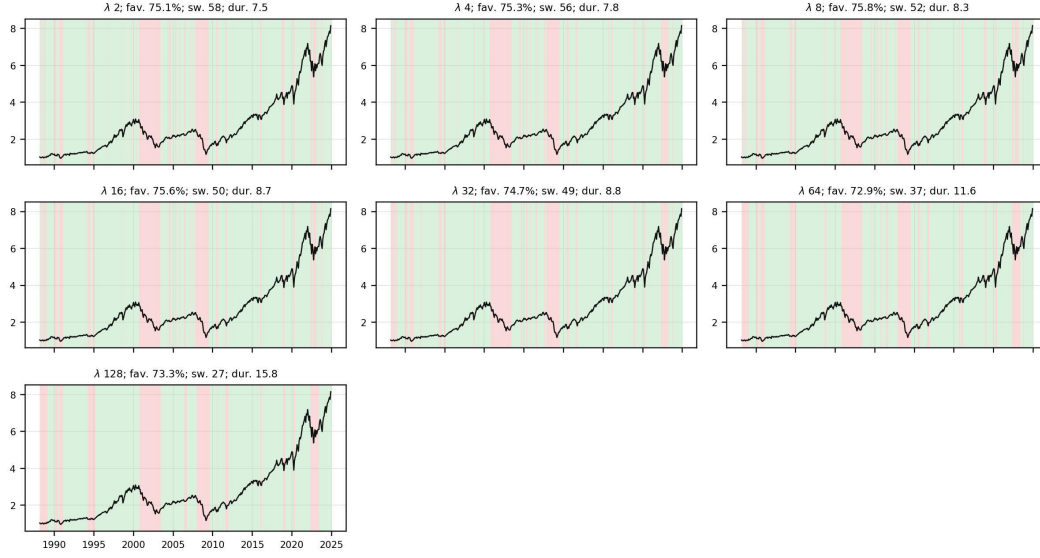


Figure C.4: Equity regime label:  $L^2$  teacher paths.

Equity regime label:  $L^1$ -medoid teacher paths

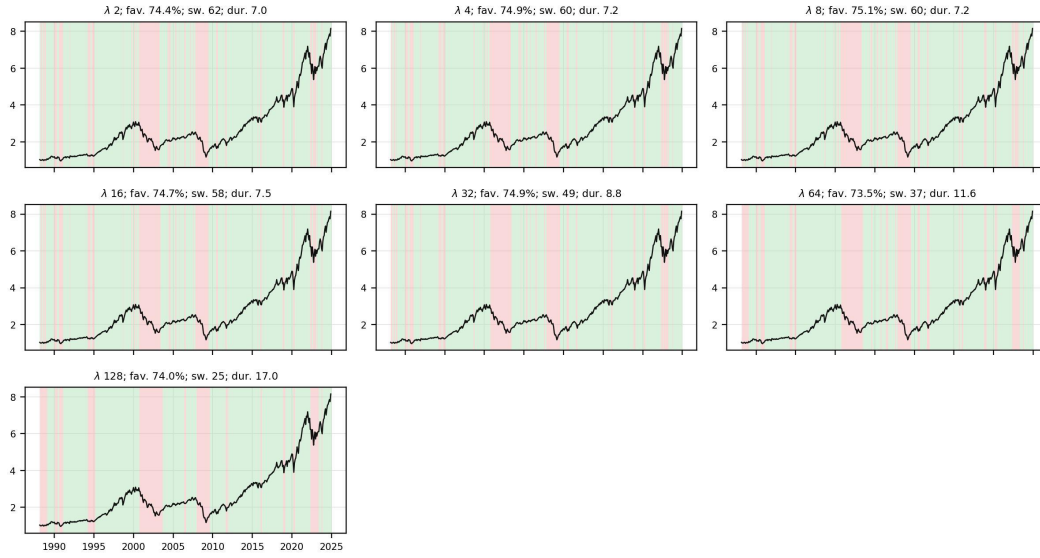


Figure C.5: Equity regime label:  $L^1$ -medoid teacher paths.

Bond regime label: L2 teacher paths

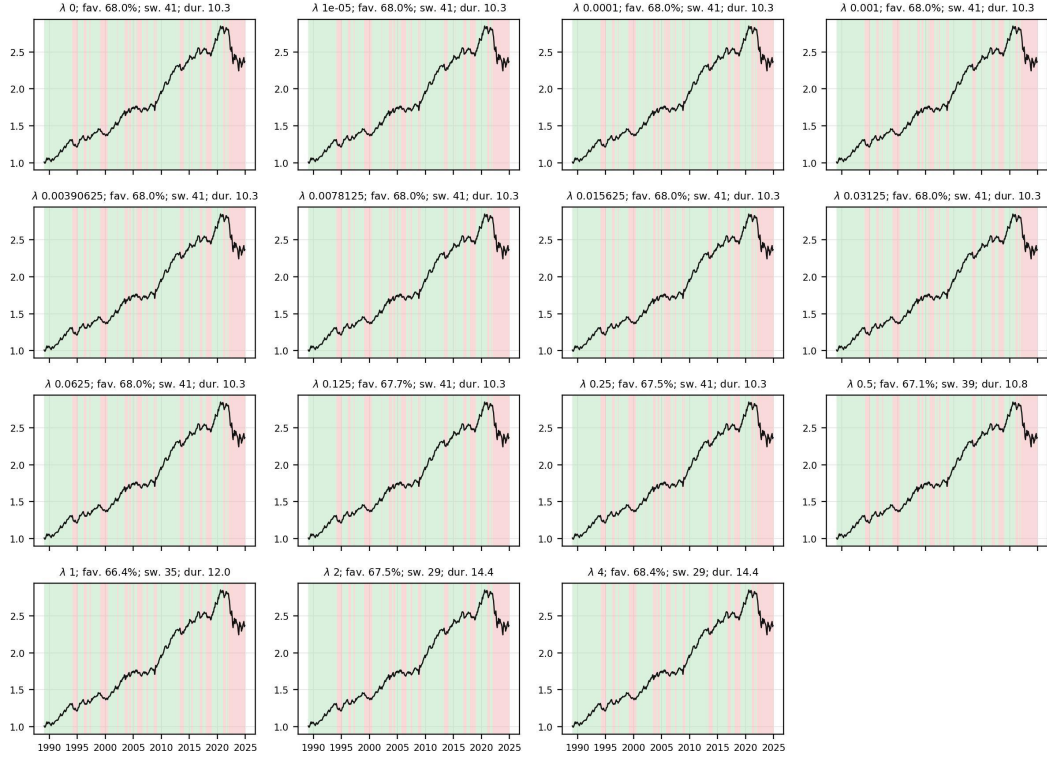


Figure C.6: Bond regime label:  $L^2$  teacher paths.

Bond regime label:  $L^1$ -medoid teacher paths

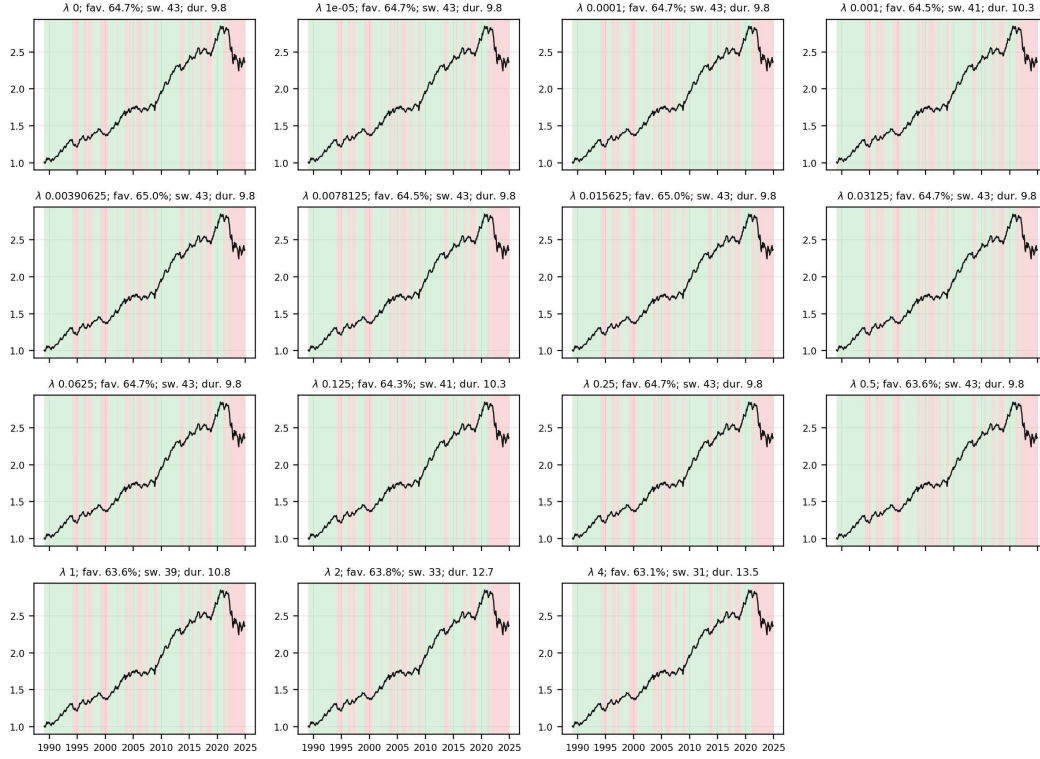


Figure C.7: Bond regime label:  $L^1$ -medoid teacher paths.

## C.2 Jump-penalty diagnostics

Figures C.8 and C.9 summarise how label properties vary across the jump-penalty grid. The panels compare the  $L^2$  and  $L^1$ -medoid teacher paths using favourable-state share, switch count, state durations, return separation, Sharpe separation, unfavourable-state volatility, and median feature-separation ratio.

### Equity regime label: jump-penalty diagnostics

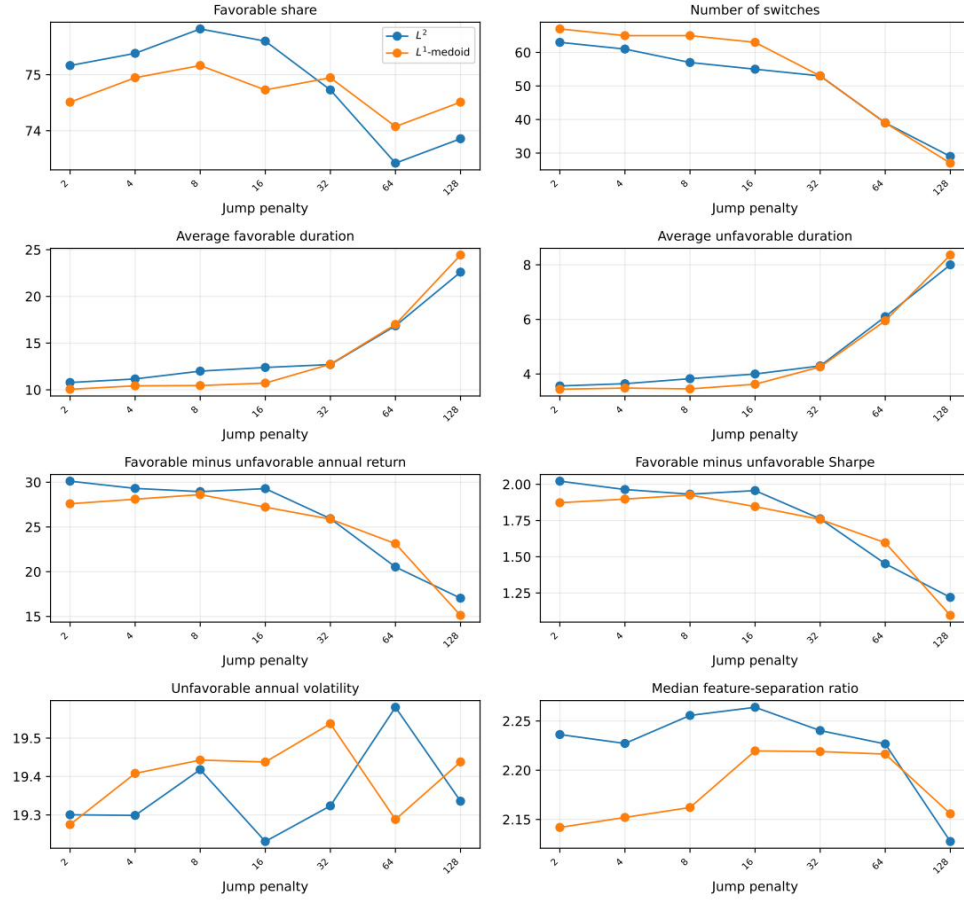


Figure C.8: Equity regime label: jump-penalty diagnostics.

Bond regime label: jump-penalty diagnostics

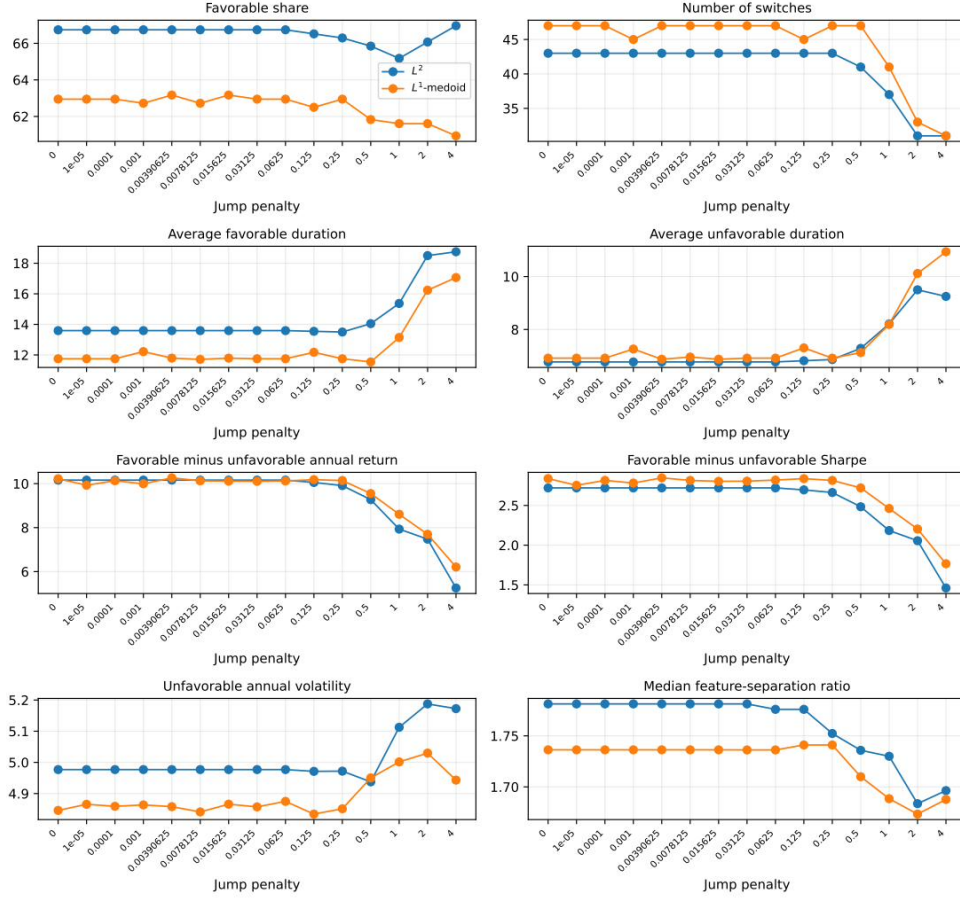


Figure C.9: Bond regime label: jump-penalty diagnostics.

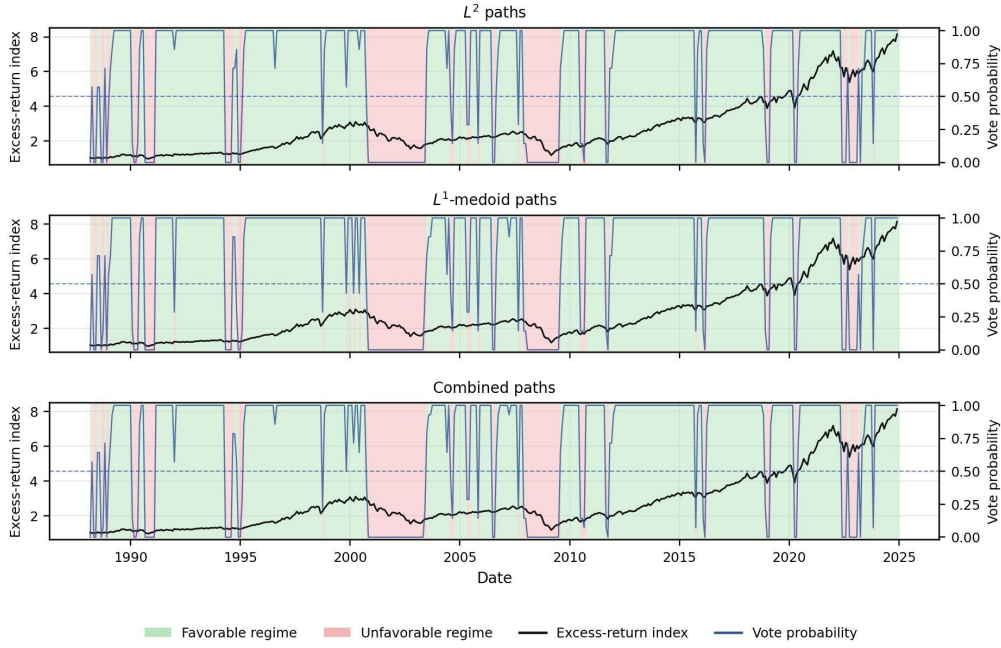
### C.3 Majority-vote construction

For asset  $a$ , the member vote share and final majority label are

$$\pi_m^a = \frac{1}{|\mathcal{P}_a|} \sum_{p \in \mathcal{P}_a} S_m^{a,p}, \quad S_m^a = \mathbf{1}\{\pi_m^a \geq 1/2\}. \quad (67)$$

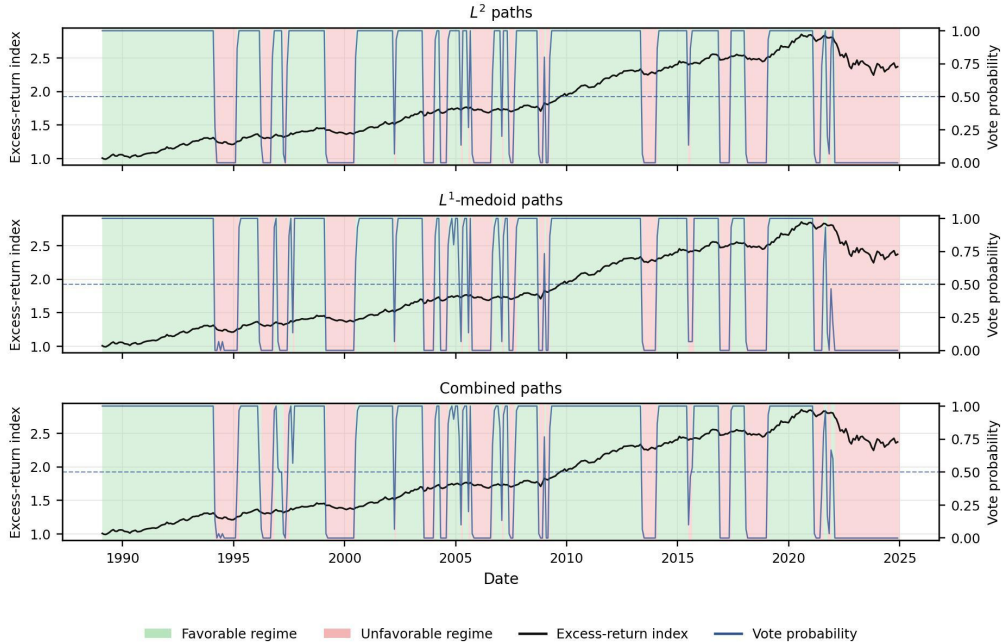
Figures C.10 and C.11 show how the final teacher labels are formed from the  $L^2$ ,  $L^1$ -medoid, and combined teacher paths. Each panel plots the labelled asset's excess-return index, shades the majority-vote regime, and overlays the vote probability. The dashed horizontal line marks the one-half threshold used to form the majority label.

#### Equity regime label: majority-vote construction



**Figure C.10:** Equity regime label: majority-vote construction.

#### Bond regime label: majority-vote construction



**Figure C.11:** Bond regime label: majority-vote construction.

### C.4 Transitions, persistence, and crisis windows

Table C.9 reports transition probabilities and expected state durations for the final combined teacher labels.



**Table C.9:** Teacher regime label transition summary

| Label  | $P(U \rightarrow U)$ | $P(U \rightarrow F)$ | $P(F \rightarrow U)$ | $P(F \rightarrow F)$ | $E[D_U]$ | $E[D_F]$ |
|--------|----------------------|----------------------|----------------------|----------------------|----------|----------|
| Equity | 75.9%                | 24.1%                | 7.5%                 | 92.5%                | 4.2      | 13.3     |
| Bond   | 85.8%                | 14.2%                | 7.3%                 | 92.7%                | 7.1      | 13.8     |

Notes:  $F$  denotes the favourable regime and  $U$  denotes the unfavourable regime. The transition probabilities are estimated from the final monthly combined majority-vote label for each asset. Expected duration is computed as  $1/(1 - p_{ss})$  months for state  $s$ .

Table C.10 reports path-level persistence and label-conditional Sharpe statistics by asset, teacher model, and jump-penalty value.

**Table C.10:** Teacher path persistence summary

| Label  | Model         | $\lambda$  | Fav. share | Sw. | Avg. dur. | Fav. dur. | Unfav. dur. | Fav. Sharpe | Unfav. Sharpe |
|--------|---------------|------------|------------|-----|-----------|-----------|-------------|-------------|---------------|
| Equity | $L^1$ -medoid | 2          | 74.4%      | 62  | 7.0       | 10.3      | 3.6         | 1.1         | -0.68         |
| Equity | $L^1$ -medoid | 4          | 74.9%      | 60  | 7.2       | 10.7      | 3.7         | 1.1         | -0.7          |
| Equity | $L^1$ -medoid | 8          | 75.1%      | 60  | 7.2       | 10.7      | 3.7         | 1.1         | -0.72         |
| Equity | $L^1$ -medoid | 16         | 74.7%      | 58  | 7.5       | 11.0      | 3.9         | 1.1         | -0.66         |
| Equity | $L^1$ -medoid | 32         | 74.9%      | 49  | 8.8       | 13.2      | 4.4         | 1.1         | -0.61         |
| Equity | $L^1$ -medoid | 64         | 73.5%      | 37  | 11.6      | 17.1      | 6.2         | 1.1         | -0.55         |
| Equity | $L^1$ -medoid | 128        | 74.0%      | 25  | 17.0      | 25.2      | 8.8         | 0.86        | -0.24         |
| Equity | $L^2$         | 2          | 75.1%      | 58  | 7.5       | 11.1      | 3.8         | 1.2         | -0.79         |
| Equity | $L^2$         | 4          | 75.3%      | 56  | 7.8       | 11.5      | 3.9         | 1.1         | -0.76         |
| Equity | $L^2$         | 8          | 75.8%      | 52  | 8.3       | 12.4      | 4.1         | 1.1         | -0.75         |
| Equity | $L^2$         | 16         | 75.6%      | 50  | 8.7       | 12.8      | 4.3         | 1.1         | -0.76         |
| Equity | $L^2$         | 32         | 74.7%      | 49  | 8.8       | 13.2      | 4.5         | 1.1         | -0.61         |
| Equity | $L^2$         | 64         | 72.9%      | 37  | 11.6      | 16.9      | 6.3         | 1           | -0.44         |
| Equity | $L^2$         | 128        | 73.3%      | 27  | 15.8      | 23.1      | 8.4         | 0.92        | -0.31         |
| Bond   | $L^1$ -medoid | 0          | 64.8%      | 43  | 9.8       | 12.7      | 6.9         | 2           | -0.9          |
| Bond   | $L^1$ -medoid | 1e-05      | 64.8%      | 43  | 9.8       | 12.7      | 6.9         | 1.9         | -0.85         |
| Bond   | $L^1$ -medoid | 0.0001     | 64.8%      | 43  | 9.8       | 12.7      | 6.9         | 2           | -0.88         |
| Bond   | $L^1$ -medoid | 0.001      | 64.6%      | 41  | 10.3      | 13.3      | 7.3         | 2           | -0.86         |
| Bond   | $L^1$ -medoid | 0.00390625 | 65.0%      | 43  | 9.8       | 12.8      | 6.9         | 2           | -0.91         |
| Bond   | $L^1$ -medoid | 0.0078125  | 64.6%      | 43  | 9.8       | 12.7      | 7.0         | 2           | -0.88         |
| Bond   | $L^1$ -medoid | 0.015625   | 65.0%      | 43  | 9.8       | 12.8      | 6.9         | 2           | -0.88         |
| Bond   | $L^1$ -medoid | 0.03125    | 64.8%      | 43  | 9.8       | 12.7      | 6.9         | 2           | -0.88         |
| Bond   | $L^1$ -medoid | 0.0625     | 64.8%      | 43  | 9.8       | 12.7      | 6.9         | 2           | -0.88         |
| Bond   | $L^1$ -medoid | 0.125      | 64.4%      | 41  | 10.3      | 13.2      | 7.3         | 2           | -0.88         |
| Bond   | $L^1$ -medoid | 0.25       | 64.8%      | 43  | 9.8       | 12.7      | 6.9         | 2           | -0.89         |
| Bond   | $L^1$ -medoid | 0.5        | 63.7%      | 43  | 9.8       | 12.5      | 7.1         | 2           | -0.76         |
| Bond   | $L^1$ -medoid | 1          | 63.7%      | 39  | 10.8      | 13.8      | 7.8         | 1.9         | -0.63         |
| Bond   | $L^1$ -medoid | 2          | 63.9%      | 33  | 12.7      | 16.2      | 9.2         | 1.7         | -0.53         |
| Bond   | $L^1$ -medoid | 4          | 63.2%      | 31  | 13.5      | 17.1      | 9.9         | 1.5         | -0.33         |
| Bond   | $L^2$         | 0          | 68.1%      | 41  | 10.3      | 14.0      | 6.6         | 1.8         | -0.95         |
| Bond   | $L^2$         | 1e-05      | 68.1%      | 41  | 10.3      | 14.0      | 6.6         | 1.8         | -0.95         |
| Bond   | $L^2$         | 0.0001     | 68.1%      | 41  | 10.3      | 14.0      | 6.6         | 1.8         | -0.95         |
| Bond   | $L^2$         | 0.001      | 68.1%      | 41  | 10.3      | 14.0      | 6.6         | 1.8         | -0.95         |
| Bond   | $L^2$         | 0.00390625 | 68.1%      | 41  | 10.3      | 14.0      | 6.6         | 1.8         | -0.95         |
| Bond   | $L^2$         | 0.0078125  | 68.1%      | 41  | 10.3      | 14.0      | 6.6         | 1.8         | -0.95         |
| Bond   | $L^2$         | 0.015625   | 68.1%      | 41  | 10.3      | 14.0      | 6.6         | 1.8         | -0.95         |
| Bond   | $L^2$         | 0.03125    | 68.1%      | 41  | 10.3      | 14.0      | 6.6         | 1.8         | -0.95         |
| Bond   | $L^2$         | 0.0625     | 68.1%      | 41  | 10.3      | 14.0      | 6.6         | 1.8         | -0.95         |
| Bond   | $L^2$         | 0.125      | 67.8%      | 41  | 10.3      | 14.0      | 6.6         | 1.8         | -0.93         |
| Bond   | $L^2$         | 0.25       | 67.6%      | 41  | 10.3      | 13.9      | 6.7         | 1.8         | -0.91         |
| Bond   | $L^2$         | 0.5        | 67.1%      | 39  | 10.8      | 14.5      | 7.1         | 1.7         | -0.81         |
| Bond   | $L^2$         | 1          | 66.4%      | 35  | 12.0      | 15.9      | 8.1         | 1.6         | -0.59         |
| Bond   | $L^2$         | 2          | 67.6%      | 29  | 14.4      | 19.5      | 9.3         | 1.6         | -0.54         |

| Label | Model | $\lambda$ | Fav. share | Sw. | Avg. dur. | Fav. dur. | Unfav. dur. | Fav. Sharpe | Unfav. Sharpe |
|-------|-------|-----------|------------|-----|-----------|-----------|-------------|-------------|---------------|
| Bond  | $L^2$ | 4         | 68.5%      | 29  | 14.4      | 19.7      | 9.1         | 1.2         | -0.25         |

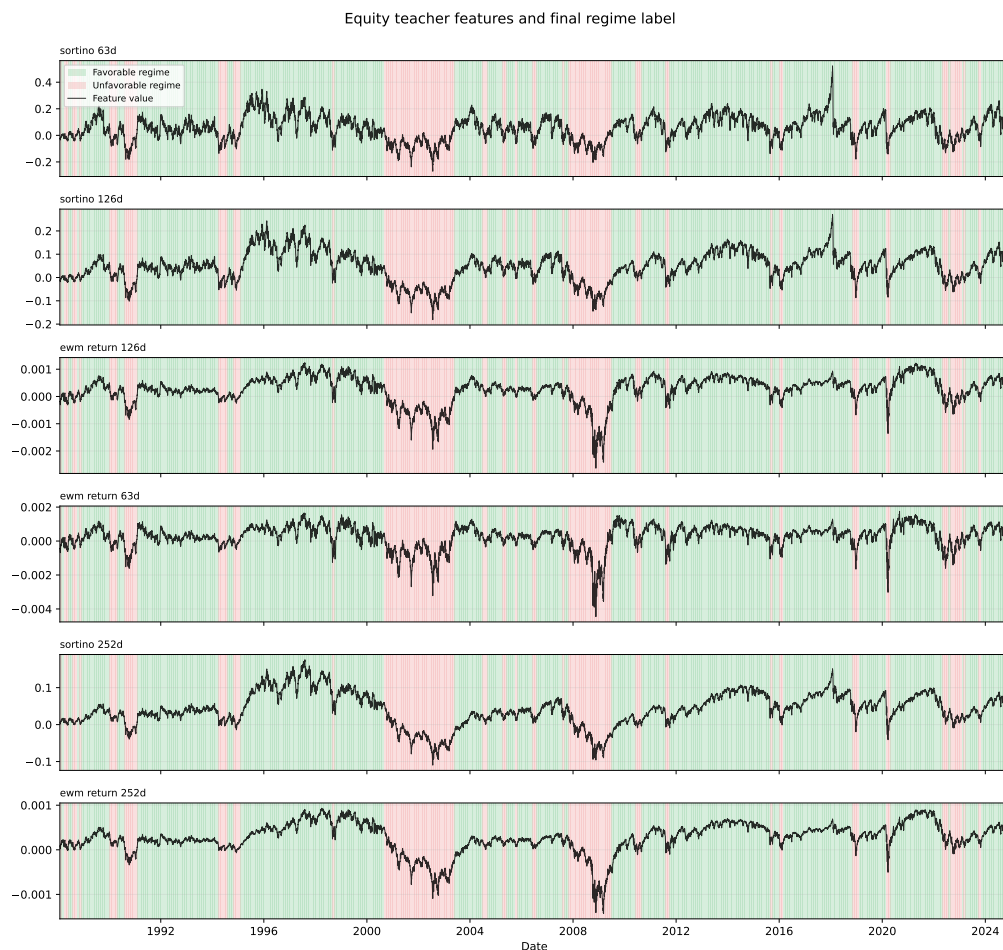
Table C.11 reports crisis-window behaviour for the final combined teacher labels. Favourable share is computed from the final monthly label within each window; return statistics use the labelled asset's monthly excess returns over the same window.

**Table C.11:** Teacher regime label crisis-window summary

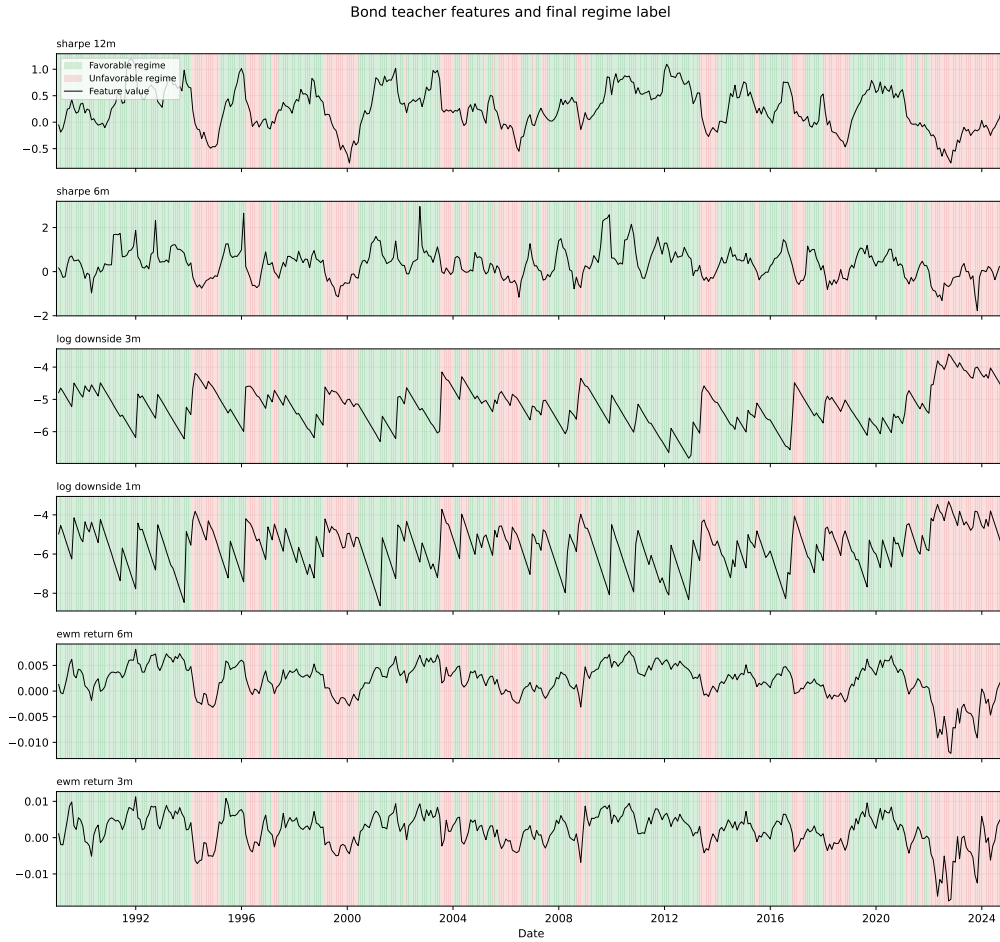
| Label  | Window              | Months | Fav. share | Ann. ret. | Ann. vol. | Sharpe | Worst  | Max. DD |
|--------|---------------------|--------|------------|-----------|-----------|--------|--------|---------|
| Equity | 1990 recession      | 9      | 33.3%      | 0.8%      | 18.3%     | 0.045  | -10.0% | -16.3%  |
| Equity | 1994 bond shock     | 12     | 50.0%      | -4.9%     | 10.6%     | -0.46  | -4.8%  | -9.0%   |
| Equity | Dot-com bust        | 32     | 0.0%       | -19.9%    | 18.9%     | -1.1   | -11.1% | -46.9%  |
| Equity | GFC                 | 18     | 0.0%       | -29.6%    | 24.6%     | -1.2   | -17.0% | -50.7%  |
| Equity | Euro sovereign 2011 | 8      | 75.0%      | -10.6%    | 18.7%     | -0.57  | -7.2%  | -15.9%  |
| Equity | Covid crash         | 3      | 33.3%      | -33.8%    | 47.0%     | -0.72  | -12.6% | -12.6%  |
| Equity | Inflation 2022      | 12     | 41.7%      | -20.4%    | 22.9%     | -0.89  | -9.5%  | -21.3%  |
| Bond   | 1990 recession      | 9      | 100.0%     | 4.5%      | 3.4%      | 1.3    | -2.0%  | -2.0%   |
| Bond   | 1994 bond shock     | 12     | 8.3%       | -6.7%     | 4.3%      | -1.5   | -2.7%  | -7.8%   |
| Bond   | Dot-com bust        | 32     | 96.9%      | 6.5%      | 3.3%      | 1.9    | -1.8%  | -2.6%   |
| Bond   | GFC                 | 18     | 72.2%      | 3.3%      | 5.2%      | 0.64   | -2.4%  | -4.8%   |
| Bond   | Euro sovereign 2011 | 8      | 100.0%     | 8.8%      | 2.6%      | 3.5    | -0.3%  | -0.3%   |
| Bond   | Covid crash         | 3      | 100.0%     | 11.0%     | 4.9%      | 2.3    | -0.7%  | -0.7%   |
| Bond   | Inflation 2022      | 12     | 0.0%       | -14.9%    | 8.2%      | -1.8   | -4.5%  | -14.5%  |

## C.5 Teacher features and state separation

Figures C.12 and C.13 show the standardised teacher features used to fit the equity and bond teacher paths. Standardisation is used only for visual comparability across features.



**Figure C.12:** Equity teacher features.



**Figure C.13:** Bond teacher features.

Table C.12 reports the features with the strongest separation between favourable and unfavourable teacher states.

**Table C.12:** Teacher feature state-separation summary

| Label  | Model         | Feature                     | Median rank | Median sep. |
|--------|---------------|-----------------------------|-------------|-------------|
| Equity | $L^1$ -medoid | Sortino ratio, 63 days      | 1.0         | 2.53        |
| Equity | $L^1$ -medoid | Sortino ratio, 126 days     | 2.0         | 2.49        |
| Equity | $L^1$ -medoid | EWM return, 126 days        | 3.0         | 2.39        |
| Equity | $L^1$ -medoid | EWM return, 63 days         | 4.0         | 2.32        |
| Equity | $L^1$ -medoid | Sortino ratio, 252 days     | 5.0         | 2.08        |
| Equity | $L^1$ -medoid | EWM return, 252 days        | 6.0         | 1.99        |
| Equity | $L^1$ -medoid | Sortino ratio, 21 days      | 7.0         | 1.62        |
| Equity | $L^1$ -medoid | EWM return, 21 days         | 8.0         | 1.44        |
| Equity | $L^2$         | Sortino ratio, 63 days      | 2.0         | 2.59        |
| Equity | $L^2$         | Sortino ratio, 126 days     | 1.0         | 2.55        |
| Equity | $L^2$         | EWM return, 126 days        | 3.0         | 2.44        |
| Equity | $L^2$         | EWM return, 63 days         | 4.0         | 2.4         |
| Equity | $L^2$         | Sortino ratio, 252 days     | 5.0         | 2.13        |
| Equity | $L^2$         | EWM return, 252 days        | 6.0         | 2.03        |
| Equity | $L^2$         | Sortino ratio, 21 days      | 7.0         | 1.64        |
| Equity | $L^2$         | EWM return, 21 days         | 8.0         | 1.46        |
| Bond   | $L^1$ -medoid | Sharpe ratio, 12 months     | 1.0         | 2.25        |
| Bond   | $L^1$ -medoid | Sharpe ratio, 6 months      | 2.0         | 1.98        |
| Bond   | $L^1$ -medoid | Log downside risk, 3 months | 3.0         | 1.94        |
| Bond   | $L^1$ -medoid | Log downside risk, 1 month  | 4.0         | 1.93        |
| Bond   | $L^1$ -medoid | EWM return, 6 months        | 5.0         | 1.9         |
| Bond   | $L^1$ -medoid | EWM return, 3 months        | 6.0         | 1.74        |
| Bond   | $L^1$ -medoid | EWM return, 12 months       | 7.0         | 1.71        |
| Bond   | $L^1$ -medoid | Log downside risk, 6 months | 8.0         | 1.62        |
| Bond   | $L^2$         | Sharpe ratio, 12 months     | 1.0         | 2.14        |
| Bond   | $L^2$         | Sharpe ratio, 6 months      | 2.0         | 2.0         |
| Bond   | $L^2$         | Log downside risk, 3 months | 3.0         | 1.99        |
| Bond   | $L^2$         | Log downside risk, 1 month  | 4.0         | 1.99        |
| Bond   | $L^2$         | EWM return, 6 months        | 5.0         | 1.91        |
| Bond   | $L^2$         | EWM return, 3 months        | 6.0         | 1.78        |
| Bond   | $L^2$         | EWM return, 12 months       | 7.0         | 1.69        |
| Bond   | $L^2$         | Log downside risk, 6 months | 8.0         | 1.65        |

Notes: For each fitted teacher path, features are ranked by their separation between favourable and unfavourable state centroids. Median rank is the median of that within-path rank over all jump-penalty paths in the model family. Median separation is the median standardised centroid-separation ratio over those paths; larger values mean stronger separation between the two regimes.

## D Coupled teacher: estimation details and diagnostics

The coupled teacher is estimated on the return panel beginning in 1970; after the fixed 132-month warm-up its first causal monthly label is 1989-01, coinciding with the start of the presented independent label. The pre-1989 history therefore enters only the rolling standardisation window; because labelling is strictly causal and performed one month at a time, it never enters any labelled month, forecast, or out-of-sample allocation result, so the two approaches remain directly comparable.

### Objective, coupling logistic, and fitting

For standardised joint features  $(z_t^{\text{eq}}, z_t^{\text{bd}})$  the per-month cost is (4) and the path minimises the total of emission and jump costs by Viterbi. The  $L^2$  member uses Euclidean leg distances with mean centroids; the  $L^1$  member uses absolute leg distances with median centroids (two centroids per leg, indexed by the benign/adverse bit). The coupling probability  $\kappa_t$  in (5) is fitted by  $L^2$ -penalised logistic regression ( $C = 1$ ) with balanced class weights and clipped to  $[0, 1]$ . Each fit alternates: (i) re-estimate centroids and refit the coupling logistic given the path; (ii) update the path by Viterbi given the centroids and  $\kappa_t$ . At most ten iterations, a single random restart plus a warm start from the previous month's path; the lowest-objective solution is kept. Per leg, the lower-mean state is set to the adverse bit (a model-internal label for the lower-mean centroid, not a negative-return state); for the equity leg this bit spans both the normal and stressed regimes, and only in stressed does equity realise genuinely negative risk-adjusted returns (Table 1). A stressed-share cap then relabels to normal the stressed months whose equity moving-average crossover exceeds the 18% quantile of its trailing window, so only the weakest-trend months remain stressed. After a

132-month warm-up, each month is refit on the expanding history;  $(\lambda, \eta)$  is selected per month from  $\lambda \in \{2, 3, 4, 6\}$  and  $\eta \in \{1, 2, 4, 8\}$  to maximise the regime separation score subject to share, stressed-cap and minimum-run-length constraints (Table D.13). The jump model is unsupervised; the coupling logistic is supervised, but its target is the leg-agreement indicator generated by that same unsupervised model rather than any external or forward-looking label. The logistic is therefore refit jointly with the path at every month (refit cadence one month) using only data available through the labelled month, so the whole teacher stage is unsupervised with respect to future returns and strictly causal.

**Table D.13:** Coupled teacher panel and refitting schedule.

| Setting                       | Value                                               |
|-------------------------------|-----------------------------------------------------|
| Legs                          | Equity (S&P 500 excess) and bond (aggregate excess) |
| Frequency                     | Monthly                                             |
| Warm-up (initial sample)      | 132 months (11 years)                               |
| Refit cadence                 | Every month                                         |
| Standardisation window        | 120 months, winsorised at 1%/99%                    |
| Random restarts per fit       | 1, plus a warm start from the previous month        |
| Coordinate-descent iterations | up to 10                                            |
| Jump-penalty grid $\lambda$   | $\{2, 3, 4, 6\}$                                    |
| Coupling-scale grid $\eta$    | $\{1, 2, 4, 8\}$                                    |
| Ensemble members              | 22 (11 feature families $\times \{L^1, L^2\}$ )     |
| Combination                   | Equal-weight vote (argmax of class counts)          |

### The coupling logistic $\kappa_t$ in detail

This subsection specifies the coupling logistic completely, so that its inputs, its target, its estimation, its effect on the state cost, and its causal status leave no open question.

**What  $\kappa_t$  represents.** The three admissible joint states split into *agree* states, in which the two legs share the same bit (calm = (0, 0) and stressed = (1, 1)), and the single *disagree* state, in which the bits differ (normal = (1, 0)). The scalar  $\kappa_t \in [0, 1]$  is the probability that, in month  $t$ , the equity and bond legs are in an agree state, conditional on observable market conditions. It is not a free parameter and not a portfolio weight; it is recomputed each month from the market-state features below.

**Target.** Given the current fitted state path, the binary target is the realised leg-agreement indicator

$$y_t = \mathbf{1}\{s_t^{\text{eq}} = s_t^{\text{bd}}\} \in \{0, 1\},$$

which equals 1 in calm or stressed months (both bits equal) and 0 in normal months. The target is thus produced by the unsupervised jump model itself; it uses no external label and no future return.

**Features.** The regressors are the five-dimensional market-state vector

$$x_t^{\text{mkt}} = \{\rho_{12,t}^{\text{eq,bd}}, \text{trend}_t^{\text{eq}}, \text{trend}_t^{\text{bd}}, \text{dd}_t^{\text{eq}}, \text{dd}_t^{\text{bd}}\},$$

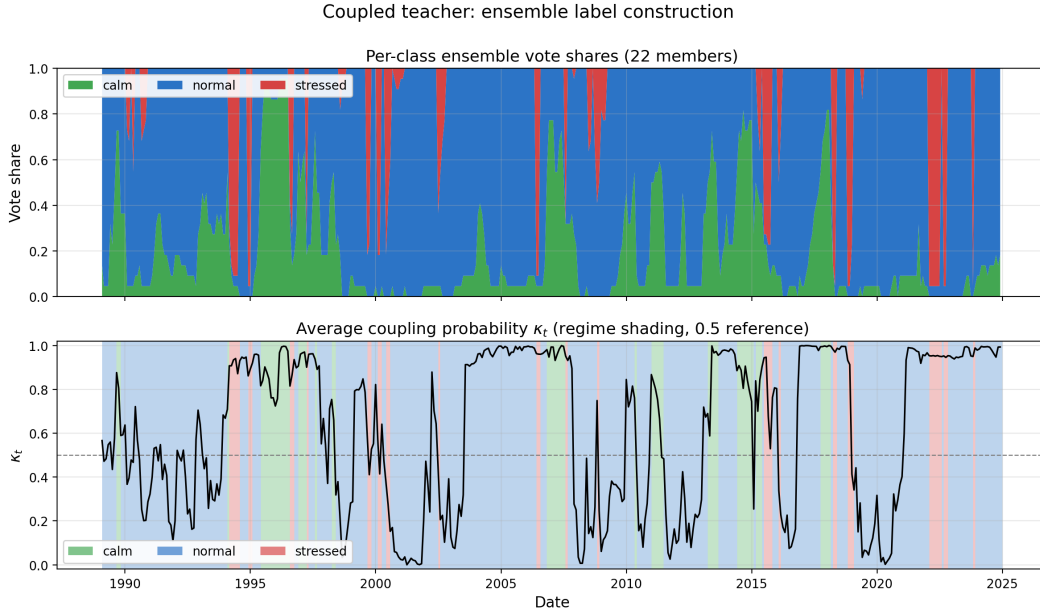
namely the trailing twelve-month equity–bond return correlation, each leg’s trend, and each leg’s drawdown level (Appendix A). These are the descriptors that drive equity–bond co-movement, so the regression learns the mapping from the current market state to the probability that the two legs share a regime. The features are standardised on the same trailing 120-month, 1%/99% winsorised window used for the leg features, so no future information enters.

**Estimator.**  $\kappa_t = \sigma(\beta_0 + \beta^\top x_t^{\text{mkt}})$  is an  $L^2$ -penalised (ridge) logistic regression with inverse-regularisation strength  $C = 1$  and balanced class weights; the fitted probability is clipped to  $[0, 1]$ . Balanced class weights matter here: agree months (calm + stressed) are the majority, and without re-weighting  $\kappa_t$  would be biased upward and would structurally suppress the disagree (normal) state; balancing keeps the normal state attainable.

**Effect on the state cost.**  $\kappa_t$  enters the per-month emission cost (4) as  $\eta[\kappa_t \mathbf{1}\{j \in D\} + (1 - \kappa_t) \mathbf{1}\{j \in A\}]$ : the disagree state is charged  $\eta\kappa_t$  and each agree state  $\eta(1 - \kappa_t)$ . A high  $\kappa_t$  therefore lowers the relative cost of calm and stressed, and a low  $\kappa_t$  lowers the relative cost of normal. The scalar  $\eta$  sets how strongly this prior tilts the Viterbi path;  $\eta$  and the jump penalty  $\lambda$  are selected per month from  $\{1, 2, 4, 8\}$  and  $\{2, 3, 4, 6\}$  (Table D.13).

**Joint estimation and refit cadence.** The logistic is not a post-hoc add-on: it is refit inside the coordinate descent that fits the teacher. Starting from the previous month’s warm-started path, each iteration (i) re-estimates the leg centroids given the current path, (ii) refits the logistic of  $y_t$  on  $x_t^{\text{mkt}}$  to obtain  $\kappa_t$ , and (iii) recomputes the path by dynamic programming under (4); the iteration runs up to ten steps and the lowest-objective solution is kept. This is repeated at *every* labelled month (refit cadence one month, no delay): at month  $t$  the entire fit — centroids, coefficients  $\beta$ , and path — uses only the expanding sample through month  $t$ .

**Causal, unsupervised status.** Two facts together establish that the teacher is causal and unsupervised with respect to future returns. First, every quantity used at month  $t$  (features, target, centroids, coefficients) is computed from data available through  $t$  only, so the label carries no look-ahead. Second, the only supervised component — the coupling logistic — is trained on a target  $y_t$  that is itself generated by the unsupervised jump model, not by any human annotation or realised future return; it is best read as a self-consistency step that sharpens the model’s own agree/disagree structure. The teacher therefore remains unsupervised in the sense relevant to leakage: no future return and no external label ever enters the fit.



**Figure D.14:** COUPLED ensemble label construction (causal teacher labelling). The top panel reports per-class vote shares across the 22 coupled members. The bottom panel reports the average coupling probability  $\kappa_t$ , with regime shading and the 0.5 reference line.

**Table D.14:** Coupled teacher: regime transition probabilities (%), row = from, column = to, with expected durations  $1/(1 - p_{ss})$  in months.

| From \ To | Calm | Normal | Stressed | $E[\text{dur}]$ |
|-----------|------|--------|----------|-----------------|
| Calm      | 78.7 | 14.8   | 6.6      | 4.7             |
| Normal    | 4.0  | 91.8   | 4.3      | 12.2            |
| Stressed  | 0.0  | 45.0   | 55.0     | 2.2             |

Table D.15 reports crisis-window behaviour for the final combined coupled labels. The



stressed share is the fraction of months assigned to the stressed joint regime within each window; return statistics use each leg’s monthly excess returns over the same window. This is the coupled counterpart of the independent crisis summary in Table C.11.

**Table D.15:** Coupled teacher: crisis-window summary by leg. Stressed share is the fraction of window months in the stressed joint regime; return statistics use the labelled leg’s monthly excess return over the window.

| Leg    | Window              | Months | Stressed sh. | Ann. ret. | Ann. vol. | Sharpe | Worst  | Max. DD |
|--------|---------------------|--------|--------------|-----------|-----------|--------|--------|---------|
| Equity | 1990 recession      | 9      | 0%           | +0.8%     | 17.4%     | +0.05  | −10.1% | −16.4%  |
| Equity | 1994 bond shock     | 12     | 58%          | −4.9%     | 10.1%     | −0.48  | −4.8%  | −9.0%   |
| Equity | Dot-com bust        | 32     | 3%           | −20.0%    | 18.6%     | −1.07  | −11.1% | −46.9%  |
| Equity | GFC                 | 18     | 6%           | −36.4%    | 23.1%     | −1.58  | −17.0% | −51.3%  |
| Equity | Euro sovereign 2011 | 8      | 0%           | −10.6%    | 17.5%     | −0.61  | −7.2%  | −15.9%  |
| Equity | Covid crash         | 3      | 0%           | −33.9%    | 38.4%     | −0.88  | −12.6% | −12.6%  |
| Equity | Inflation 2022      | 12     | 67%          | −20.4%    | 21.9%     | −0.93  | −9.5%  | −21.3%  |
| Bond   | 1990 recession      | 9      | 0%           | +4.5%     | 3.2%      | +1.41  | −2.0%  | −2.0%   |
| Bond   | 1994 bond shock     | 12     | 58%          | −6.7%     | 4.2%      | −1.61  | −2.7%  | −7.8%   |
| Bond   | Dot-com bust        | 32     | 3%           | +6.5%     | 3.3%      | +1.97  | −1.8%  | −2.6%   |
| Bond   | GFC                 | 18     | 6%           | +3.8%     | 5.2%      | +0.74  | −2.4%  | −4.8%   |
| Bond   | Euro sovereign 2011 | 8      | 0%           | +8.8%     | 2.4%      | +3.69  | −0.3%  | −0.3%   |
| Bond   | Covid crash         | 3      | 0%           | +11.0%    | 4.0%      | +2.76  | −0.7%  | −0.7%   |
| Bond   | Inflation 2022      | 12     | 67%          | −14.9%    | 7.9%      | −1.90  | −4.5%  | −14.5%  |

### Label construction and state summaries

Table D.16 records how the joint regime is assembled from the two per-leg state bits, and Table D.17 reports the label-conditional economics for both legs across the three joint regimes (the body reports a condensed version in Table 1).

**Table D.16:** Coupled teacher: label construction. The equity and bond per-leg state bits are estimated jointly through the coupling term, and their combination defines the three regimes.

| Component        | Teacher features                                                                                                             | Frequency | Initial sample |
|------------------|------------------------------------------------------------------------------------------------------------------------------|-----------|----------------|
| Equity state bit | Equity Sortino, drawdown and trend features                                                                                  | Monthly   | 11 years       |
| Bond state bit   | Bond Sortino, drawdown and trend features                                                                                    | Monthly   | 11 years       |
| Joint regime     | Combination (eq bit, bd bit) mapped to calm / normal / stressed, with the coupling probability $\kappa_t$ tying the two bits | Monthly   | 11 years       |

**Table D.17:** Coupled teacher: three-regime summary with spell counts (1989-01 to 2024-11, causal teacher labelling). Share is the fraction of labelled months in each joint regime; equity and bond statistics are same-month annualised excess returns conditional on the regime; spells is the number of distinct runs.

| Regime   | Share | Equity (same-month) |       |        | Bond (same-month) |      |        | Avg. dur. | Spells |
|----------|-------|---------------------|-------|--------|-------------------|------|--------|-----------|--------|
|          |       | Ret.                | Vol.  | Sharpe | Ret.              | Vol. | Sharpe |           |        |
| Calm     | 14.2% | +21.0%              | 8.1%  | +2.60  | +0.8%             | 3.5% | +0.22  | 4.7       | 13     |
| Normal   | 76.6% | +10.1%              | 14.9% | +0.68  | +4.1%             | 3.9% | +1.06  | 11.8      | 28     |
| Stressed | 9.3%  | −40.9%              | 13.7% | −2.99  | −8.4%             | 5.0% | −1.69  | 2.2       | 18     |

### Member-level and feature-level diagnostics

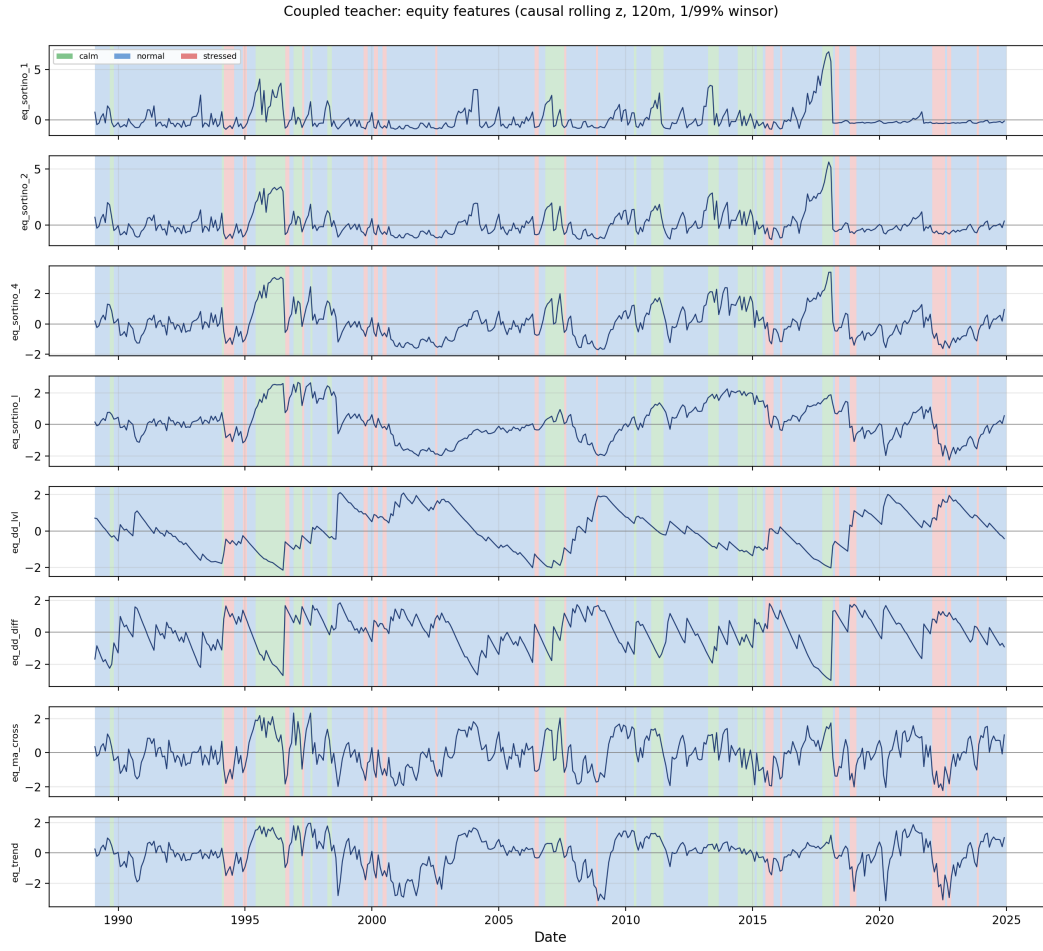
Table D.18 reports the twenty-two members. Most of the members select the smallest jump penalty  $\lambda = 2$ : the persistence in the final labels comes from the coupling term and the monthly refit, not from a large  $\lambda$ . Members separate well (most separation scores between 4 and 7), with Sortino- and momentum-based families separating best. Table D.19 ranks the per-leg features by how strongly their standardised mean separates across the three regimes.

**Table D.18:** Coupled teacher: member persistence and separation, one row per feature family and distance norm (22 members), sorted by the separation score.  $(\lambda, \eta)$  is the modal selected pair.

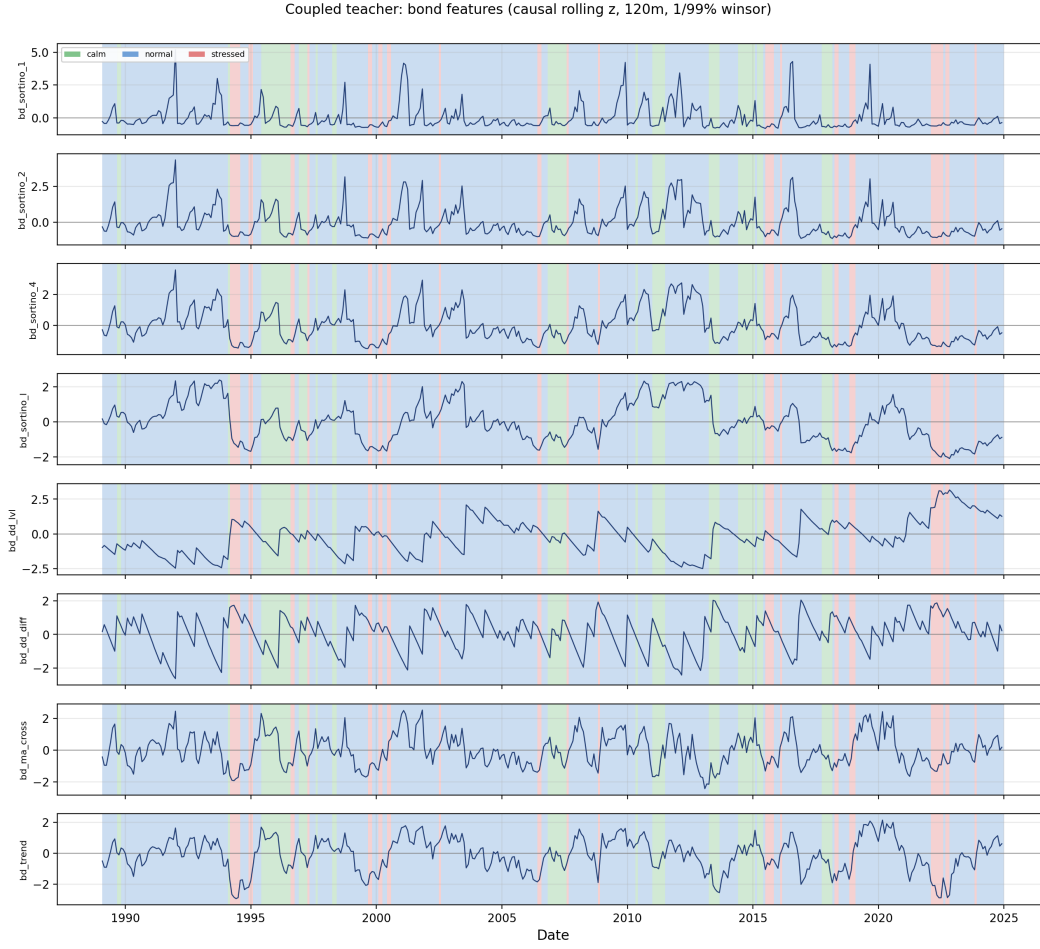
| Family        | Norm  | Calm | Normal | Stress. | Switches | Avg. dur. | Sep.  | $\lambda$ | $\eta$ |
|---------------|-------|------|--------|---------|----------|-----------|-------|-----------|--------|
| sortino       | $L^2$ | 0.03 | 0.84   | 0.13    | 59       | 7.2       | +7.07 | 2         | 8      |
| all           | $L^1$ | 0.20 | 0.70   | 0.10    | 85       | 5.0       | +6.24 | 2         | 1      |
| sortino_trend | $L^2$ | 0.06 | 0.83   | 0.11    | 55       | 7.7       | +5.96 | 2         | 4      |
| sortino       | $L^1$ | 0.19 | 0.69   | 0.11    | 77       | 5.5       | +5.89 | 2         | 2      |
| mom           | $L^2$ | 0.05 | 0.83   | 0.13    | 55       | 7.7       | +5.83 | 2         | 4      |
| core          | $L^1$ | 0.13 | 0.77   | 0.10    | 52       | 8.1       | +5.70 | 2         | 1      |
| sortino_dd    | $L^1$ | 0.18 | 0.71   | 0.10    | 62       | 6.8       | +5.69 | 2         | 1      |
| sortino_dd    | $L^2$ | 0.08 | 0.81   | 0.12    | 53       | 8.0       | +5.67 | 2         | 1      |
| sortino_short | $L^1$ | 0.20 | 0.71   | 0.09    | 71       | 6.0       | +5.60 | 2         | 2      |
| sortino_trend | $L^1$ | 0.28 | 0.60   | 0.11    | 77       | 5.5       | +5.29 | 3         | 2      |
| all           | $L^2$ | 0.23 | 0.70   | 0.07    | 51       | 8.3       | +5.10 | 2         | 2      |
| mom           | $L^1$ | 0.31 | 0.59   | 0.09    | 81       | 5.3       | +5.03 | 2         | 1      |
| risk          | $L^2$ | 0.22 | 0.71   | 0.08    | 40       | 10.5      | +4.60 | 2         | 2      |
| dd_trend      | $L^1$ | 0.11 | 0.81   | 0.08    | 46       | 9.2       | +4.59 | 2         | 8      |
| risk          | $L^1$ | 0.21 | 0.71   | 0.08    | 39       | 10.8      | +4.54 | 2         | 1      |
| core          | $L^2$ | 0.23 | 0.68   | 0.09    | 48       | 8.8       | +4.47 | 3         | 4      |
| trend         | $L^2$ | 0.25 | 0.68   | 0.07    | 51       | 8.3       | +4.46 | 2         | 1      |
| dd_trend      | $L^2$ | 0.21 | 0.74   | 0.05    | 31       | 13.5      | +4.41 | 2         | 1      |
| sortino_short | $L^2$ | 0.01 | 0.83   | 0.15    | 55       | 7.7       | +4.21 | 2         | 4      |
| dd            | $L^1$ | 0.21 | 0.71   | 0.08    | 41       | 10.3      | +4.07 | 2         | 4      |
| dd            | $L^2$ | 0.16 | 0.75   | 0.09    | 42       | 10.0      | +3.98 | 2         | 1      |
| trend         | $L^1$ | 0.67 | 0.32   | 0.01    | 62       | 6.8       | +2.20 | 2         | 8      |

**Table D.19:** Coupled teacher: feature state-separation. For each feature the standardised value is averaged within each regime; “Sep.” is the spread (max – min) of the three group means.

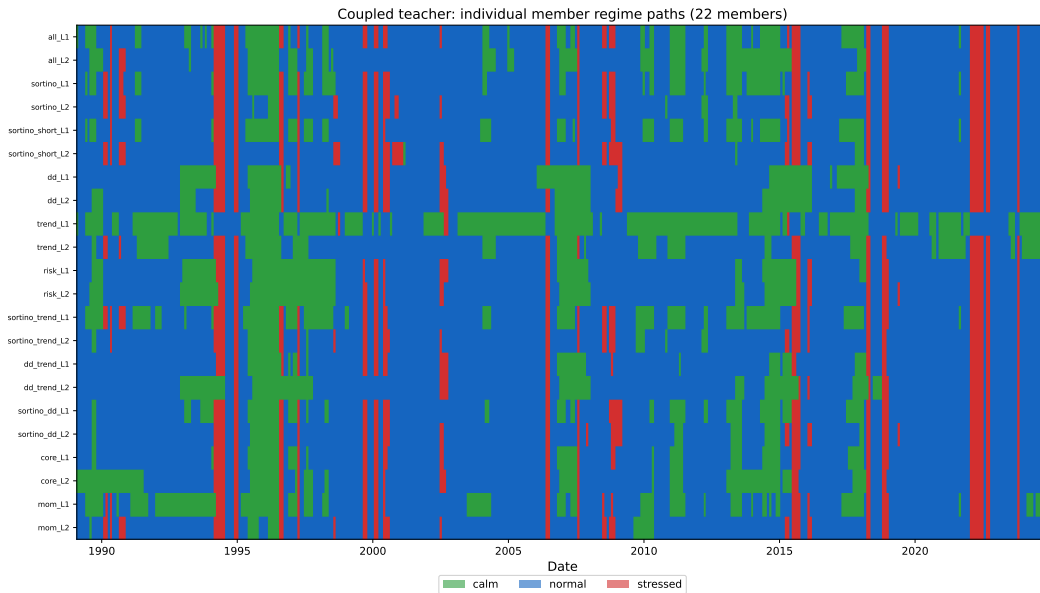
| Feature      | Sep. | Calm  | Normal | Stressed |
|--------------|------|-------|--------|----------|
| eq_sortino_4 | 2.62 | +1.72 | −0.11  | −0.90    |
| eq_sortino_2 | 2.54 | +1.72 | −0.11  | −0.82    |
| eq_ma_cross  | 2.30 | +0.89 | −0.02  | −1.41    |
| eq_sortino_1 | 2.17 | +1.56 | −0.09  | −0.61    |
| eq_dd_diff   | 2.08 | −1.05 | +0.02  | +1.03    |
| eq_trend     | 2.02 | +0.82 | −0.07  | −1.20    |
| eq_sortino_1 | 1.99 | +1.60 | −0.07  | −0.39    |
| bd_trend     | 1.62 | −0.09 | +0.09  | −1.52    |
| bd_sortino_1 | 1.43 | −0.06 | +0.13  | −1.30    |
| eq_dd_lvl    | 1.39 | −1.18 | +0.19  | +0.22    |
| bd_dd_lvl    | 1.29 | −0.45 | −0.29  | +0.84    |
| bd_sortino_4 | 1.26 | −0.13 | +0.13  | −1.13    |
| bd_ma_cross  | 1.07 | −0.11 | +0.12  | −0.95    |
| bd_dd_diff   | 1.03 | +0.05 | −0.05  | +0.98    |
| bd_sortino_2 | 0.96 | −0.17 | +0.09  | −0.87    |
| bd_sortino_1 | 0.63 | −0.23 | +0.05  | −0.58    |



**Figure D.15:** Coupled teacher: equity features after the causal rolling z-score (120-month window, 1%/99% winsorisation).

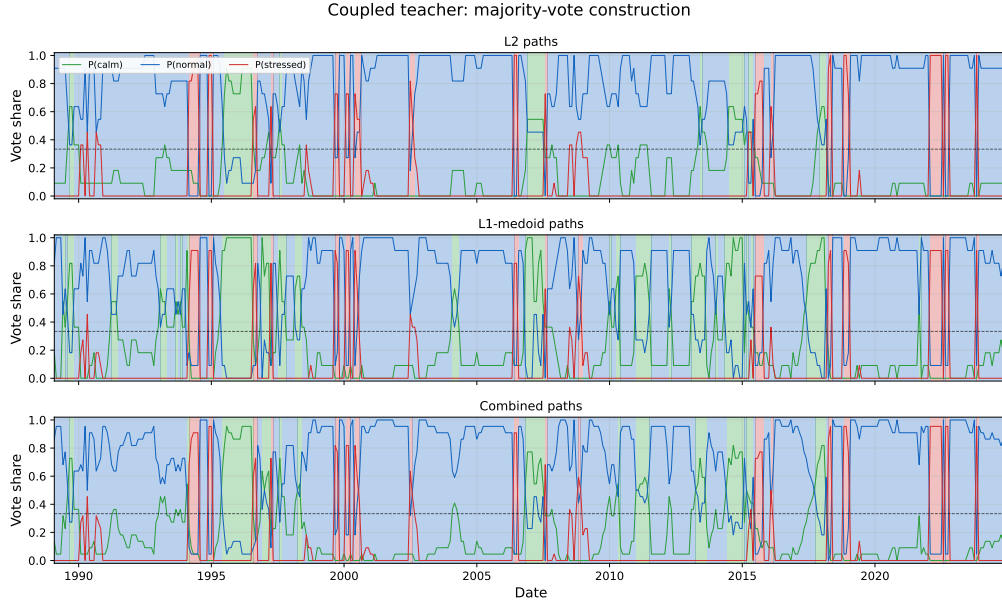


**Figure D.16:** Coupled teacher: bond features after the same causal rolling z-score. The flat left edge is the warm-up region before the standardisation window is populated.



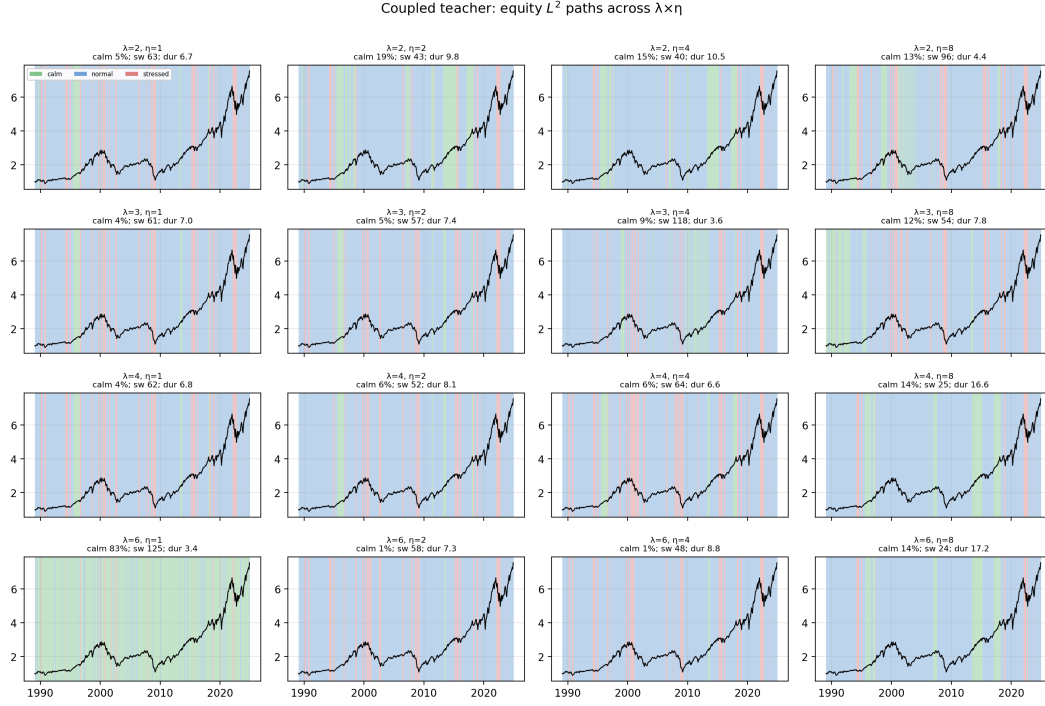
**Figure D.17:** Coupled teacher: per-member regime labels across time for the 22 ensemble members (rows), illustrating the agreement that produces the majority-vote label.

Figure D.18 shows how the final coupled label is formed from the candidate member paths under three scopes ( $L^2$  members,  $L^1$ -medoid members, and all 22 combined), the coupled counterpart of the per-leg independent majority-vote construction in Figures C.10–C.11.

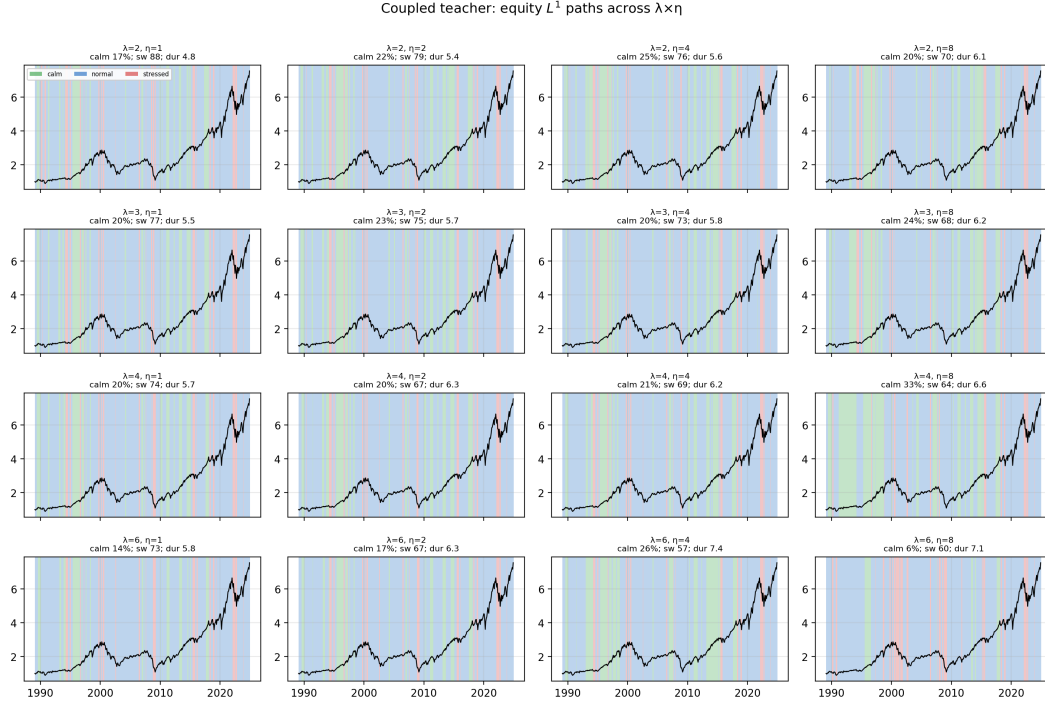


**Figure D.18:** Coupled teacher: majority-vote construction. Each panel restricts the ensemble to a scope ( $L^2$  members,  $L^1$ -medoid members, or all 22 combined), shades the equity excess-return index by the scope’s majority joint regime, and overlays the majority-class vote share on the right axis. The dashed line marks an absolute majority (0.5); below it the winning class is a plurality among the three regimes.

**Jump-penalty diagnostics.** To isolate the effect of persistence, the jump penalty is held fixed across a wider grid  $\lambda \in \{2, 3, 4, 6, 8, 12, 16\}$  at  $\eta = 2$  for the all-feature member under each norm (these paths form a sensitivity diagnostic, not the reported labels). A larger penalty produces longer, stickier regimes: switch counts fall and durations rise with  $\lambda$  while the calm share shrinks, and the separation score is highest at small penalties—consistent with the reported specification selecting  $\lambda = 2$  for most of the members (Figures D.19–D.25).



**Figure D.19:** Coupled teacher: equity  $L^2$  paths across the jump penalty (each panel fixes  $\lambda, \eta = 2$ , and shades the equity excess-return index by the fitted regime).



**Figure D.20:** Coupled teacher: equity  $L^1$  paths across the jump penalty.

Coupled teacher: bond  $L^2$  paths across  $\lambda \times \eta$

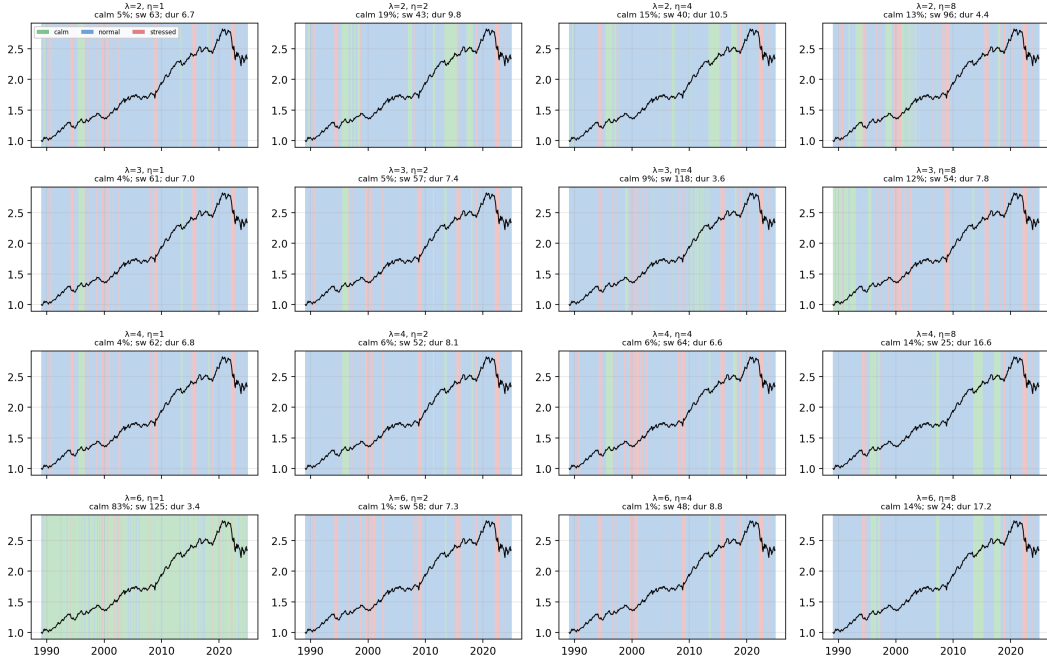


Figure D.21: Coupled teacher: bond  $L^2$  paths across the jump penalty.

Coupled teacher: bond  $L^1$  paths across  $\lambda \times \eta$

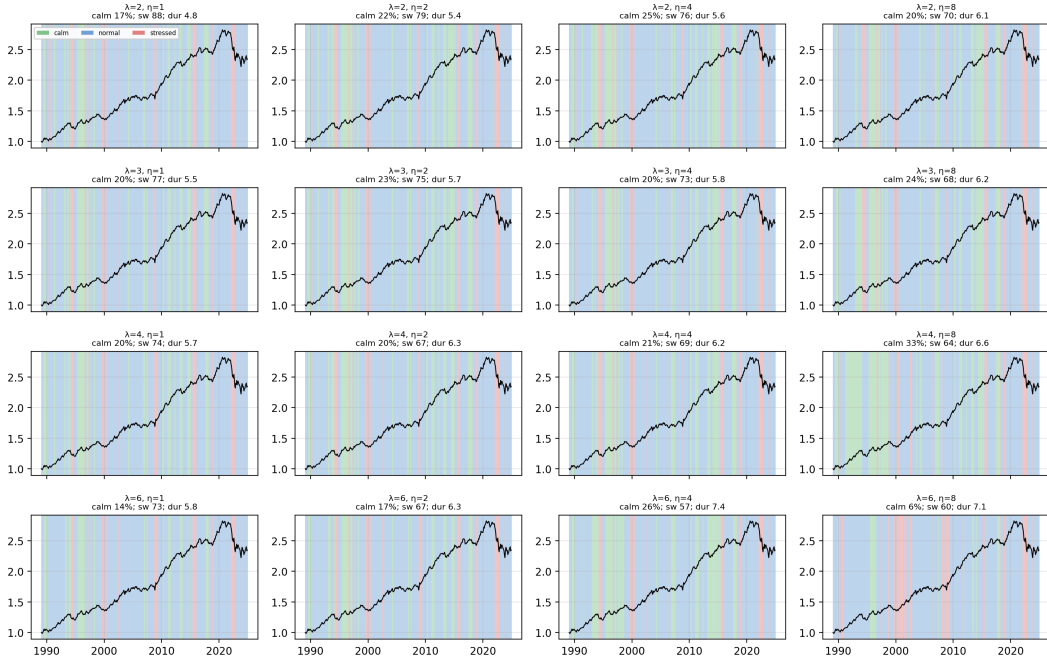
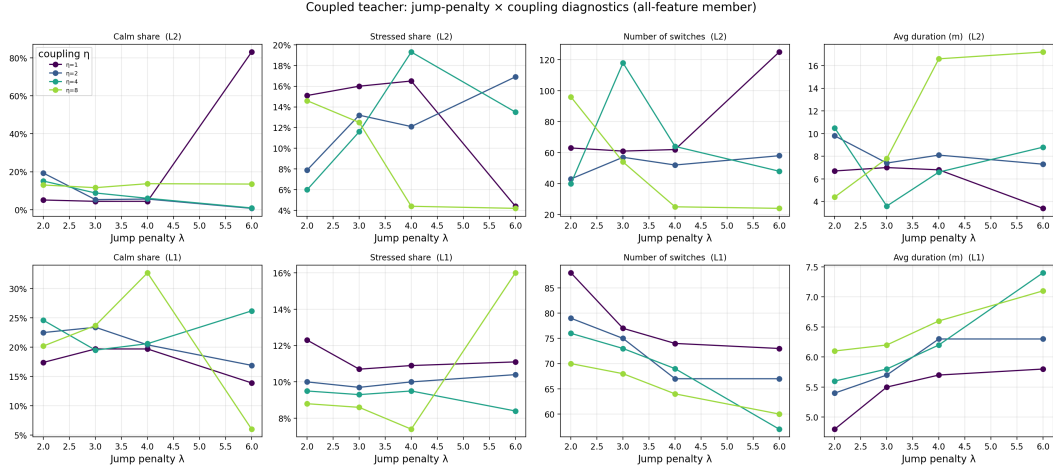
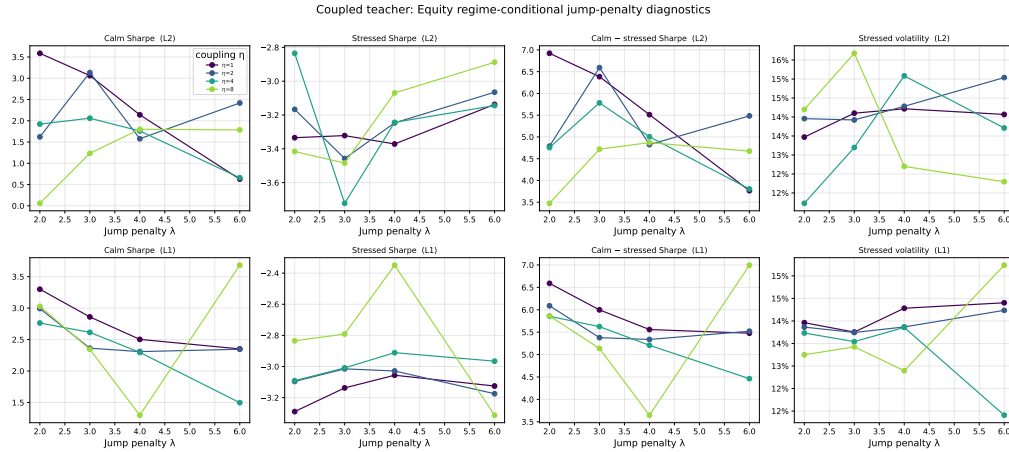


Figure D.22: Coupled teacher: bond  $L^1$  paths across the jump penalty.



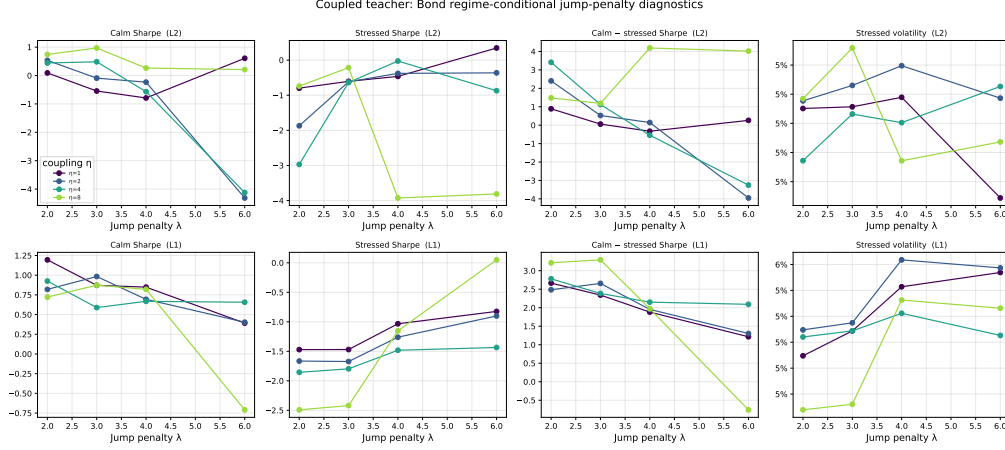
**Figure D.23:** Coupled teacher: jump-penalty diagnostics for the all-feature member—regime shares, switch count, average duration and separation score as functions of  $\lambda$ , for the  $L^1$  and  $L^2$  norms.

Figures D.24 and D.25 report the regime-conditional economics as a function of the jump penalty for each leg separately, the coupled counterpart of the per-leg independent jump-penalty diagnostics in Figures C.8 and C.9.



**Figure D.24:** Coupled teacher: equity regime-conditional jump-penalty diagnostics. For each norm ( $L^2$ ,  $L^1$ -medoid) and coupling value  $\eta$ , the panels report the equity calm-state Sharpe, stressed-state Sharpe, calm-minus-stressed Sharpe separation, and stressed-state volatility as functions of the jump penalty  $\lambda$ .





**Figure D.25:** Coupled teacher: bond regime-conditional jump-penalty diagnostics, defined as in Figure D.24 using the bond excess return.

## E Student Prediction Details

### E.1 Targets, features, and validation

The student models predict the one-month-ahead combined majority labels for the equity regime and the bond regime. The feature vector contains the z-score and smoothed z-score transformation of each monthly base predictor.

For each target and model family, prediction is performed with a rolling validation design. The initial training sample is 96 monthly observations, followed by a 36-month validation window. Candidate hyperparameters are selected on the validation data, and the selected setting is estimated on the full available sample before the forecast month. Validation ranking uses AUC, balanced accuracy, Brier score, net Sharpe ratio, drawdown improvement, volatility reduction, and turnover.

Predicted probabilities are averaged over seeds  $\{101, 111, 121\}$ .

### E.2 Model grids

The logistic student model uses median imputation, feature standardisation, balanced class weights, and  $L^2$  regularisation. The regularisation grid is

$$C \in \{0.03, 0.05, 0.10, 0.25, 0.50, 1.00\}. \quad (68)$$

The random-forest student model uses median imputation, 500 trees, bootstrap sampling, square-root feature subsampling, and balanced subsample class weights. The grid is

$$(d, \ell) \in \{(2, 10), (2, 20), (3, 10), (3, 20), (4, 20)\}, \quad (69)$$

where  $d$  is tree depth and  $\ell$  is the minimum leaf size.

## F Student regime prediction diagnostics

**Table F.20:** Out-of-sample student prediction summary

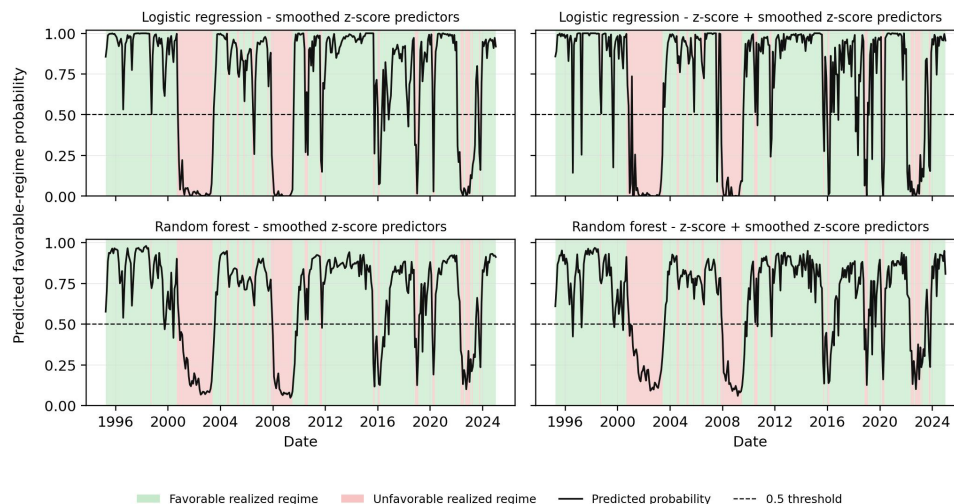
| Label         | Predictor set              | Model               | OOS sample      | Fav. share | Mean $\hat{p}$ | AUC   | Bal. acc. | Brier | Log loss |
|---------------|----------------------------|---------------------|-----------------|------------|----------------|-------|-----------|-------|----------|
| Bond regime   | z-score + smoothed z-score | Logistic regression | 1996-02–2024-12 | 62.2%      | 0.594          | 0.879 | 0.841     | 0.125 | 0.515    |
| Bond regime   | z-score + smoothed z-score | Random forest       | 1996-02–2024-12 | 62.2%      | 0.601          | 0.873 | 0.800     | 0.147 | 0.471    |
| Bond regime   | smoothed z-score           | Logistic regression | 1996-02–2024-12 | 62.2%      | 0.574          | 0.892 | 0.831     | 0.128 | 0.462    |
| Bond regime   | smoothed z-score           | Random forest       | 1996-02–2024-12 | 62.2%      | 0.590          | 0.849 | 0.784     | 0.153 | 0.484    |
| Equity regime | z-score + smoothed z-score | Logistic regression | 1995-03–2024-12 | 76.3%      | 0.707          | 0.929 | 0.846     | 0.088 | 0.296    |
| Equity regime | z-score + smoothed z-score | Random forest       | 1995-03–2024-12 | 76.3%      | 0.658          | 0.903 | 0.831     | 0.107 | 0.361    |
| Equity regime | smoothed z-score           | Logistic regression | 1995-03–2024-12 | 76.3%      | 0.700          | 0.918 | 0.862     | 0.086 | 0.307    |
| Equity regime | smoothed z-score           | Random forest       | 1995-03–2024-12 | 76.3%      | 0.660          | 0.902 | 0.837     | 0.105 | 0.355    |

*Note.* The table reports out-of-sample diagnostics from the selected rolling student models. AUC is the probability that a favourable-regime month receives a higher predicted probability than an unfavourable-regime month. Balanced accuracy is  $\frac{1}{2}\{TP/(TP + FN) + TN/(TN + FP)\}$ . The Brier score is  $n^{-1} \sum_i (\hat{p}_i - y_i)^2$ , and log loss is  $-n^{-1} \sum_i \{y_i \log(\hat{p}_i) + (1 - y_i) \log(1 - \hat{p}_i)\}$ , where  $y_i$  is the realised teacher label and  $\hat{p}_i$  is the predicted favourable-regime probability.

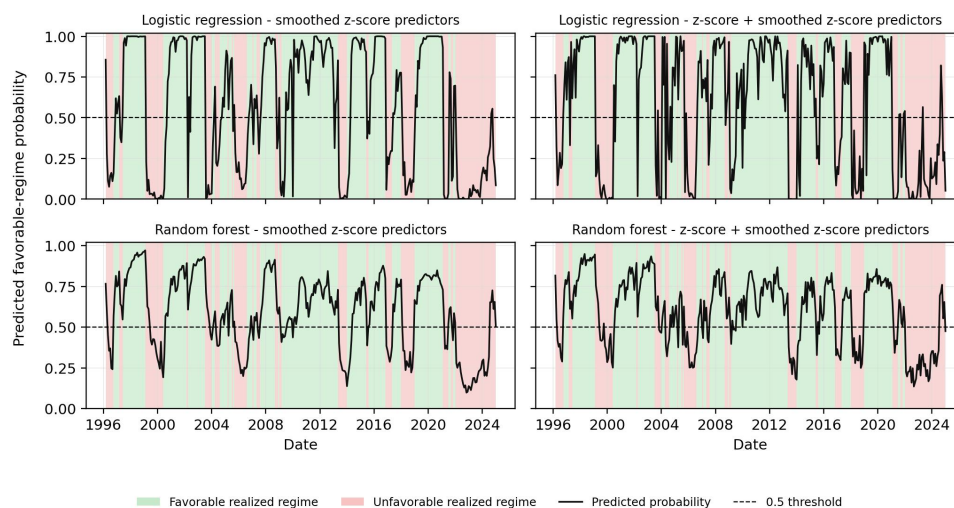
Equity-regime prediction is strongest for logistic regression. The logistic model with the combined z-score and smoothed z-score predictor set reaches an AUC of 0.929, balanced accuracy of 0.846, Brier score of 0.088, and log loss of 0.296. The smoothed-only logistic model gives the highest balanced accuracy for the equity label, 0.862, with a similar AUC of 0.918. Bond-regime prediction is strongest for logistic regression with the smoothed-only predictor set, which reaches an AUC of 0.892 and balanced accuracy of 0.831. Random forests also predict both labels, but their AUC and balanced-accuracy values are below the corresponding logistic-regression values in most specifications.

Figures F.26 and F.26 plot the out-of-sample probability paths. The shaded background marks the realised teacher regime, and the black line is the student-model probability assigned to the favourable regime.

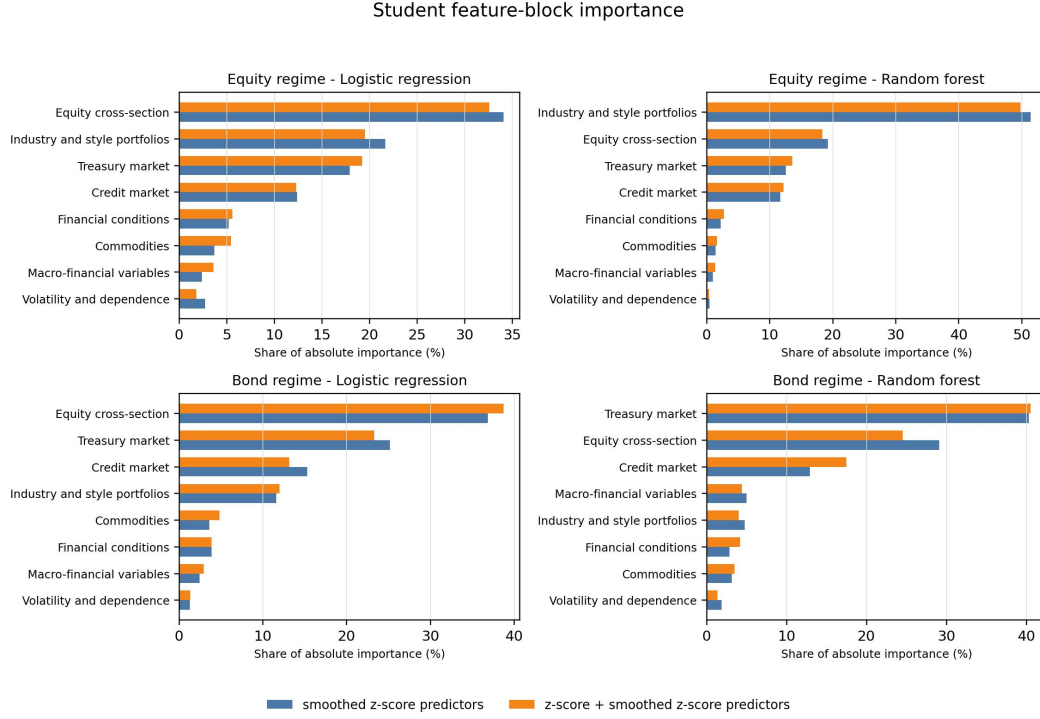
### Equity regime prediction: out-of-sample probabilities



### Bond regime prediction: out-of-sample probabilities



**Figure F.26:** INDEPENDENT student out-of-sample probability paths. The top panel reports equity favourable-regime probabilities, and the bottom panel reports bond favourable-regime probabilities. Shading marks the realised teacher label, the black line reports the predicted favourable-regime probability, and the dashed line marks the 0.5 threshold.



**Figure F.27:** INDEPENDENT student feature-block relevance. Logistic-regression relevance is measured by absolute coefficient magnitude. Random-forest relevance is measured by impurity-based feature importance. Values are aggregated within predictor blocks.

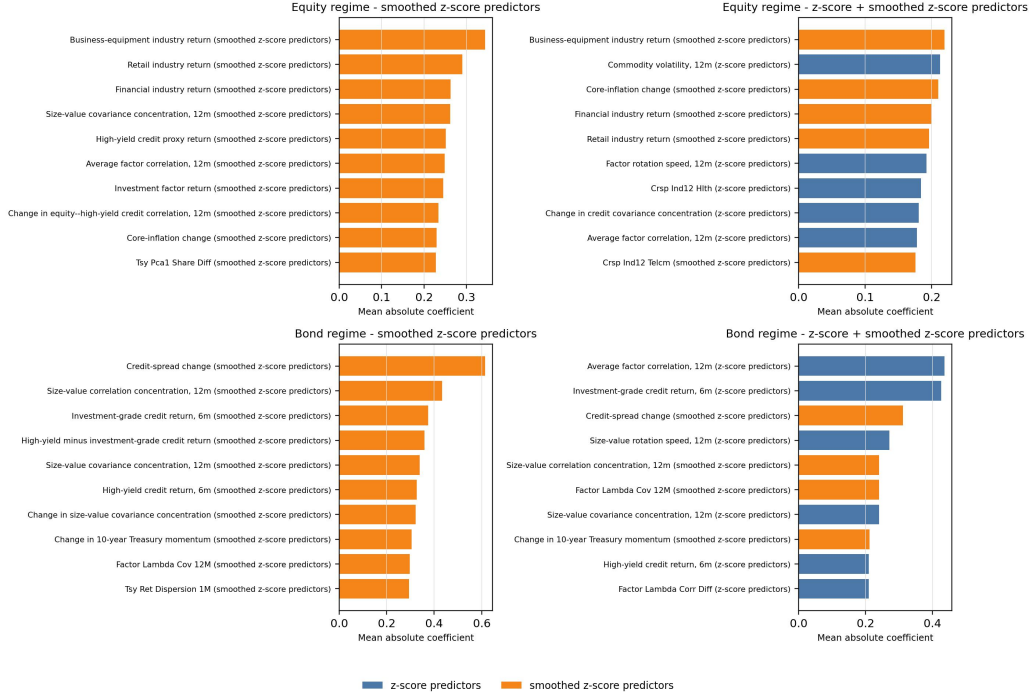
Table F.21 reports the most frequently selected hyperparameter setting over the rolling out-of-sample test months. The table reports which specification is most often selected within each target, predictor set, and model class.

**Table F.21:** Most frequent selected student-model specification

| Label         | Predictor set                         | Model               | Most frequent setting                          | Sel. share | Months |
|---------------|---------------------------------------|---------------------|------------------------------------------------|------------|--------|
| Bond regime   | z-score + smoothed z-score predictors | Logistic regression | $C = 0.03$ , class weight=balanced             | 56.2%      | 347    |
| Bond regime   | z-score + smoothed z-score predictors | Random forest       | trees=500, depth=3, leaf=10, max features=sqrt | 32.3%      | 347    |
| Bond regime   | smoothed z-score predictors           | Logistic regression | $C = 0.03$ , class weight=balanced             | 38.6%      | 347    |
| Bond regime   | smoothed z-score predictors           | Random forest       | trees=500, depth=2, leaf=10, max features=sqrt | 30.8%      | 347    |
| Equity regime | z-score + smoothed z-score predictors | Logistic regression | $C = 0.03$ , class weight=balanced             | 60.1%      | 358    |
| Equity regime | z-score + smoothed z-score predictors | Random forest       | trees=500, depth=3, leaf=10, max features=sqrt | 67.6%      | 358    |
| Equity regime | smoothed z-score predictors           | Logistic regression | $C = 0.03$ , class weight=balanced             | 56.4%      | 358    |
| Equity regime | smoothed z-score predictors           | Random forest       | trees=500, depth=3, leaf=10, max features=sqrt | 70.7%      | 358    |

*Note.* The table reports the most frequently selected hyperparameter setting across out-of-sample rolling test months. The selection share is the fraction of out-of-sample months in which that setting is selected for the corresponding regime label, model class, and predictor set.

### Individual predictor importance, logistic regression



**Figure F.28:** Student prediction individual predictor importance.

*Note.* Individual predictor importance uses model-level importances from the rolling prediction exercise. For logistic regression, importance is the absolute value of the fitted coefficient after preprocessing. For random forests, importance is impurity-based feature importance. Mean absolute importance is averaged across rolling selected-model fits and seeds. When both the z-score and smoothed z-score transformations of the same predictor are available within a predictor set, the plotted value is the larger mean absolute importance for that economic predictor. Each panel shows the eight largest economic predictors by the maximum mean absolute importance across the two predictor sets for the same label and model.

The individual-predictor diagnostics are consistent with the block-level figure. Equity-regime prediction relies heavily on industry returns, style-portfolio returns, equity-factor information, and cross-market credit correlation. Bond-regime prediction relies on credit-spread changes, investment-grade and high-yield credit returns, Treasury returns, and equity-factor dependence measures. These predictors explain why the equity cross-section, Treasury market, and credit market blocks receive high importance shares in the block-level results.

The feature-block definitions are: equity cross-section contains equity factors, size-value variables, and equity dispersion or dependence measures; Treasury market contains Treasury returns, yield-curve and rate changes, bond momentum, and bond-market dependence; credit market contains investment-grade and high-yield credit returns, credit spreads, dispersion, and credit dependence; commodities contain commodity returns, momentum, volatility, Sharpe ratios, and commodity dependence; financial conditions contain NFCI, adjusted NFCI, their changes, and their spread; macro-financial variables contain activity, inflation, and sentiment variables; volatility and dependence contains volatility and remaining correlation or concentration measures; and industry and style portfolios contain remaining industry and style-portfolio returns.

## G Coupled student: estimation details

**Coupled student.** The target is the next-month three-class regime. At each refit the five predictors most correlated with the next-month regime over the trailing window are kept,

and the current-regime indicators and run-length are appended. Two classifiers are fitted with balanced class weights: an  $L_2$ -penalised multinomial logistic regression and a random forest. The logistic regularisation grid is  $C \in \{0.03, 0.1, 0.3, 1.0\}$  and the random-forest grid is  $\{(n=300, d=4, \ell=5), (n=400, d=6, \ell=3)\}$ , where  $n$  is the number of trees,  $d$  the maximum depth and  $\ell$  the minimum leaf size; random-forest probabilities are averaged over seeds  $\{101, 111, 121\}$  and the logistic solver is deterministic. Hyper-parameters are selected by balanced accuracy on the validation window. Estimation is walk-forward (132-month training, 60-month validation by balanced accuracy, 6-month test), giving the out-of-sample span 1996-02 to 2024-10. The out-of-sample span is anchored at 1996-02 for both approaches; the earliest coupled windows use an expanding training sample that reaches the full 132-month width by 2005, after which the 132/60/6 rolling design applies unchanged. This affects only the size of the earliest training sets, not the common out-of-sample window.

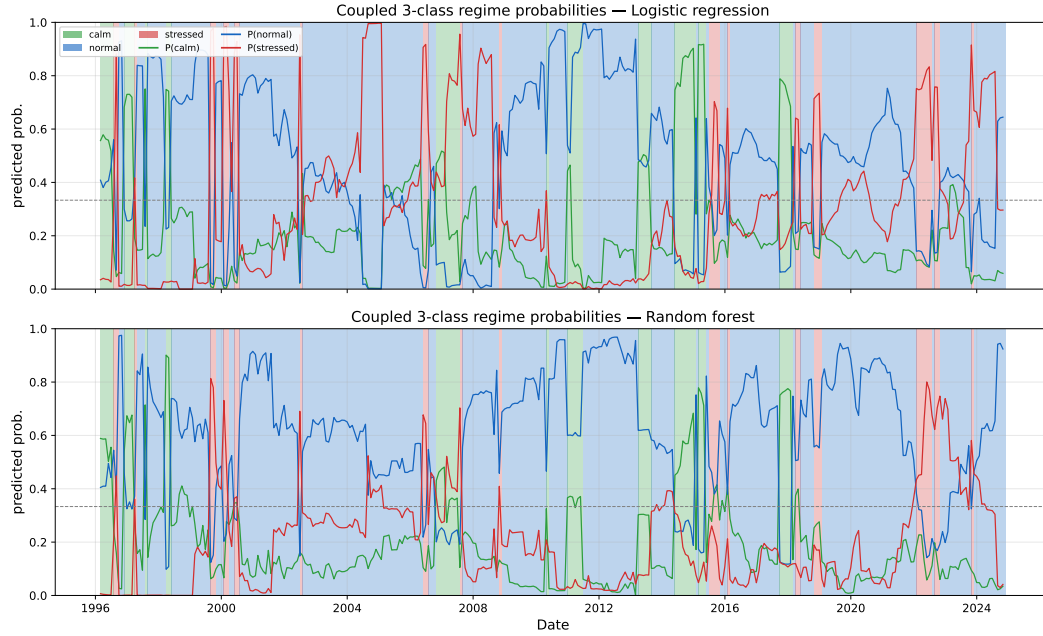
**Coupled student forecast diagnostics.** Table G.22 reports the out-of-sample probabilistic metrics for the two coupled students, and Table G.23 reports the most frequently selected hyper-parameter setting across the walk-forward windows. The predicted three-class probability paths are shown in Figure G.29. Figure G.30 reports block-level relevance, and Figure G.31 reports individual-predictor importance.

**Table G.22:** Coupled student: out-of-sample three-class forecast metrics (OOS 1996-02 to 2024-10). Balanced accuracy averages per-class recall; AUC is the macro one-vs-rest area; Brier and log loss are probabilistic scores (lower is better).

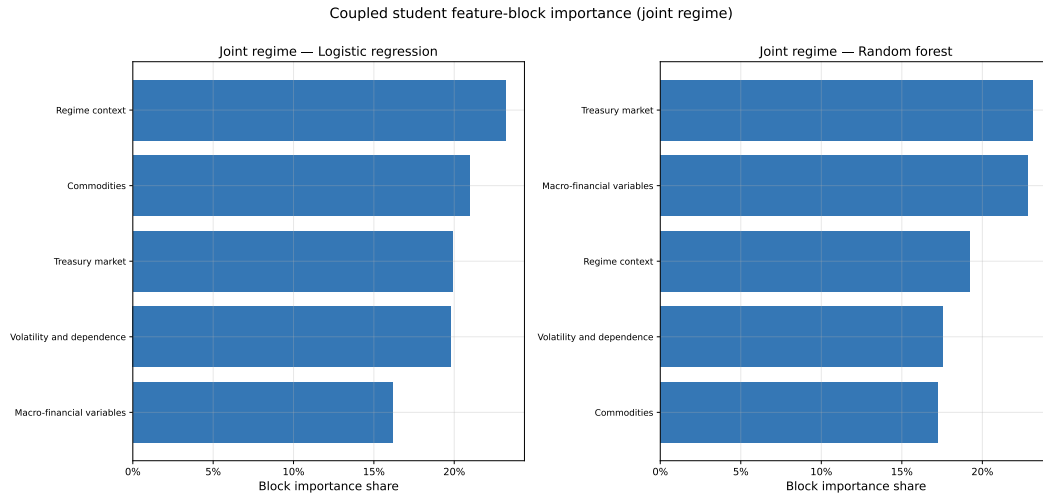
| Model               | Accuracy | Balanced acc. | AUC (macro OVR) | Brier | Log loss |
|---------------------|----------|---------------|-----------------|-------|----------|
| Logistic regression | 0.672    | 0.620         | 0.724           | 0.553 | 1.074    |
| Random forest       | 0.791    | 0.593         | 0.759           | 0.367 | 0.747    |

**Table G.23:** Coupled student: most frequently selected hyper-parameter setting across the walk-forward out-of-sample windows, with the share of windows in which it was chosen.

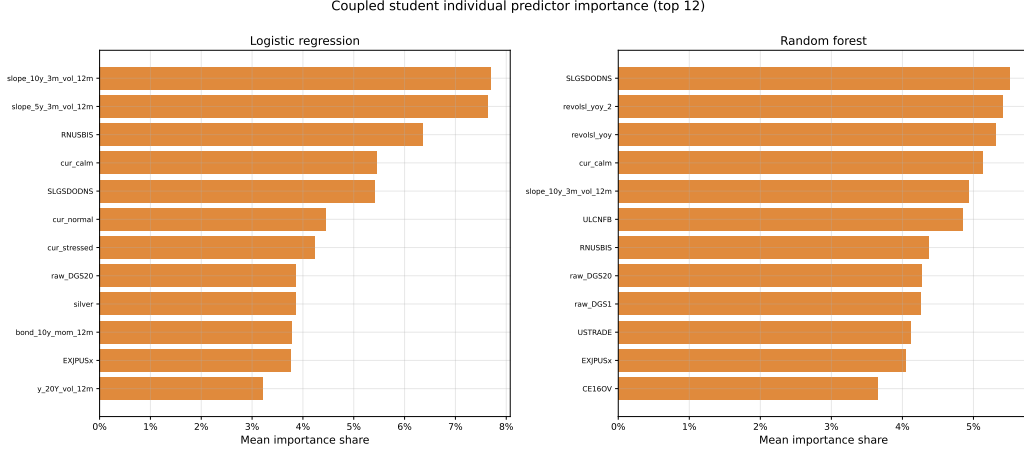
| Model               | Most frequent setting         | Selection share | OOS windows |
|---------------------|-------------------------------|-----------------|-------------|
| Logistic regression | $C = 0.03$                    | 74.1%           | 58          |
| Random forest       | $d = 4$ , leaf = 5, $n = 300$ | 70.7%           | 58          |



**Figure G.29:** COUPLED student out-of-sample probability paths for the three joint regimes (calm, normal, stressed), shown for the logistic and random-forest students. Shading marks the realised teacher regime, the black lines report the predicted regime probabilities, and the dashed line marks the 0.5 threshold.



**Figure G.30:** COUPLED student feature-block relevance for the three-class next-month regime forecast. Logistic-regression relevance is measured by absolute coefficient magnitude. Random-forest relevance is measured by impurity-based feature importance. Values are aggregated within predictor blocks.



**Figure G.31:** Coupled student: individual-predictor importance for the next-month regime forecast.

## H Allocation Details

### H.1 Probability smoothing and shrinkage

Let  $p_{i,m}$  be the predicted favourable-regime probability for asset  $i \in \{E, B\}$ . Predicted probabilities are used directly or smoothed with exponential half-life

$$h \in \{0, 3, 6\}. \quad (70)$$

For each asset and state, historical next-month excess-return moments are estimated from past observations and shrunk toward pre-prediction unconditional moments:

$$a_{i,s,m} = \frac{n_{i,s,m}}{n_{i,s,m} + 36}, \quad (71)$$

$$\tilde{\mu}_{i,s,m} = a_{i,s,m}\hat{\mu}_{i,s,m} + (1 - a_{i,s,m})\mu_{i,0}, \quad (72)$$

$$\tilde{\sigma}_{i,s,m}^2 = a_{i,s,m}\hat{\sigma}_{i,s,m}^2 + (1 - a_{i,s,m})\sigma_{i,0}^2. \quad (73)$$

### H.2 Probability-to-moment mapping

The probability-mixture mean and variance are

$$\mu_{i,m} = (1 - p_{i,m})\tilde{\mu}_{i,0,m} + p_{i,m}\tilde{\mu}_{i,1,m}, \quad (74)$$

$$\sigma_{i,m}^2 = (1 - p_{i,m})(\tilde{\sigma}_{i,0,m}^2 + \tilde{\mu}_{i,0,m}^2) + p_{i,m}(\tilde{\sigma}_{i,1,m}^2 + \tilde{\mu}_{i,1,m}^2) - \mu_{i,m}^2. \quad (75)$$

The equity–bond correlation is estimated from expanding historical returns and shrunk toward the pre-prediction correlation using the same shrinkage form. The covariance matrix is then formed from the conditional volatilities and the shrunk equity–bond correlation.

### H.3 Portfolio grid and net returns

The reported  $L^1$  allocation uses the cash-allowed long-only budget

$$w_E + w_B \leq 1,$$

with equity and bond weights each capped at one and selected on a 0.02 grid. The reported allocation grid varies

$$\begin{aligned} \gamma &\in \{0.25, 0.5, 1, 2, 3, 5\}, \\ c &\in \{0.0005, 0.0010, 0.0025\}, \\ h &\in \{0, 3, 6\}, \end{aligned}$$

and

$$\lambda_{\text{trade}} \in \{0, 0.0005, 0.001, 0.0025, 0.005, 0.0075, 0.01, 0.015, 0.02, 0.03, 0.05\}.$$

Here  $h$  is the probability-smoothing half-life and  $\lambda_{\text{trade}}$  is the turnover penalty in the allocation objective. Monthly net portfolio return is

$$1 + r_{m+1}^p = (1 - c\|w_m - w_{m-1}\|_1) \left( 1 + w_{E,m}R_{E,m+1} + w_{B,m}R_{B,m+1} + w_{C,m}R_{m+1}^f \right). \quad (76)$$



## H.4 Allocation diagnostics and parameter sensitivity

This appendix reports the allocation diagnostics behind the dynamic allocation results in Section 5.1. Table H.24 reports the main independent allocation summary using 10 bps one-way transaction costs. Table H.25 gives the baseline parameter settings for the three dynamic strategies. Figure H.32 reports the corresponding predicted probabilities, portfolio weights, turnover, wealth, and drawdown.

**Table H.24:** Allocation performance summary

| Role                 | Model               | Predictor set              | Ann. ret. | Ann. vol. | Sharpe | Max DD | Ann. turn. | Final   |
|----------------------|---------------------|----------------------------|-----------|-----------|--------|--------|------------|---------|
| 60/40 benchmark      | --                  | --                         | 4.5%      | 9.6%      | 0.47   | -33.4% | 19.7%      | 669.78  |
| High-return dynamic  | Logistic regression | smoothed z-score           | 7.6%      | 10.6%     | 0.71   | -18.9% | 152.1%     | 1513.59 |
| Defensive dynamic    | Random forest       | smoothed z-score           | 4.4%      | 5.8%      | 0.76   | -13.0% | 59.8%      | 636.60  |
| Low-turnover dynamic | Logistic regression | z-score + smoothed z-score | 4.8%      | 7.2%      | 0.67   | -11.5% | 51.4%      | 719.99  |

*Note.* Dynamic strategy returns are net of transaction costs. The 60/40 benchmark is rebalanced monthly and reported net of the same one-way transaction-cost convention. The benchmark and dynamic strategies are evaluated using the same monthly return, drawdown, turnover, and final-wealth definitions.

**Table H.25:** Selected dynamic allocation specifications

| Role         | Model         | Predictor set              | $\gamma$ | $\tau$ | Cost   | Sm. | Ann. ret. | Sharpe | Max DD | Turn.  | Final  |
|--------------|---------------|----------------------------|----------|--------|--------|-----|-----------|--------|--------|--------|--------|
| High-return  | Logistic      | smoothed z-score           | 0.5      | 0      | 10 bps | 0   | 7.6%      | 0.71   | -18.9% | 152.1% | 1513.6 |
| Defensive    | Random forest | smoothed z-score           | 3        | 0.001  | 10 bps | 0   | 4.4%      | 0.76   | -13.0% | 59.8%  | 636.6  |
| Low-turnover | Logistic      | z-score + smoothed z-score | 3        | 0.001  | 10 bps | 3   | 4.8%      | 0.67   | -11.5% | 51.4%  | 720.0  |

*Note.* Parameter settings for the three dynamic strategies shown in the main allocation figures. “Role” abbreviates High-return/Defensive/Low-turnover dynamic; “Logistic” is logistic regression; predictor sets are smoothed z-score and z-score + smoothed z-score.  $\gamma$  is risk aversion,  $\tau$  the turnover penalty, “Sm.” the probability-smoothing half-life (months), “Turn.” annual turnover, “Final” terminal wealth (start = 100).

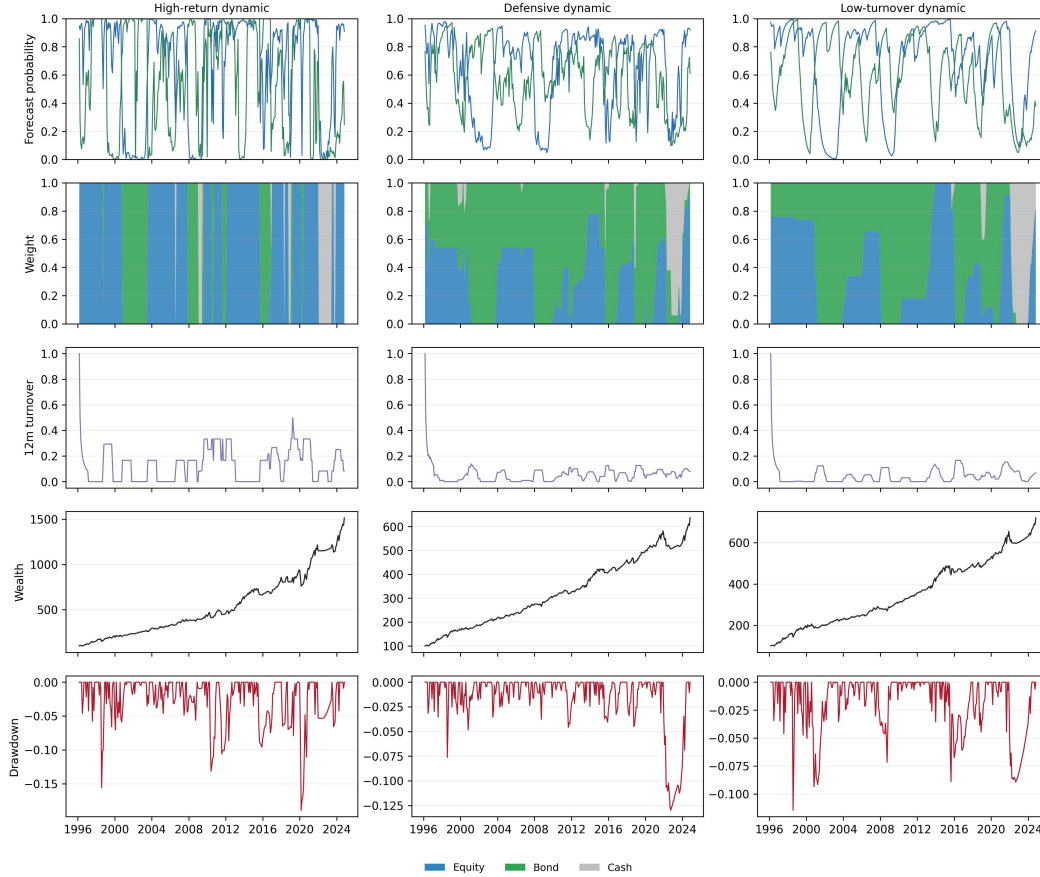
**Headline-row interpretation.** The three independent dynamic rows are representative points from the allocation grid. They are selected to illustrate distinct allocation trade-offs. The high-return row shows a logistic-regression specification with high terminal wealth and materially lower drawdown than the 60/40 benchmark. The defensive row shows a random-forest specification with lower volatility, lower drawdown, and a higher Sharpe ratio. The low-turnover row shows a logistic-regression specification with probability smoothing and annual turnover near 51%. The full parameter sweeps around these rows are reported below.

## H.5 Additional independent benchmarks

The additional benchmarks use the same monthly excess-return columns as the independent allocation sample ending in 2024-10. Cash is treated as zero excess return because the independent allocation stage uses equity and bond excess returns but no separate cash-return column. These benchmarks are included to compare the regime-aware rules with simple dynamic or single-asset alternatives that do not use regime forecasts.

| Benchmark                 | Ann. return | Ann. vol. | Sharpe | Max drawdown | Ann. turnover | Final wealth |
|---------------------------|-------------|-----------|--------|--------------|---------------|--------------|
| 60/40 benchmark           | 4.5%        | 9.6%      | 0.47   | -36.2%       | 19.7%         | 359.35       |
| Equity-only               | 6.9%        | 15.4%     | 0.44   | -62.2%       | 0.0%          | 505.52       |
| Bond-only                 | 2.1%        | 4.1%      | 0.51   | -21.3%       | 0.0%          | 177.25       |
| Volatility-targeted 60/40 | 3.6%        | 8.9%      | 0.41   | -35.8%       | 8.0%          | 229.92       |

**Table H.26:** Additional independent dynamic-allocation benchmarks. Returns are monthly excess returns over the independent allocation sample ending in 2024-10. Cash is treated as zero excess return. The volatility-targeted 60/40 rule uses a trailing 36-month realised volatility estimate, a 10% annual volatility target, and leverage capped at one.



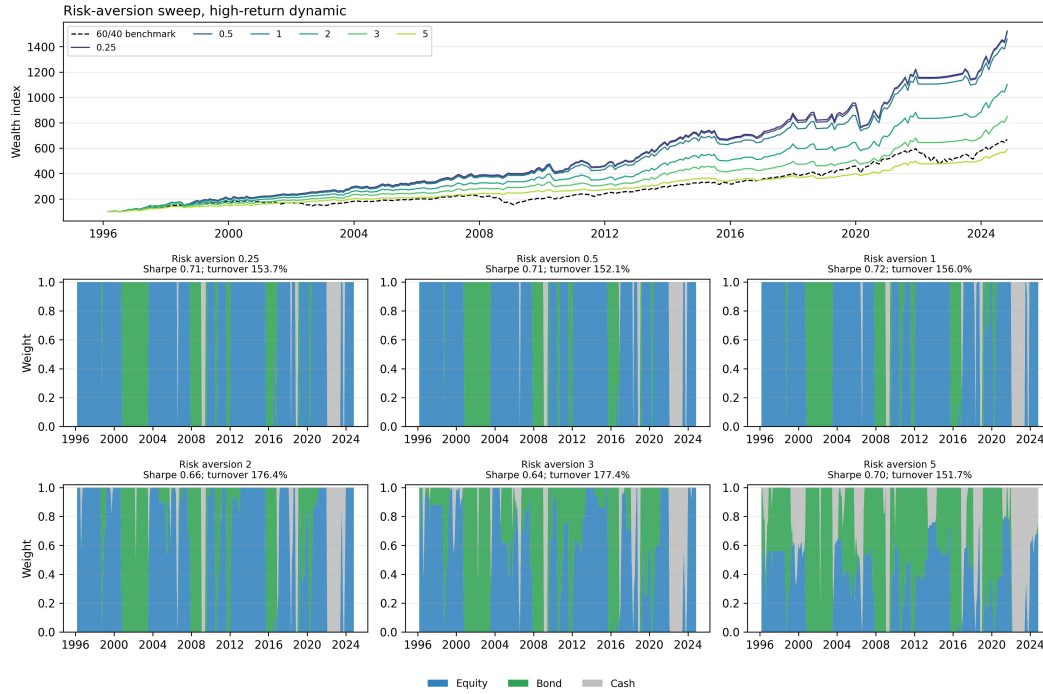
**Figure H.32:** Allocation mechanics for selected dynamic strategies

Note: The panels report predicted favourable-regime probabilities, portfolio weights, rolling 12-month turnover, wealth, and drawdown for the three selected dynamic strategies.

The mechanics figure shows that the three dynamic strategies use the same regime forecasts in different ways. The high-return strategy takes large equity positions and accepts higher turnover. The defensive strategy uses more bond exposure and reduces equity exposure when forecasts weaken. The low-turnover strategy changes allocations more slowly and uses probability smoothing to reduce trading.

### H.5.1 Parameter sensitivity around selected strategies

The sensitivity checks vary allocation parameters around the three selected dynamic strategies. In each one-parameter sweep, the reported parameter varies and the remaining allocation parameters are fixed at the strategy-specific baseline values in Table H.25. The transaction-cost and turnover-penalty table varies these two parameters jointly and keeps the remaining parameters fixed at the same baseline values.



**Figure H.33:** Risk-aversion sweep, high-return dynamic

Note: Risk aversion varies across panels. The model, predictor set, turnover penalty, transaction cost, and smoothing half-life are fixed at the strategy-specific baseline values in Table H.25.

**Table H.27:** Risk-aversion sweep, high-return dynamic

| Risk aversion | Ann. return | Ann. vol. | Sharpe | Max. drawdown | Monthly CVaR 1% | Final wealth | Ann. turnover |
|---------------|-------------|-----------|--------|---------------|-----------------|--------------|---------------|
| 0.25          | 7.6%        | 10.7%     | 0.71   | -20.0%        | -11.1%          | 1523.40      | 153.7%        |
| 0.5           | 7.6%        | 10.6%     | 0.71   | -18.9%        | -10.8%          | 1513.59      | 152.1%        |
| 1             | 7.4%        | 10.4%     | 0.72   | -15.6%        | -9.7%           | 1460.98      | 156.0%        |
| 2             | 6.4%        | 9.7%      | 0.66   | -15.6%        | -9.3%           | 1101.27      | 176.4%        |
| 3             | 5.4%        | 8.5%      | 0.64   | -13.5%        | -8.5%           | 850.63       | 177.4%        |
| 5             | 4.1%        | 5.8%      | 0.70   | -8.0%         | -5.6%           | 588.05       | 151.7%        |



**Figure H.34:** Risk-aversion sweep, defensive dynamic

Note: Risk aversion varies across panels. The model, predictor set, turnover penalty, transaction cost, and smoothing half-life are fixed at the strategy-specific baseline values in Table H.25.

**Table H.28:** Risk-aversion sweep, defensive dynamic

| Risk aversion | Ann. return | Ann. vol. | Sharpe | Max. drawdown | Monthly CVaR 1% | Final wealth | Ann. turnover |
|---------------|-------------|-----------|--------|---------------|-----------------|--------------|---------------|
| 0.25          | 6.1%        | 10.6%     | 0.58   | -18.2%        | -10.5%          | 1035.50      | 99.2%         |
| 0.5           | 5.9%        | 10.2%     | 0.58   | -17.0%        | -10.0%          | 976.60       | 100.3%        |
| 1             | 6.0%        | 9.7%      | 0.62   | -17.1%        | -9.9%           | 1001.81      | 91.1%         |
| 2             | 5.4%        | 7.6%      | 0.70   | -12.1%        | -7.5%           | 837.17       | 77.9%         |
| 3             | 4.4%        | 5.8%      | 0.76   | -13.0%        | -5.5%           | 636.60       | 59.8%         |
| 5             | 3.5%        | 4.4%      | 0.78   | -12.4%        | -4.2%           | 495.49       | 43.1%         |



**Figure H.35:** Risk-aversion sweep, low-turnover dynamic

Note: Risk aversion varies across panels. The model, predictor set, turnover penalty, transaction cost, and smoothing half-life are fixed at the strategy-specific baseline values in Table H.25.

**Table H.29:** Risk-aversion sweep, low-turnover dynamic

| Risk aversion | Ann. return | Ann. vol. | Sharpe | Max. drawdown | Monthly CVaR 1% | Final wealth | Ann. turnover |
|---------------|-------------|-----------|--------|---------------|-----------------|--------------|---------------|
| 0.25          | 5.8%        | 11.9%     | 0.49   | -23.9%        | -11.7%          | 936.18       | 66.1%         |
| 0.5           | 5.5%        | 11.1%     | 0.49   | -22.6%        | -11.1%          | 865.89       | 68.2%         |
| 1             | 5.5%        | 10.0%     | 0.55   | -20.4%        | -10.6%          | 862.11       | 67.2%         |
| 2             | 5.4%        | 8.9%      | 0.61   | -15.6%        | -9.0%           | 849.15       | 61.4%         |
| 3             | 4.8%        | 7.2%      | 0.67   | -11.5%        | -7.1%           | 719.99       | 51.4%         |
| 5             | 3.8%        | 5.0%      | 0.76   | -9.9%         | -4.5%           | 545.07       | 35.5%         |

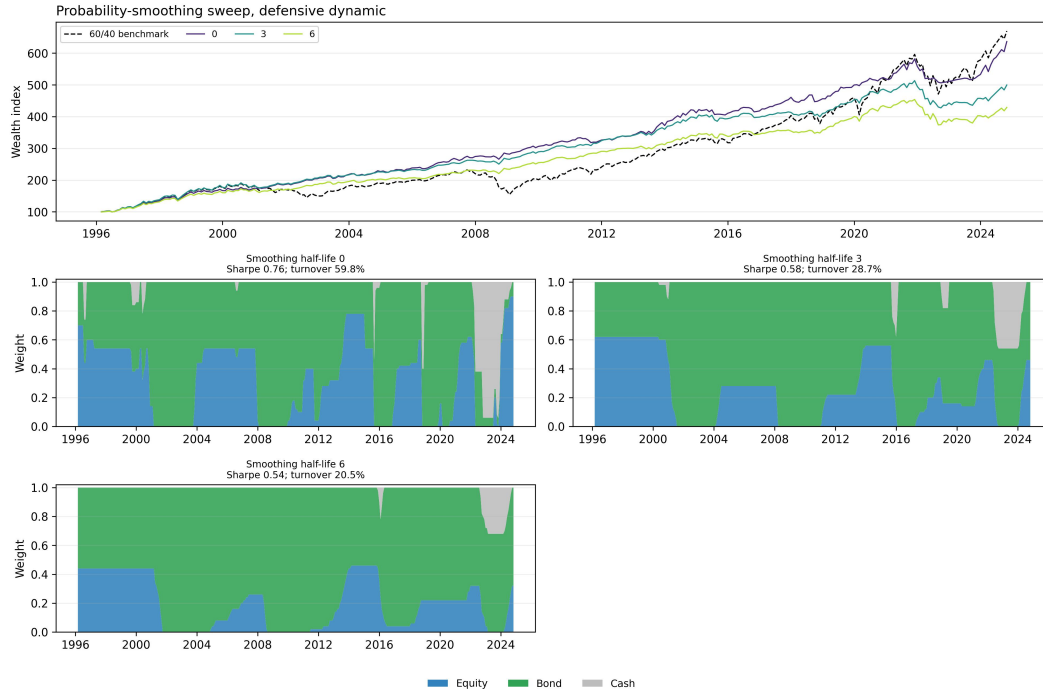


**Figure H.36:** Probability-smoothing sweep, high-return dynamic

Note: Probability smoothing varies across panels. The model, predictor set, risk aversion, turnover penalty, and transaction cost are fixed at the strategy-specific baseline values in Table H.25.

**Table H.30:** Probability-smoothing sweep, high-return dynamic

| Smoothing half-life | Ann. return | Ann. vol. | Sharpe | Max. drawdown | Monthly CVaR 1% | Final wealth | Ann. turnover |
|---------------------|-------------|-----------|--------|---------------|-----------------|--------------|---------------|
| 0                   | 7.6%        | 10.6%     | 0.71   | -18.9%        | -10.8%          | 1513.59      | 152.1%        |
| 3                   | 6.0%        | 11.3%     | 0.53   | -20.0%        | -11.5%          | 1007.50      | 82.5%         |
| 6                   | 6.3%        | 12.0%     | 0.52   | -23.4%        | -11.7%          | 1073.65      | 47.4%         |

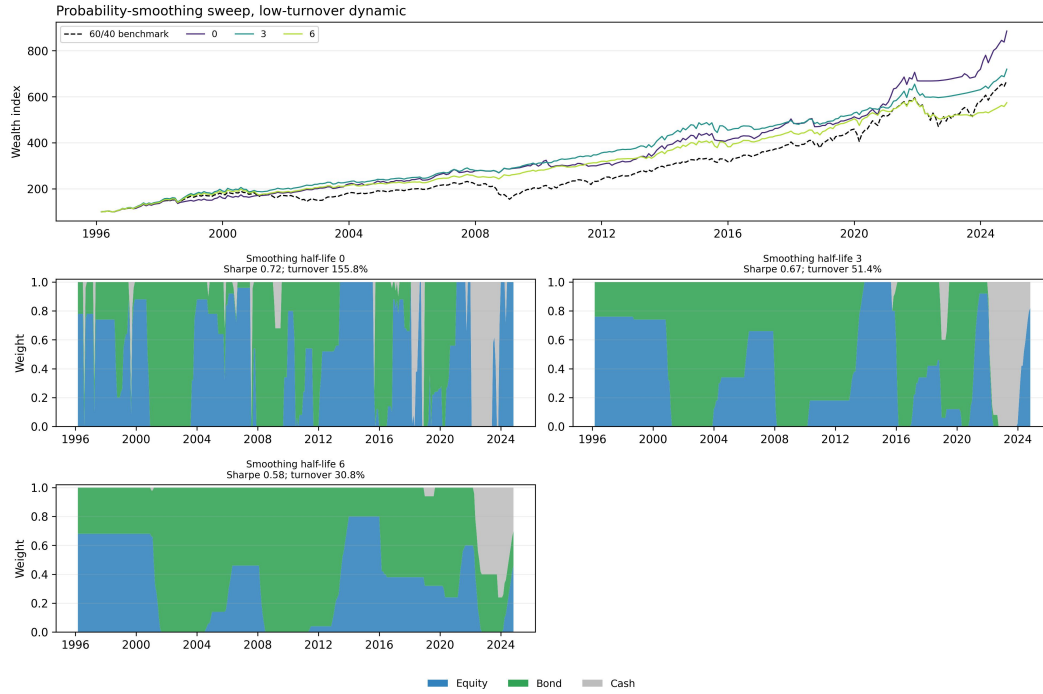


**Figure H.37:** Probability-smoothing sweep, defensive dynamic

Note: Probability smoothing varies across panels. The model, predictor set, risk aversion, turnover penalty, and transaction cost are fixed at the strategy-specific baseline values in Table H.25.

**Table H.31:** Probability-smoothing sweep, defensive dynamic

| Smoothing half-life | Ann. return | Ann. vol. | Sharpe | Max. drawdown | Monthly CVaR 1% | Final wealth | Ann. turnover |
|---------------------|-------------|-----------|--------|---------------|-----------------|--------------|---------------|
| 0                   | 4.4%        | 5.8%      | 0.76   | -13.0%        | -5.5%           | 636.60       | 59.8%         |
| 3                   | 3.5%        | 6.0%      | 0.58   | -16.7%        | -6.0%           | 499.69       | 28.7%         |
| 6                   | 2.9%        | 5.4%      | 0.54   | -17.6%        | -5.2%           | 429.04       | 20.5%         |



**Figure H.38:** Probability-smoothing sweep, low-turnover dynamic

Note: Probability smoothing varies across panels. The model, predictor set, risk aversion, turnover penalty, and transaction cost are fixed at the strategy-specific baseline values in Table H.25.

**Table H.32:** Probability-smoothing sweep, low-turnover dynamic

| Smoothing half-life | Ann. return | Ann. vol. | Sharpe | Max. drawdown | Monthly CVaR 1% | Final wealth | Ann. turnover |
|---------------------|-------------|-----------|--------|---------------|-----------------|--------------|---------------|
| 0                   | 5.6%        | 7.7%      | 0.72   | -11.1%        | -7.6%           | 885.67       | 155.8%        |
| 3                   | 4.8%        | 7.2%      | 0.67   | -11.5%        | -7.1%           | 719.99       | 51.4%         |
| 6                   | 4.0%        | 6.9%      | 0.58   | -15.2%        | -7.0%           | 574.31       | 30.8%         |



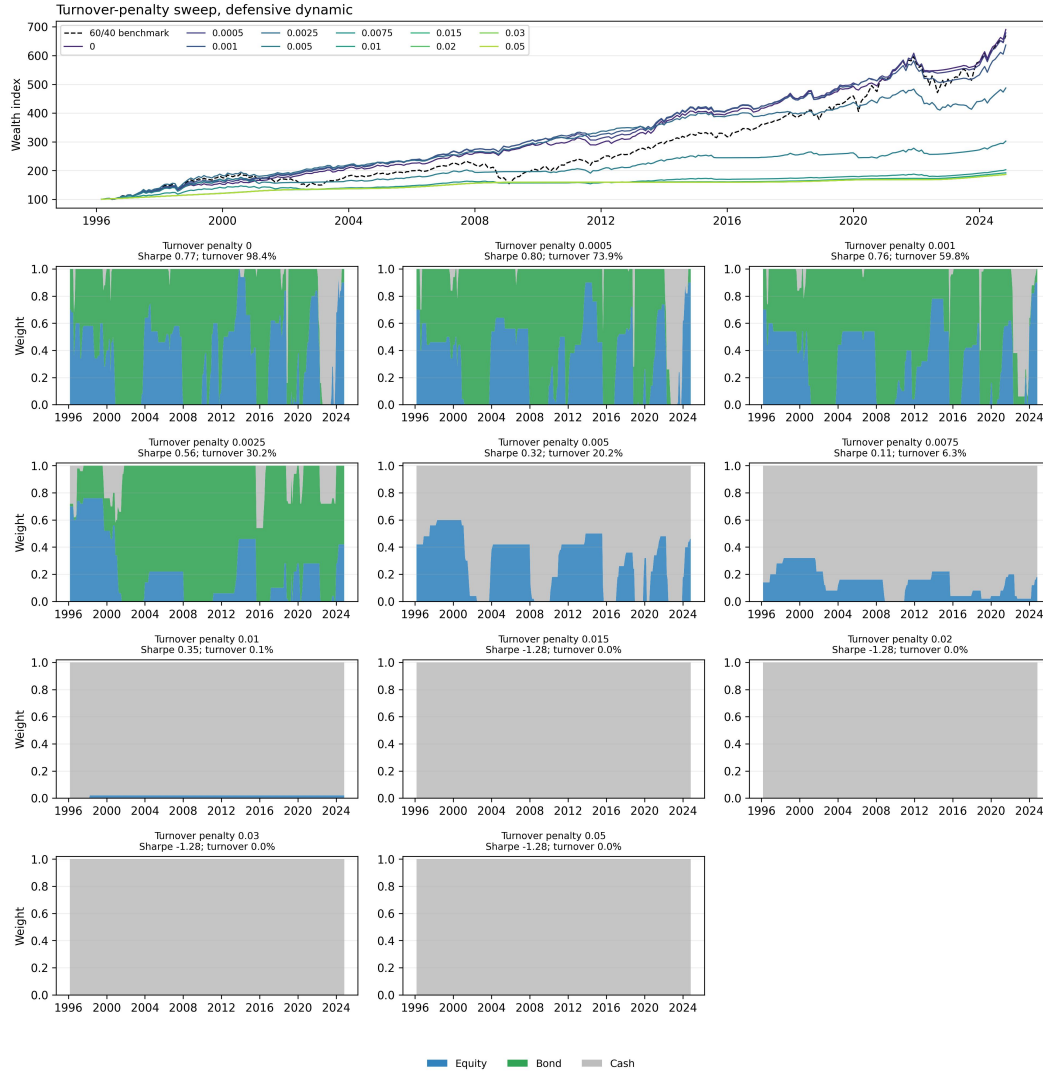


**Figure H.39:** Turnover-penalty sweep, high-return dynamic

Note: Turnover penalty varies across panels. The model, predictor set, risk aversion, transaction cost, and smoothing half-life are fixed at the strategy-specific baseline values in Table H.25.

**Table H.33:** Turnover-penalty sweep, high-return dynamic

| Turnover penalty | Ann. return | Ann. vol. | Sharpe | Max. drawdown | Monthly CVaR 1% | Final wealth | Ann. turnover |
|------------------|-------------|-----------|--------|---------------|-----------------|--------------|---------------|
| 0                | 7.6%        | 10.6%     | 0.71   | -18.9%        | -10.8%          | 1513.59      | 152.1%        |
| 0.0005           | 7.6%        | 10.7%     | 0.71   | -20.0%        | -11.1%          | 1525.04      | 135.5%        |
| 0.001            | 7.3%        | 10.7%     | 0.69   | -20.0%        | -11.1%          | 1412.70      | 125.9%        |
| 0.0025           | 5.2%        | 10.2%     | 0.51   | -20.0%        | -11.2%          | 796.46       | 114.5%        |
| 0.005            | 5.8%        | 11.2%     | 0.51   | -21.1%        | -11.3%          | 936.68       | 59.1%         |
| 0.0075           | 4.2%        | 12.1%     | 0.35   | -33.5%        | -11.7%          | 608.48       | 48.1%         |
| 0.01             | 0.8%        | 5.5%      | 0.14   | -22.2%        | -5.9%           | 232.06       | 8.8%          |
| 0.015            | -0.0%       | 0.0%      | -1.28  | 0.0%          | -0.0%           | 186.68       | 0.0%          |
| 0.02             | -0.0%       | 0.0%      | -1.28  | 0.0%          | -0.0%           | 186.68       | 0.0%          |
| 0.03             | -0.0%       | 0.0%      | -1.28  | 0.0%          | -0.0%           | 186.68       | 0.0%          |
| 0.05             | -0.0%       | 0.0%      | -1.28  | 0.0%          | -0.0%           | 186.68       | 0.0%          |

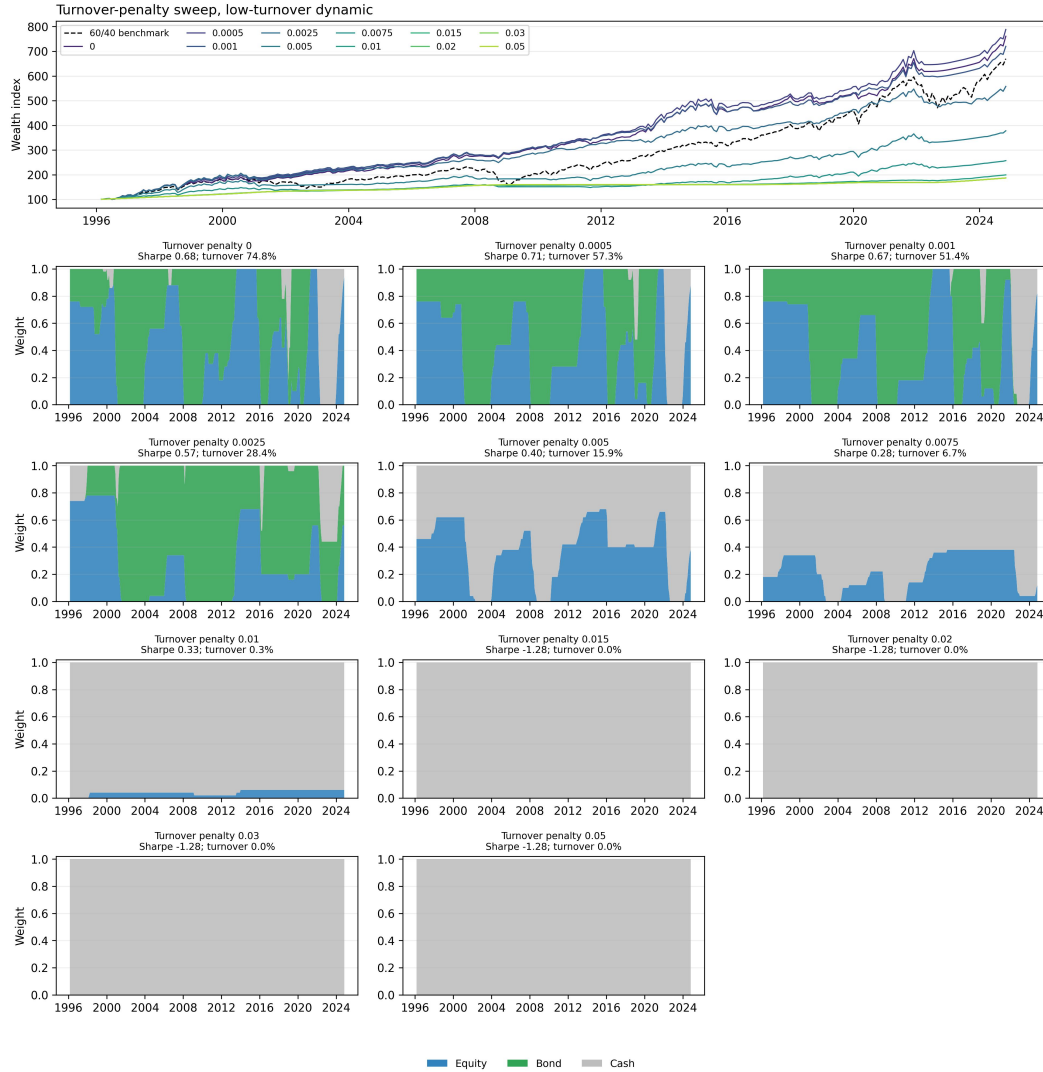


**Figure H.40:** Turnover-penalty sweep, defensive dynamic

Note: Turnover penalty varies across panels. The model, predictor set, risk aversion, transaction cost, and smoothing half-life are fixed at the strategy-specific baseline values in Table H.25.

**Table H.34:** Turnover-penalty sweep, defensive dynamic

| Turnover penalty | Ann. return | Ann. vol. | Sharpe | Max. drawdown | Monthly CVaR 1% | Final wealth | Ann. turnover |
|------------------|-------------|-----------|--------|---------------|-----------------|--------------|---------------|
| 0                | 4.7%        | 6.1%      | 0.77   | -10.2%        | -5.9%           | 690.57       | 98.4%         |
| 0.0005           | 4.6%        | 5.8%      | 0.80   | -11.1%        | -5.2%           | 678.95       | 73.9%         |
| 0.001            | 4.4%        | 5.8%      | 0.76   | -13.0%        | -5.5%           | 636.60       | 59.8%         |
| 0.0025           | 3.4%        | 6.1%      | 0.56   | -15.1%        | -6.4%           | 487.25       | 30.2%         |
| 0.005            | 1.7%        | 5.2%      | 0.32   | -12.0%        | -5.9%           | 302.54       | 20.2%         |
| 0.0075           | 0.3%        | 2.7%      | 0.11   | -8.8%         | -3.3%           | 203.22       | 6.3%          |
| 0.01             | 0.1%        | 0.3%      | 0.35   | -0.5%         | -0.3%           | 192.37       | 0.1%          |
| 0.015            | -0.0%       | 0.0%      | -1.28  | 0.0%          | -0.0%           | 186.68       | 0.0%          |
| 0.02             | -0.0%       | 0.0%      | -1.28  | 0.0%          | -0.0%           | 186.68       | 0.0%          |
| 0.03             | -0.0%       | 0.0%      | -1.28  | 0.0%          | -0.0%           | 186.68       | 0.0%          |
| 0.05             | -0.0%       | 0.0%      | -1.28  | 0.0%          | -0.0%           | 186.68       | 0.0%          |



**Figure H.41:** Turnover-penalty sweep, low-turnover dynamic

Note: Turnover penalty varies across panels. The model, predictor set, risk aversion, transaction cost, and smoothing half-life are fixed at the strategy-specific baseline values in Table H.25.

**Table H.35:** Turnover-penalty sweep, low-turnover dynamic

| Turnover penalty | Ann. return | Ann. vol. | Sharpe | Max. drawdown | Monthly CVaR 1% | Final wealth | Ann. turnover |
|------------------|-------------|-----------|--------|---------------|-----------------|--------------|---------------|
| 0                | 5.0%        | 7.4%      | 0.68   | -10.8%        | -7.1%           | 760.69       | 74.8%         |
| 0.0005           | 5.1%        | 7.2%      | 0.71   | -11.5%        | -7.2%           | 788.23       | 57.3%         |
| 0.001            | 4.8%        | 7.2%      | 0.67   | -11.5%        | -7.1%           | 719.99       | 51.4%         |
| 0.0025           | 3.9%        | 6.8%      | 0.57   | -13.3%        | -6.9%           | 557.59       | 28.4%         |
| 0.005            | 2.5%        | 6.2%      | 0.40   | -13.8%        | -6.4%           | 377.90       | 15.9%         |
| 0.0075           | 1.1%        | 4.0%      | 0.28   | -8.2%         | -4.2%           | 257.09       | 6.7%          |
| 0.01             | 0.2%        | 0.7%      | 0.33   | -1.6%         | -0.7%           | 199.35       | 0.3%          |
| 0.015            | -0.0%       | 0.0%      | -1.28  | 0.0%          | -0.0%           | 186.68       | 0.0%          |
| 0.02             | -0.0%       | 0.0%      | -1.28  | 0.0%          | -0.0%           | 186.68       | 0.0%          |
| 0.03             | -0.0%       | 0.0%      | -1.28  | 0.0%          | -0.0%           | 186.68       | 0.0%          |
| 0.05             | -0.0%       | 0.0%      | -1.28  | 0.0%          | -0.0%           | 186.68       | 0.0%          |

**Table H.36:** Transaction-cost and turnover-penalty sensitivity

| Strategy             | Turnover penalty | 5 bps | 10 bps | 25 bps |
|----------------------|------------------|-------|--------|--------|
| High-return dynamic  | 0.0000           | 0.77  | 0.74   | 0.67   |
| High-return dynamic  | 0.0005           | 0.75  | 0.74   | 0.62   |
| High-return dynamic  | 0.0010           | 0.75  | 0.72   | 0.53   |
| High-return dynamic  | 0.0025           | 0.64  | 0.55   | 0.44   |
| High-return dynamic  | 0.0050           | 0.59  | 0.56   | 0.47   |
| High-return dynamic  | 0.0100           | 0.15  | 0.17   |        |
| Defensive dynamic    | 0.0000           | 0.75  | 0.78   | 0.69   |
| Defensive dynamic    | 0.0005           | 0.79  | 0.81   | 0.61   |
| Defensive dynamic    | 0.0010           | 0.81  | 0.77   | 0.57   |
| Defensive dynamic    | 0.0025           | 0.63  | 0.58   | 0.43   |
| Defensive dynamic    | 0.0050           | 0.38  | 0.35   | 0.20   |
| Defensive dynamic    | 0.0100           | 0.15  | 0.35   |        |
| Low-turnover dynamic | 0.0000           | 0.71  | 0.70   | 0.63   |
| Low-turnover dynamic | 0.0005           | 0.71  | 0.73   | 0.61   |
| Low-turnover dynamic | 0.0010           | 0.74  | 0.69   | 0.59   |
| Low-turnover dynamic | 0.0025           | 0.62  | 0.60   | 0.50   |
| Low-turnover dynamic | 0.0050           | 0.43  | 0.43   | 0.38   |
| Low-turnover dynamic | 0.0100           | 0.23  | 0.33   |        |

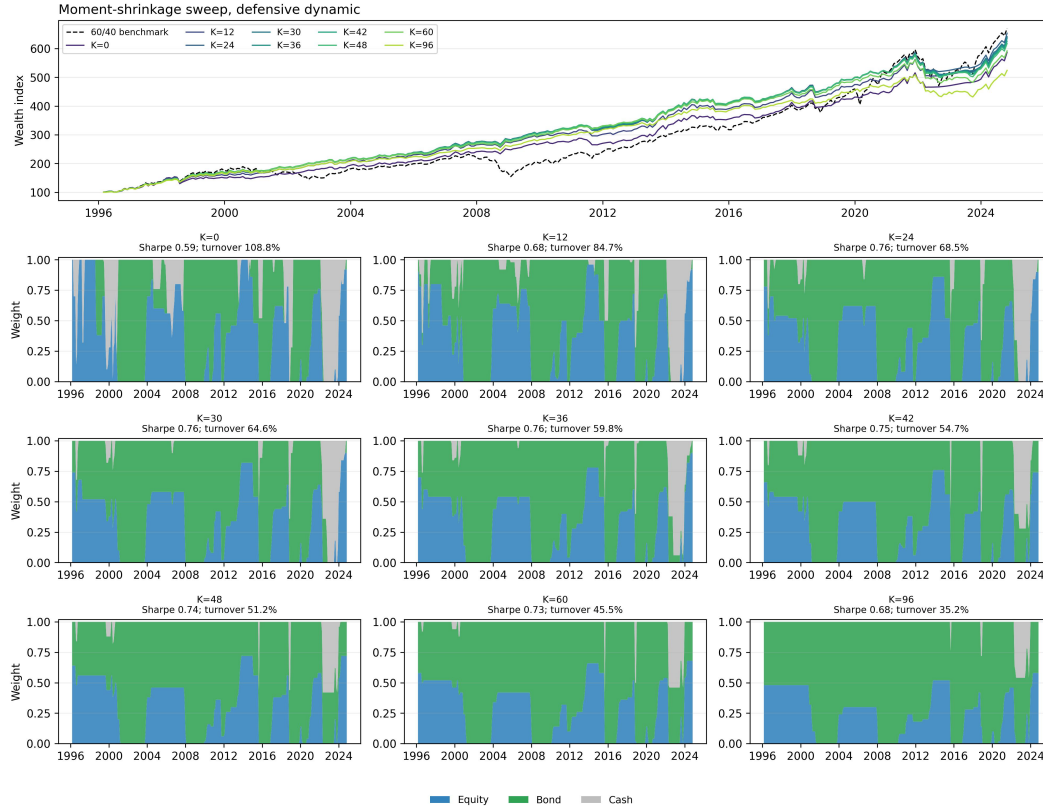
*Note.* Entries are Sharpe ratios. For each dynamic strategy, the model, predictor set, risk aversion, and smoothing half-life are fixed at the strategy-specific baseline values in Table H.25.

**Figure H.42:** Moment-shrinkage sweep, high-return dynamic

*Note:* The moment-shrinkage constant  $K$  varies across panels. The model, predictor set, risk aversion, turnover penalty, transaction cost, and smoothing half-life are fixed at the strategy-specific baseline values in Table H.25.

**Table H.37:** Moment-shrinkage sweep, high-return dynamic

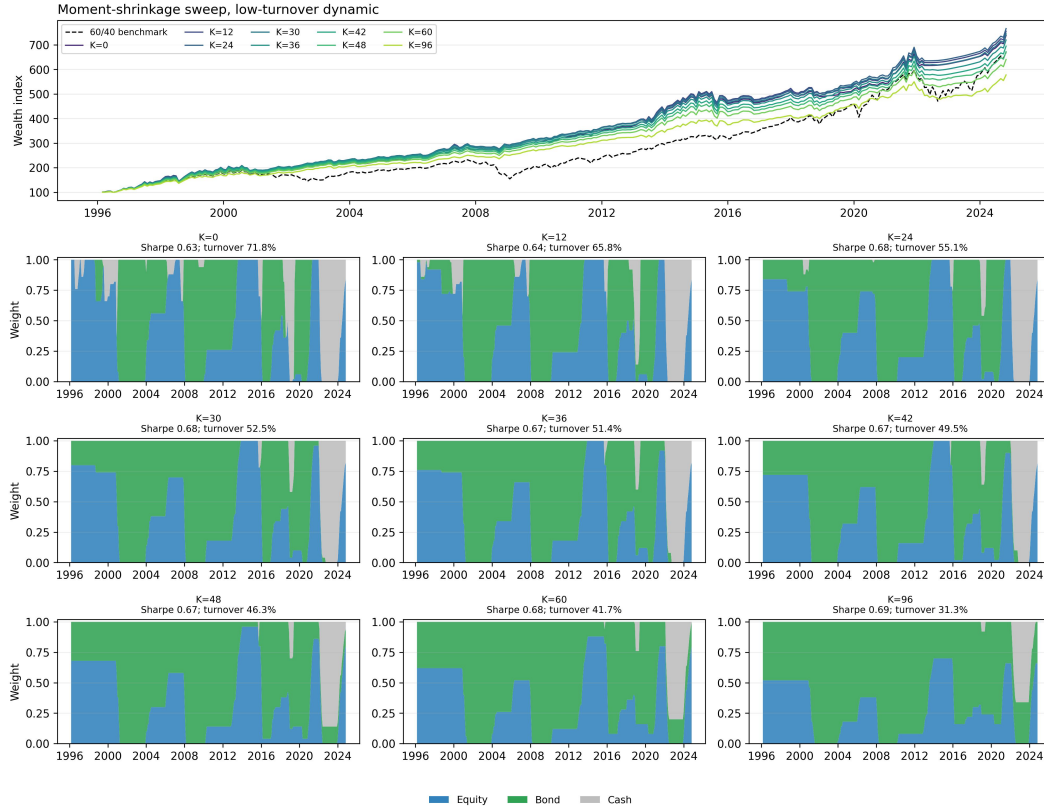
| $K$ | Ann. ret. | Ann. vol. | Sharpe | Max DD | CVaR 1% | Final   | Ann. turn. |
|-----|-----------|-----------|--------|--------|---------|---------|------------|
| 0   | 7.3%      | 10.4%     | 0.71   | -16.3% | -9.7%   | 1419.16 | 173.4%     |
| 12  | 7.6%      | 10.5%     | 0.72   | -16.3% | -9.7%   | 1523.20 | 160.1%     |
| 24  | 7.5%      | 10.6%     | 0.71   | -17.3% | -10.3%  | 1496.05 | 159.9%     |
| 30  | 7.5%      | 10.6%     | 0.71   | -18.2% | -10.5%  | 1484.06 | 160.4%     |
| 36  | 7.6%      | 10.6%     | 0.71   | -19.2% | -10.8%  | 1513.59 | 152.1%     |
| 42  | 7.6%      | 10.7%     | 0.71   | -20.1% | -11.0%  | 1523.58 | 149.6%     |
| 48  | 7.8%      | 10.7%     | 0.73   | -20.3% | -11.1%  | 1598.17 | 141.6%     |
| 60  | 7.8%      | 10.7%     | 0.73   | -20.3% | -11.1%  | 1629.29 | 135.2%     |
| 96  | 7.5%      | 11.0%     | 0.68   | -21.5% | -11.1%  | 1476.33 | 135.1%     |

**Figure H.43:** Moment-shrinkage sweep, defensive dynamic

Note: The moment-shrinkage constant  $K$  varies across panels. The model, predictor set, risk aversion, turnover penalty, transaction cost, and smoothing half-life are fixed at the strategy-specific baseline values in Table H.25.

**Table H.38:** Moment-shrinkage sweep, defensive dynamic

| $K$ | Ann. ret. | Ann. vol. | Sharpe | Max DD | CVaR 1% | Final  | Ann. turn. |
|-----|-----------|-----------|--------|--------|---------|--------|------------|
| 0   | 4.1%      | 6.9%      | 0.59   | -16.3% | -7.9%   | 587.41 | 108.8%     |
| 12  | 4.4%      | 6.4%      | 0.68   | -12.9% | -6.7%   | 636.94 | 84.7%      |
| 24  | 4.5%      | 5.9%      | 0.76   | -12.8% | -5.4%   | 653.67 | 68.5%      |
| 30  | 4.4%      | 5.8%      | 0.76   | -14.1% | -5.4%   | 641.51 | 64.6%      |
| 36  | 4.4%      | 5.8%      | 0.76   | -14.9% | -5.5%   | 636.60 | 59.8%      |
| 42  | 4.3%      | 5.7%      | 0.75   | -16.0% | -5.5%   | 625.91 | 54.7%      |
| 48  | 4.3%      | 5.8%      | 0.74   | -16.8% | -5.6%   | 617.80 | 51.2%      |
| 60  | 4.1%      | 5.6%      | 0.73   | -17.7% | -5.5%   | 592.27 | 45.5%      |
| 96  | 3.7%      | 5.3%      | 0.68   | -18.9% | -5.1%   | 524.22 | 35.2%      |

**Figure H.44:** Moment-shrinkage sweep, low-turnover dynamic

Note: The moment-shrinkage constant  $K$  varies across panels. The model, predictor set, risk aversion, turnover penalty, transaction cost, and smoothing half-life are fixed at the strategy-specific baseline values in Table H.25.

**Table H.39:** Moment-shrinkage sweep, low-turnover dynamic

| $K$ | Ann. ret. | Ann. vol. | Sharpe | Max DD | CVaR 1% | Final  | Ann. turn. |
|-----|-----------|-----------|--------|--------|---------|--------|------------|
| 0   | 4.9%      | 7.8%      | 0.63   | -16.3% | -8.6%   | 738.97 | 71.8%      |
| 12  | 5.0%      | 7.7%      | 0.64   | -15.0% | -8.2%   | 755.63 | 65.8%      |
| 24  | 5.0%      | 7.5%      | 0.68   | -13.6% | -7.6%   | 766.50 | 55.1%      |
| 30  | 4.9%      | 7.3%      | 0.68   | -12.9% | -7.3%   | 745.09 | 52.5%      |
| 36  | 4.8%      | 7.2%      | 0.67   | -13.0% | -7.1%   | 719.99 | 51.4%      |
| 42  | 4.7%      | 7.0%      | 0.67   | -13.2% | -6.9%   | 694.83 | 49.5%      |
| 48  | 4.6%      | 6.8%      | 0.67   | -12.8% | -6.5%   | 672.44 | 46.3%      |
| 60  | 4.4%      | 6.4%      | 0.68   | -12.0% | -6.0%   | 640.71 | 41.7%      |
| 96  | 4.0%      | 5.9%      | 0.69   | -14.5% | -5.2%   | 577.82 | 31.3%      |

The sensitivity checks show how the same allocation rule changes when a single parameter is varied around each selected strategy. Higher risk aversion lowers equity exposure and reduces drawdowns in the defensive and low-turnover specifications. Probability smoothing reduces trading by slowing changes in the forecast probabilities. Higher turnover penalties reduce trading and increase cash weights, which lowers turnover but can reduce the Sharpe ratio when the penalty is too large. The moment-shrinkage sweep varies the amount of shrinkage applied to regime-conditional moments and the stock–bond correlation estimate. Larger  $K$  places more weight on unconditional pre-out-of-sample moments, while smaller  $K$  gives more weight to realised state-conditional samples.

## H.6 Independent model-selection protocol

This table summarises which independent-side choices are fixed design choices, which are selected on validation or grid search, and which are used only as diagnostics. The purpose is to separate the reported independent specification from robustness checks and presentation-only diagnostics.

| Component                     | Independent approach                                                               | Fixed in advance                                                             | Selected on validation/grid                                                                        | Robustness or diagnostic only                                         |
|-------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| Teacher feature set           | Equity return–Sortino features and canonical bond features                         | Final reported feature blocks fixed before student prediction and allocation | No                                                                                                 | Volatility/downside equity labels used only as diagnostics            |
| Teacher jump penalty grid     | Separate equity and bond jump-penalty grids                                        | Grid fixed before final majority vote                                        | Majority-vote label selected from fitted teacher paths                                             | Path persistence and feature-separation plots reported as diagnostics |
| Distance norm                 | $L^2$ centroid and $L^1$ -medoid teacher paths                                     | Both norm families fixed as teacher candidates                               | Final label uses the combined majority vote                                                        | Separate $L^2$ and $L^1$ paths reported as diagnostics                |
| Teacher majority vote         | Final equity and bond labels combine teacher paths by majority vote                | Voting rule fixed before student prediction                                  | No refit after allocation outcomes                                                                 | Vote-scope figures report candidate-path agreement                    |
| Student predictor sets        | z-score predictors, smoothed z-score predictors, and their union                   | Transformations fixed before model fitting                                   | Predictor-set comparison reported out of sample                                                    | Feature-importance figures are diagnostic                             |
| Student model hyperparameters | Logistic regression and random forest specifications                               | Model classes fixed before evaluation                                        | Hyperparameters selected within the rolling validation design                                      | Model-frequency table reports selected specifications                 |
| Student validation metrics    | AUC, balanced accuracy, Brier score, log loss, and allocation-relevant diagnostics | Metrics fixed before reporting                                               | Best reported specifications selected from rolling outputs                                         | Full prediction diagnostics kept in appendix                          |
| Allocation rule               | Long-only equity–bond allocation with cash allowed and an $L^1$ turnover penalty   | Asset universe and allocation objective fixed before grid search             | Candidate grid varies risk aversion, turnover penalty, transaction cost, and probability smoothing | Parameter-sensitivity appendix reports one-way and joint checks       |
| Risk aversion                 | Values from the allocation grid                                                    | Grid fixed before headline reporting                                         | Representative rows use selected grid points                                                       | Risk-aversion sweeps reported as sensitivity                          |
| Transaction cost              | One-way transaction-cost grid                                                      | Grid fixed before headline reporting                                         | Headline independent rows use 10 bps                                                               | Cost and turnover-penalty sensitivity reported in appendix            |
| Turnover penalty              | $L^1$ turnover penalty in the allocation objective                                 | Grid fixed before headline reporting                                         | Representative rows use role-specific penalty values                                               | Turnover-penalty sweeps reported as sensitivity                       |
| Headline allocation rows      | High-return, defensive, and low-turnover representative rows                       | Roles fixed for presentation                                                 | Rows selected as representative points from the allocation grid                                    | Full-grid ranking rules are reported only as sensitivity diagnostics  |

**Table H.40:** Independent-side model-selection protocol. The table separates fixed design choices, validation or grid-selected choices, and robustness or diagnostic components.



## I Coupled allocation: details and sensitivity

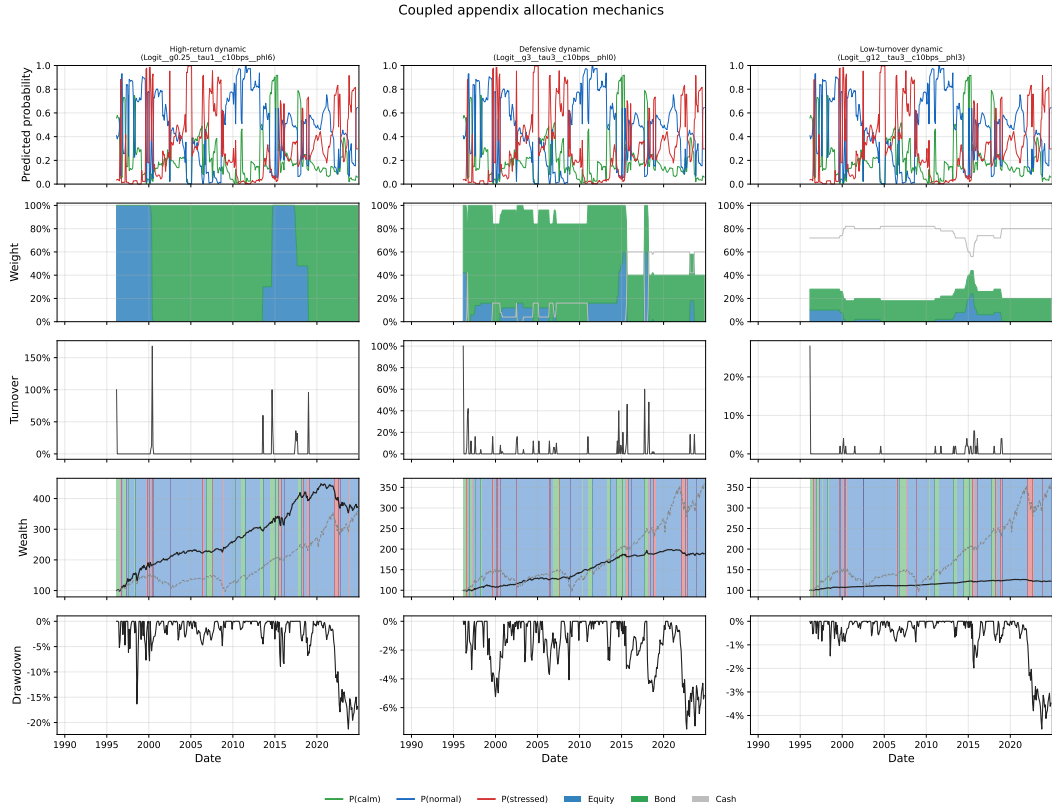
The coupled allocator is a faithful port of the INDEPENDENT allocation engine, so that the only difference between the two approaches is the regime representation that supplies the conditional moments. The investable universe is the equity leg, the bond leg and cash; weights are long-only with each risky leg capped at one and a gross risky cap of 1.0 ( $w_E + w_B \leq 1$ ), chosen on the same 0.02 grid as the INDEPENDENT side to maximise the mean–variance objective (8). Portfolios rebalance monthly and pay 10 bps on traded notional at the headline cost. The conditional moments are formed as a probability mixture over the three joint regimes: per-regime means and variances are estimated from the expanding history of returns assigned to each regime and shrunk toward the pre-out-of-sample unconditional moments with weight  $n/(n + K)$  at  $K = 36$ ; the equity–bond correlation is shrunk identically and combined with the conditional volatilities to form the covariance; the predicted probabilities are optionally smoothed with an exponential half-life  $h$  and renormalised across the three classes. The reported grid varies the risk aversion  $\gamma \in \{0.25, 0.5, 1, 2, 3, 5, 8, 12\}$ , the turnover multiplier  $\tau \in \{0, 1, 3, 5\}$  (the turnover penalty in (8) is  $\tau c \|w - w_{t-1}\|_1$ , the same  $\tau c$  convention as the INDEPENDENT side), the one-way cost  $c \in \{5, 10, 25, 50\}$  bps, and the probability-smoothing half-life  $h \in \{0, 3, 6\}$ . The three headline strategies are selected mechanically on the headline-cost grid: the defensive role maximises the Sharpe ratio subject to a maximum drawdown no worse than  $-15\%$ ; the low-turnover role minimises annual turnover subject to a Sharpe ratio at least 0.85 times the best grid Sharpe; and the high-return role maximises the annualised excess return subject to a maximum drawdown no worse than  $-35\%$ , with each role excluding the strategies already taken. Table I.41 reports the three headline strategies with the full set of metrics (CVaR and turnover, condensed in the body Table 4), and Table I.42 stresses the reference logistic probability-mixture mean–variance allocator (risk aversion 5, no smoothing) to the transaction cost and turnover multiplier.

**Table I.41:** Coupled allocation: headline results with full metrics, out of sample 1996-02 to 2024-10 (345 months). CVaR is the mean of the worst 1% of monthly excess returns; “Final” indexes 100 at the start; “Turn.” is annual turnover. The 60/40 benchmark common to both approaches is reported once in Table 4.

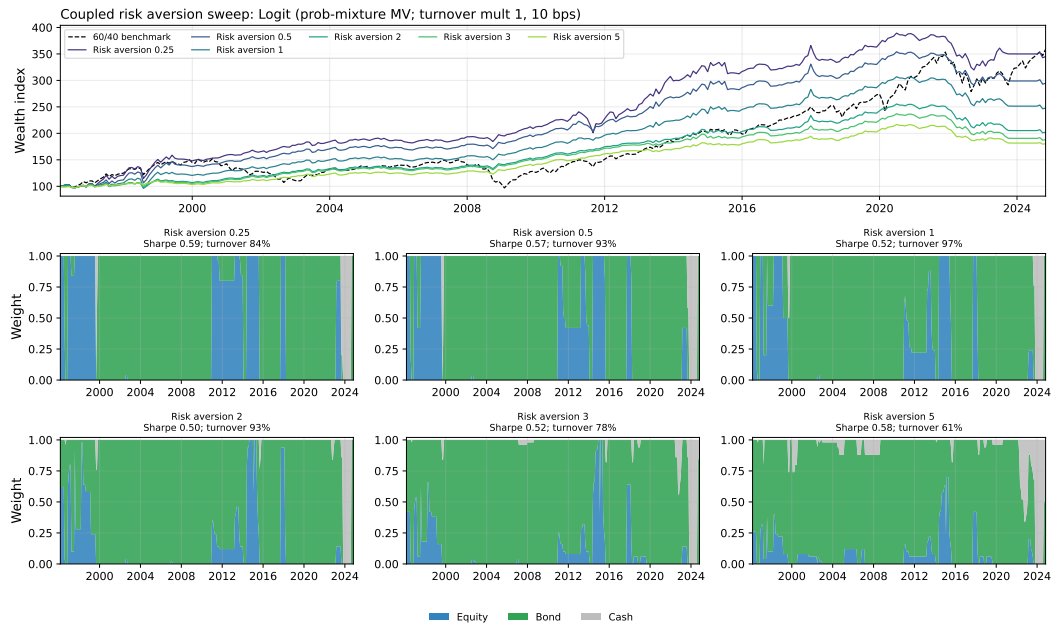
| Role         | Strategy                                   | Ann. exc. | Vol.  | Sharpe | Max DD | CVaR <sub>1%</sub> | Final | Turn. |
|--------------|--------------------------------------------|-----------|-------|--------|--------|--------------------|-------|-------|
| Defensive    | Logistic, $\gamma=3$ , $\tau=3$ , $h=0$    | +2.26%    | 3.15% | 0.719  | −7.5%  | −2.73%             | 188.8 | 23.0% |
| Low-turnover | Logistic, $\gamma=12$ , $\tau=3$ , $h=3$   | +0.70%    | 1.14% | 0.613  | −4.6%  | −1.14%             | 121.9 | 3.3%  |
| High-return  | Logistic, $\gamma=0.25$ , $\tau=1$ , $h=6$ | +4.92%    | 8.11% | 0.606  | −21.3% | −8.23%             | 372.9 | 24.3% |

**Table I.42:** Coupled allocation: robustness of the reference probability-mixture mean–variance allocator on the logistic forecast (risk aversion 5, no smoothing) to the one-way transaction cost and the turnover multiplier  $\tau$  (penalty  $\tau c \|w - w_{t-1}\|_1$ ). Cells are out-of-sample Sharpe ratios.

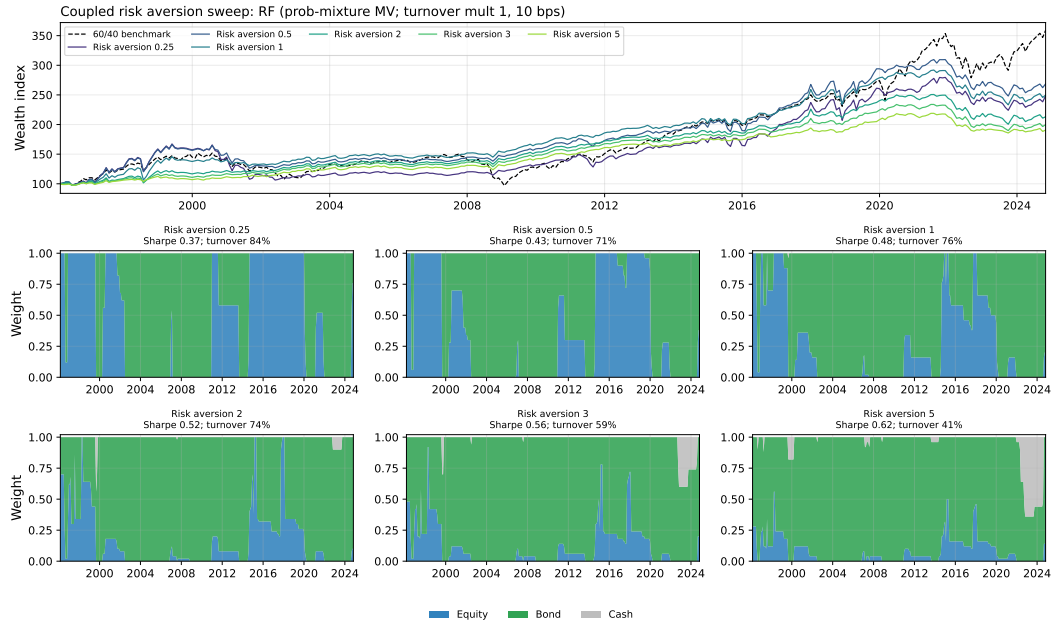
| $\tau$ | 5 bps | 10 bps | 25 bps |
|--------|-------|--------|--------|
| 0      | 0.564 | 0.545  | 0.487  |
| 1      | 0.575 | 0.580  | 0.713  |
| 3      | 0.599 | 0.651  | 0.284  |
| 5      | 0.725 | 0.420  | 0.289  |



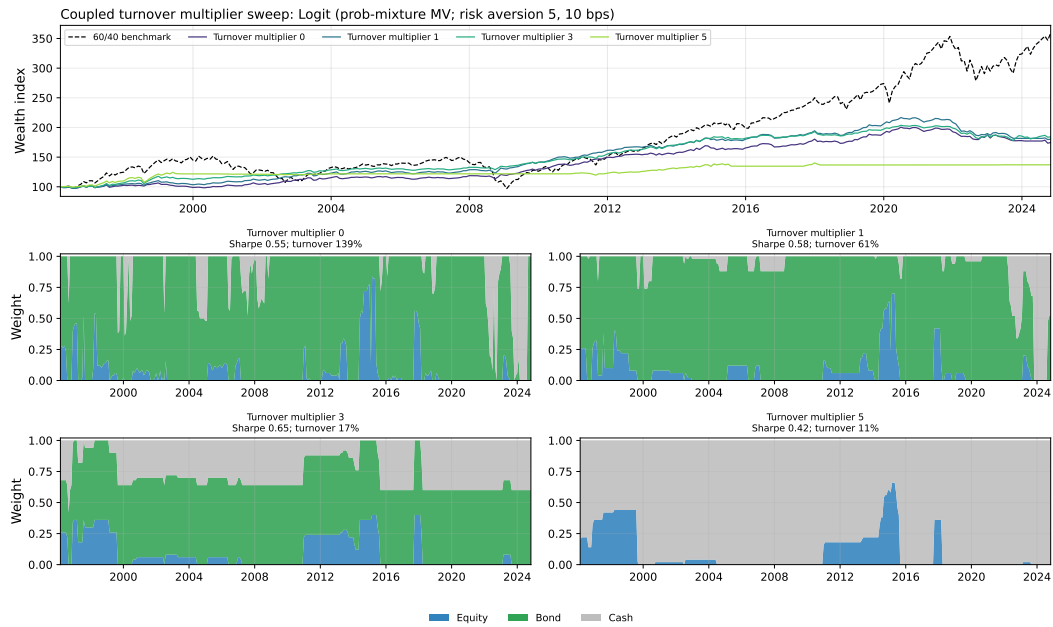
**Figure I.45:** Coupled allocation mechanics in detail for the three headline strategies. Rows, top to bottom: predicted regime probabilities; equity/bond/cash weights; monthly turnover; wealth index against the 60/40 (dashed); drawdown.



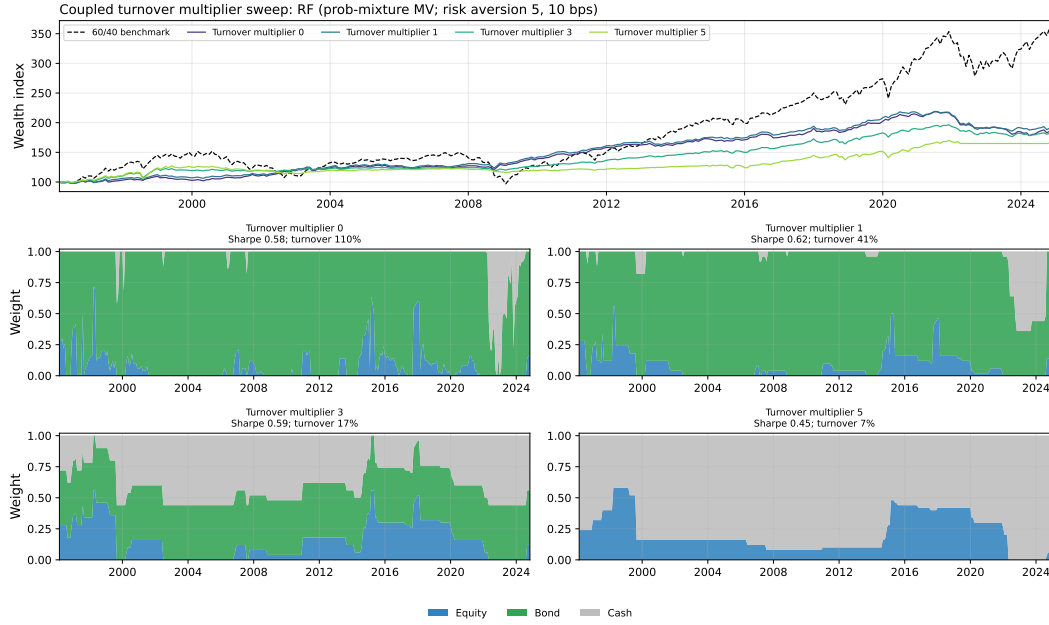
**Figure I.46:** Coupled allocation: risk-aversion sweep of the probability-mixture mean-variance allocator on the logistic forecast.



**Figure I.47:** Coupled allocation: risk-aversion sweep on the random-forest forecast.



**Figure I.48:** Coupled allocation: turnover-penalty sweep on the logistic forecast (risk aversion fixed at 5).



**Figure I.49:** Coupled allocation: turnover-penalty sweep on the random-forest forecast.

### I.1 Coupled allocation: additional benchmarks and parameter sensitivity

Table I.43 reports the coupled static and unconditional benchmarks over the out-of-sample span, and Table I.44 reports the sensitivity of the probability-mixture mean–variance allocator to the moment-shrinkage constant  $K$  (the headline strategies use  $K = 36$ ). Figures I.50–I.61 report one-parameter sweeps around the three selected strategies (high-return, defensive, low-turnover), matching the independent sweeps in Appendix H.

**Table I.43:** Coupled allocation: static and unconditional benchmarks, out of sample 1996-02 to 2024-10. Annualised excess returns net of 10 bps one-way costs; “Final” indexes 100 at the start; “Turn.” is annual turnover. The common 60/40 benchmark is reported in Table 4; the rows below are the coupled static and unconditional benchmarks.

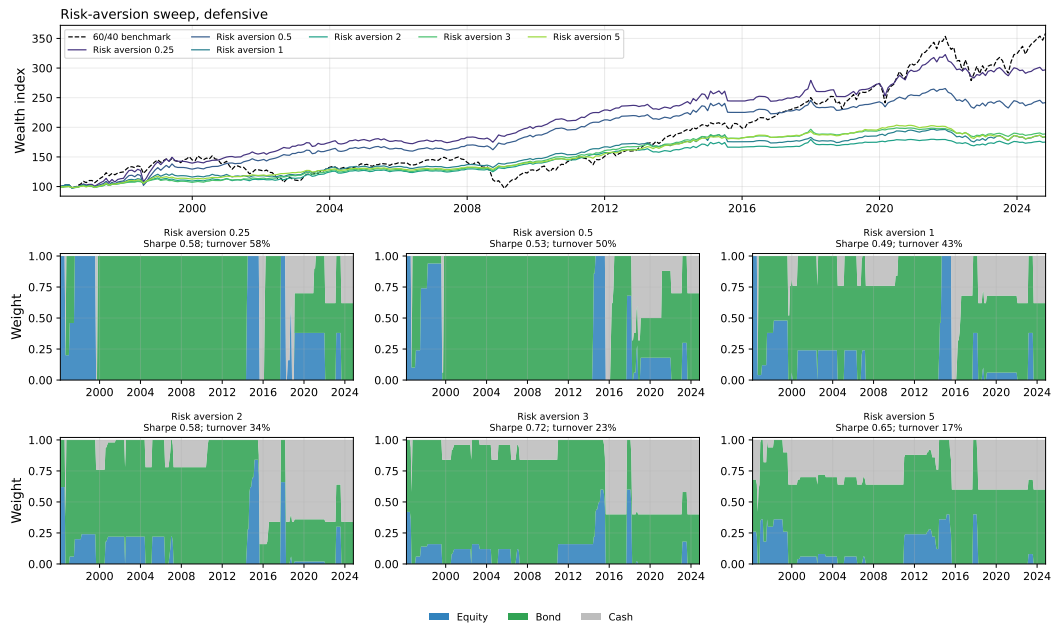
| Benchmark              | Ann. exc. | Vol.   | Sharpe | Max DD | Final | Turn. |
|------------------------|-----------|--------|--------|--------|-------|-------|
| 50/50                  | +4.46%    | 8.26%  | 0.540  | −29.9% | 326.2 | 3.5%  |
| Equity-only            | +6.85%    | 15.43% | 0.444  | −62.2% | 505.0 | 3.5%  |
| Bond-only              | +2.07%    | 4.07%  | 0.509  | −21.3% | 177.1 | 3.5%  |
| Min-variance (uncond.) | +2.20%    | 4.07%  | 0.540  | −20.2% | 183.7 | 15.5% |
| Vol-targeted 60/40     | +3.26%    | 6.92%  | 0.471  | −31.3% | 238.0 | 17.1% |

**Table I.44:** Coupled allocation: sensitivity of the probability-mixture mean–variance allocator to the moment-shrinkage constant  $K$  (annualised, net of 10 bps one-way costs). The headline strategies use  $K = 36$ .

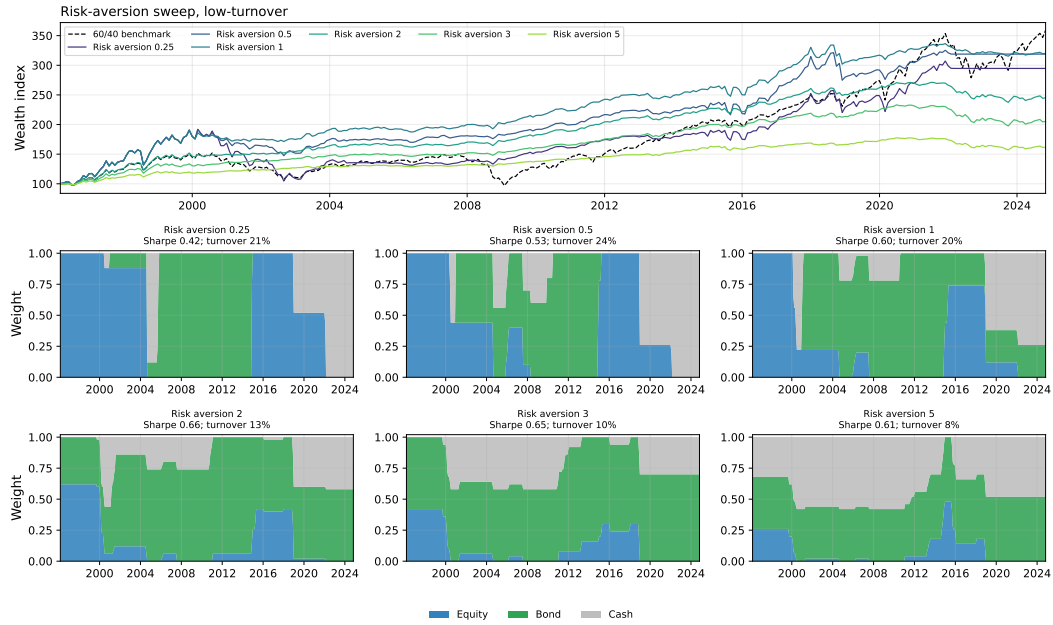
| Shrinkage $K$ | Ann. return | Ann. vol. | Sharpe | Ann. turnover |
|---------------|-------------|-----------|--------|---------------|
| 0             | +1.72%      | 3.12%     | 0.551  | 124.1%        |
| 12            | +1.93%      | 3.34%     | 0.577  | 110.1%        |
| 24            | +2.07%      | 3.49%     | 0.594  | 81.0%         |
| 30            | +2.11%      | 3.58%     | 0.590  | 69.3%         |
| 36            | +2.12%      | 3.65%     | 0.580  | 60.5%         |
| 42            | +2.12%      | 3.71%     | 0.573  | 52.9%         |
| 48            | +2.24%      | 3.77%     | 0.593  | 48.1%         |
| 60            | +2.37%      | 3.89%     | 0.607  | 40.8%         |
| 96            | +2.51%      | 4.05%     | 0.619  | 27.2%         |



**Figure I.50:** Coupled allocation: risk-aversion sweep around the high-return strategy.



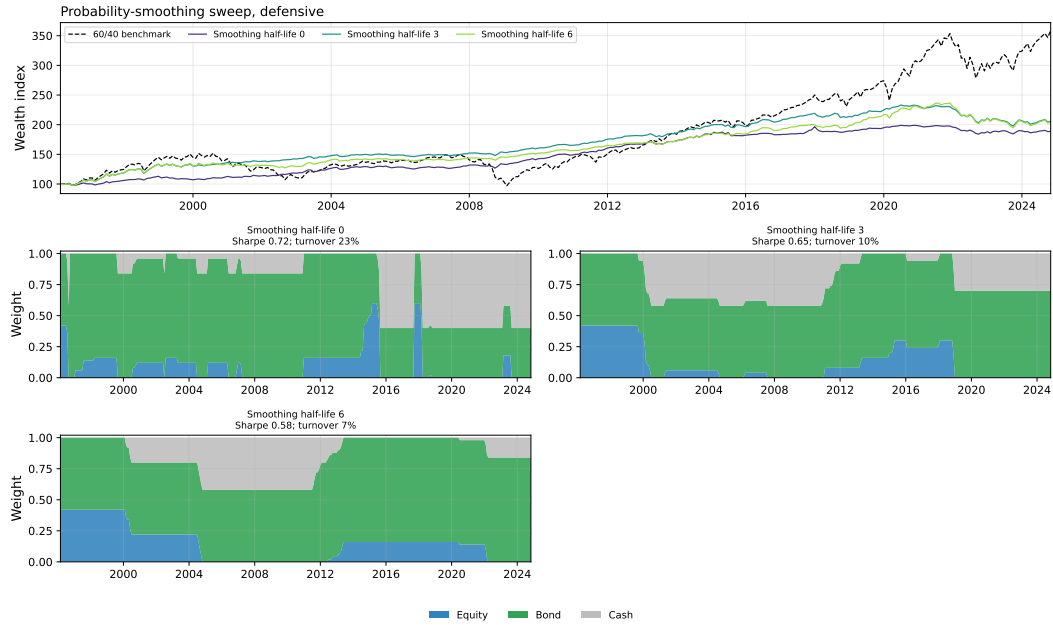
**Figure I.51:** Coupled allocation: risk-aversion sweep around the defensive strategy.



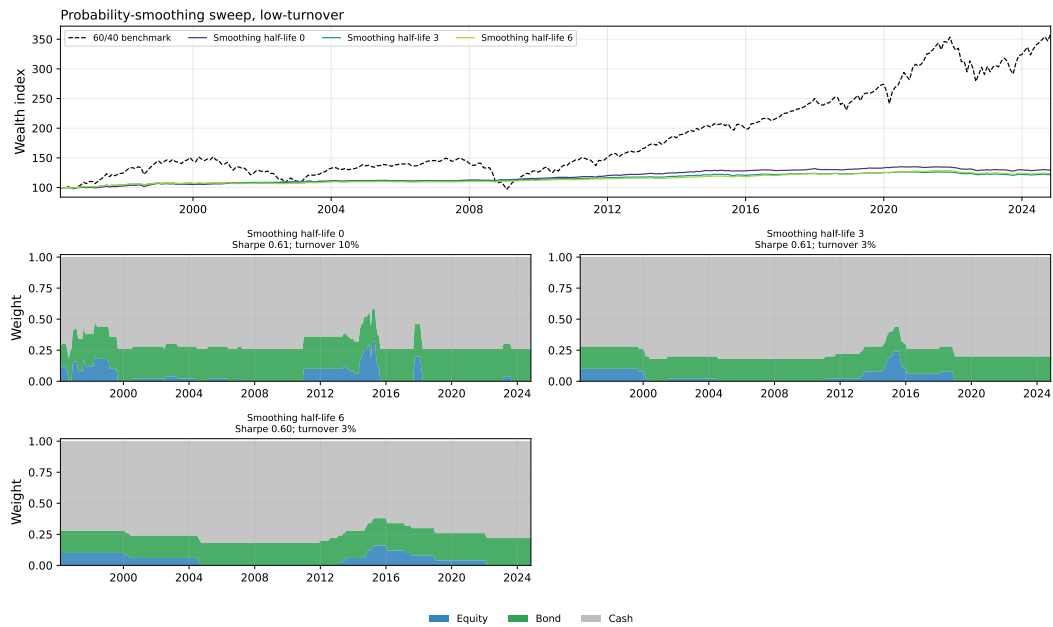
**Figure I.52:** Coupled allocation: risk-aversion sweep around the low-turnover strategy.



**Figure I.53:** Coupled allocation: probability-smoothing sweep around the high-return strategy.



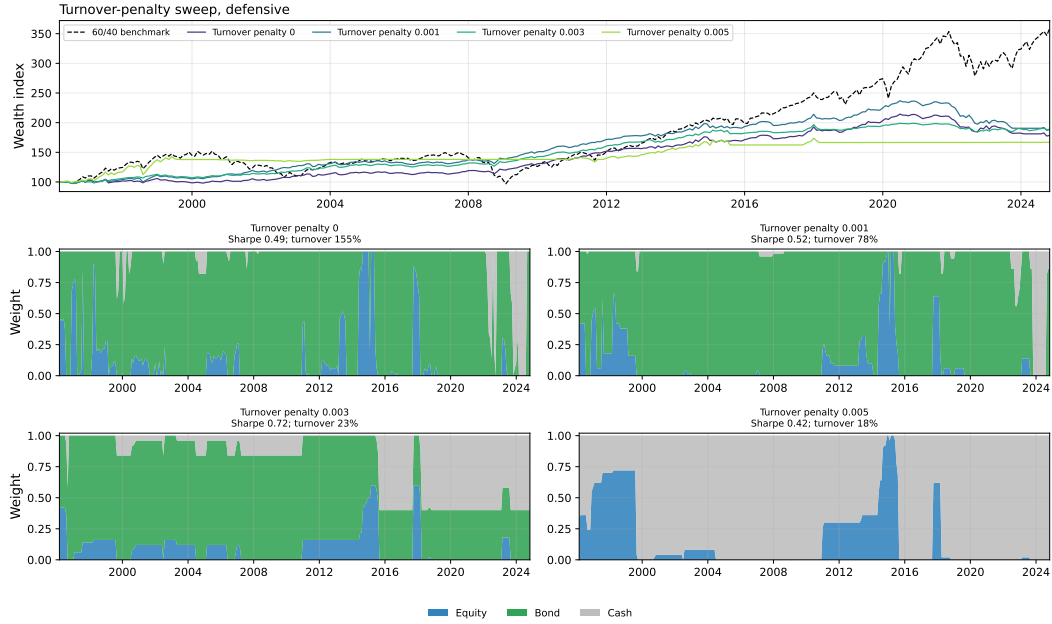
**Figure I.54:** Coupled allocation: probability-smoothing sweep around the defensive strategy.



**Figure I.55:** Coupled allocation: probability-smoothing sweep around the low-turnover strategy.

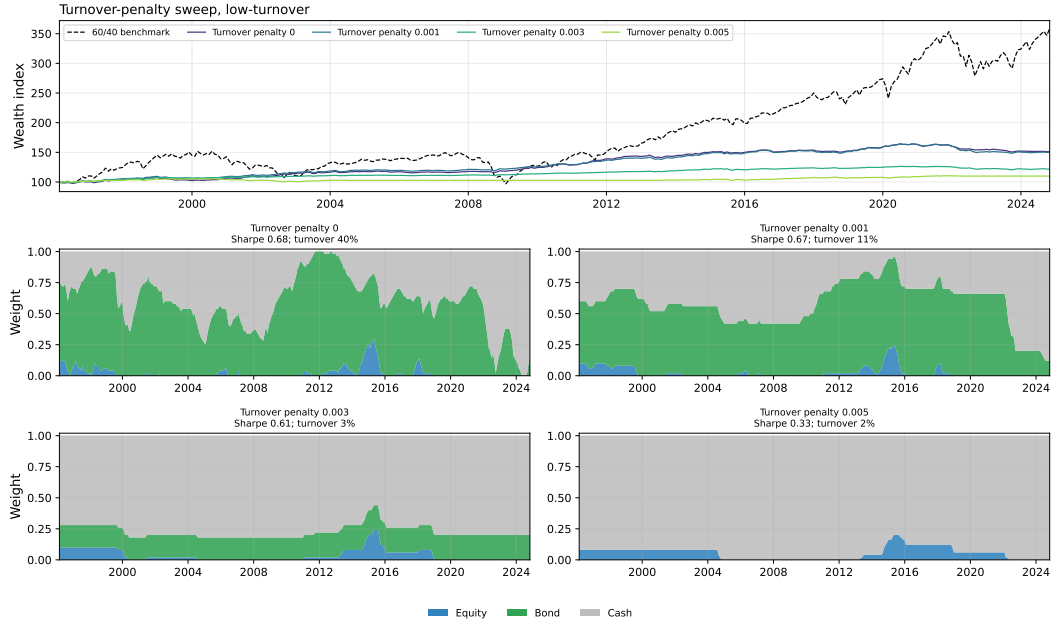


**Figure I.56:** Coupled allocation: turnover-penalty sweep around the high-return strategy. The horizontal axis is the turnover penalty  $\tau c$ , the coefficient on  $\|w - w_{t-1}\|_1$  in objective (8), using the same  $\tau c$  convention as the INDEPENDENT side.



**Figure I.57:** Coupled allocation: turnover-penalty sweep around the defensive strategy. The horizontal axis is the turnover penalty  $\tau c$ , the coefficient on  $\|w - w_{t-1}\|_1$  in objective (8), using the same  $\tau c$  convention as the INDEPENDENT side.





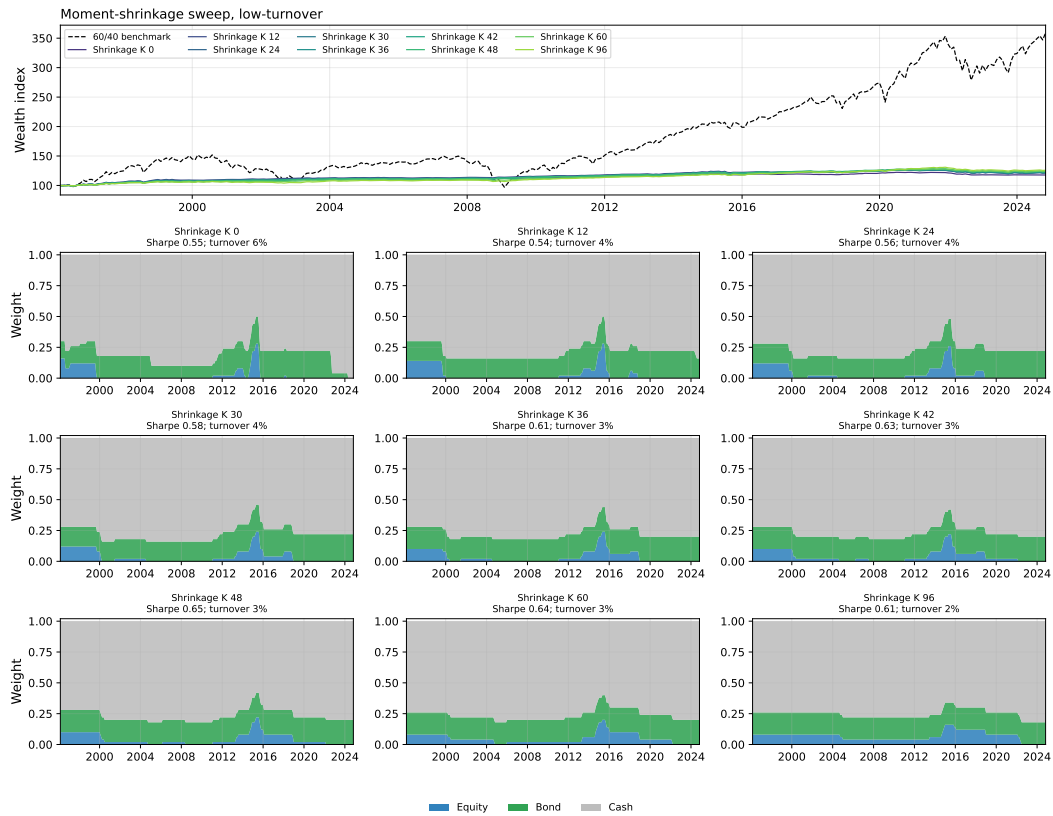
**Figure I.58:** Coupled allocation: turnover-penalty sweep around the low-turnover strategy. The horizontal axis is the turnover penalty  $\tau c$ , the coefficient on  $\|w - w_{t-1}\|_1$  in objective (8), using the same  $\tau c$  convention as the INDEPENDENT side.



**Figure I.59:** Coupled allocation: moment-shrinkage sweep around the high-return strategy.



**Figure I.60:** Coupled allocation: moment-shrinkage sweep around the defensive strategy.



**Figure I.61:** Coupled allocation: moment-shrinkage sweep around the low-turnover strategy.

## **I.2 Coupled model-selection protocol**

This table summarises which coupled-side choices are fixed design choices, which are selected on validation or grid search, and which are used only as diagnostics. As with the independent protocol in Table H.40, the purpose is to separate the reported coupled specification from robustness checks and presentation-only diagnostics. The coupled teacher produces a single joint three-state label (calm, normal, stressed), and the student forecasts that joint label directly.

| Component                             | Coupled approach                                                                                                                                                                                                                                                        | Fixed in advance                                                             | Selected on validation/grid                                                                                  | Robustness or diagnostic only                                            |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| Teacher feature set                   | Joint equity and bond return-based features feeding one three-state label                                                                                                                                                                                               | Final reported feature blocks fixed before student prediction and allocation | No                                                                                                           | Per-leg feature-separation summaries used only as diagnostics            |
| Teacher state structure               | Three joint regimes (calm, normal, stressed) built from the two per-leg state bits                                                                                                                                                                                      | Three-state joint structure fixed before estimation                          | No; the mapping from per-leg bits to the joint label is fixed                                                | Per-leg state economics reported as diagnostics                          |
| Teacher jump penalty grid             | Shared jump-penalty grid $\lambda$ with coupling parameter $\eta$ linking the legs                                                                                                                                                                                      | Grid fixed before final majority vote                                        | Majority-vote label selected from fitted teacher paths                                                       | Jump-penalty diagnostics reported across $\lambda$ and $\eta$            |
| Distance norm                         | $L^2$ centroid and $L^1$ -medoid teacher paths                                                                                                                                                                                                                          | Both norm families fixed as teacher candidates                               | Final label uses the combined majority vote across members                                                   | Separate $L^2$ and $L^1$ paths reported as diagnostics                   |
| Teacher ensemble and majority vote    | Final joint label combines the 22 ensemble members by majority vote                                                                                                                                                                                                     | Voting rule fixed before student prediction                                  | No refit after allocation outcomes                                                                           | Member raster and vote-scope figures report candidate-path agreement     |
| Student predictor sets                | Lagged macro-financial predictors for the next-month joint regime                                                                                                                                                                                                       | Transformations fixed before model fitting                                   | Single reported predictor design evaluated out of sample                                                     | Individual-predictor importance figures are diagnostic                   |
| Student model hyperparameters         | Three-class logistic regression and random forest specifications                                                                                                                                                                                                        | Model classes fixed before evaluation                                        | Hyperparameters selected within the rolling validation design                                                | Most-frequent-specification table reports selected settings              |
| Student validation metrics            | Macro one-vs-rest AUC, balanced accuracy, multiclass Brier score, and log loss                                                                                                                                                                                          | Metrics fixed before reporting                                               | Best reported specifications selected from rolling outputs                                                   | Full three-class prediction diagnostics kept in appendix                 |
| Allocation rule                       | Long-only equity–bond allocation with cash allowed and an $L^1$ turnover penalty, mapping regime probabilities to moments                                                                                                                                               | Asset universe and allocation objective fixed before grid search             | Candidate grid varies risk aversion, turnover penalty, transaction cost, and probability smoothing           | Parameter-sensitivity appendix reports one-way checks                    |
| Probability-to-moment mapping         | Probability-mixture mapping over the three joint regimes, with state moments shrunk to pre-out-of-sample unconditional moments                                                                                                                                          | Mapping and shrinkage form fixed as the INDEPENDENT report                   | Headline and reported rows use the probability mixture                                                       | Shrinkage constant $K$ and probability smoothing reported as sensitivity |
| Risk aversion                         | Mean–variance allocator with risk aversion from the grid $\gamma \in \{0.25, 0.5, 1, 2, 3, 5, 8, 12\}$                                                                                                                                                                  | Grid fixed before headline reporting                                         | Headline rows use selected grid points                                                                       | Risk-aversion sweeps reported as sensitivity                             |
| Transaction cost and turnover penalty | One-way transaction-cost grid with a $\tau c$ turnover penalty (same convention as the INDEPENDENT side)                                                                                                                                                                | Grid fixed before headline reporting                                         | Headline coupled rows use 10 bps                                                                             | Cost and turnover-multiplier sensitivity reported in appendix            |
| Headline allocation rows              | Defensive, low-turnover, and high-return rows, defined mechanically on the headline-cost grid: defensive = maximum Sharpe s.t. max draw-down $\geq -15\%$ ; low-turnover = minimum annual turnover s.t. Sharpe $\geq 0.85 \times$ best grid Sharpe; high-return = maxi- | Role objectives and constraints fixed before reporting                       | Each role is the grid point optimising its objective subject to its constraint, excluding rows already taken | Full-grid sweeps reported only as sensitivity diagnostics                |

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